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An outdoor kitchen concept for Livsyta

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Ett designkoncept på utomhuskök för Livsyta

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Sammanfattning

Det här är en masteruppsats skriven under våren 2022 inom Teknisk Design vid Kungliga Tekniska Högskolan, KTH. Syftet med projektet var att utveckla ett nytt utomhuskökskoncept för företaget Livsyta AB. Företaget specialiserar sig på att sälja holistiska matlagningsupplevelser i form av utomhuskök. Företagsägaren, Petter Andersson, har som mål att skala upp sitt företag i samband med en ny iteration av hans tidigare utomhuskök. Det nya konceptet är menat att vara modulärt och grundas i skandinavisk och japansk design samtidigt som det ska erhålla de nödvändiga funktionerna för ett komplett utekök. Leverabler för projektet är en ny design för utomhusköket i form av CAD-modeller och renderingar. Utöver det ska även tekniska ritningar göras.

Projektets arbete följer double diamond metoden och är indelat i tre faser. Inledningsvis utfördes en litteraturstudie för att skapa en förståelse för standardisering inom köksdesign samt skandinavisk och japansk design. I den första fasen utfördes undersökningar i relation till hur utomhusköksmarknaden ser ut idag och den påvisade ett potentiellt glapp i nuvarande utbud. Tillsammans med marknadsanalyser gjordes även användarstudier genom både en enkät samt intervjuer för att skapa en uppfattning om behov och önskemål hos användare. Detta sammanfattades till personas och slutligen en kravspecifikation. Under den andra fasen påbörjades idegenereringsfasen där en kombination av idégenereringsmetoder användes samt en materialanalys utfördes. Det möjliggjorde att tre koncept kunde utvecklas och presenteras för företaget. I den tredje och sista fasen presenterades

det slutgiltiga konceptet som inkluderar konstruktion samt val kring material och vitvaror.

Den slutgiltiga produkten är ett modulärt utomhuskök med inspiration av både skandinavisk och japansk design. Köket presenterades genom CAD-modeller, renderingar samt tekniska ritningar.

Nyckelord: utomhuskök, modulärt, skandinavisk design, japansk design



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Abstract

This is a master thesis done during Spring 2022 within Industrial Design Engineering at the Royale Institute of Technology, KTH. The purpose of the project is to develop a new outdoor kitchen concept for the company Livsyta AB. The company focuses on selling holistic cooking experiences in the form of outdoor kitchens. The company owner, Petter Andersson, have a goal of scaling up his company with creating a new iteration of his previous outdoor kitchen designs. The new developed concept is meant to be modularized and based on Scandinavian and Japanese design and should include the necessary functions for a complete outdoor kitchen. The deliverable in the project is a new outdoor kitchen concept design presented in form of CAD models and renderings. In addition, another deliverable is technical drawings of the outdoor kitchen concept.

The project work follows the double diamond method and is divided into three phases. An initial frame of reference was made to understand standardization within kitchen design as well as the characteristics of both Scandinavian and Japanese design. In the first phase, background research was made in order to get an understanding of today's outdoor kitchen market where the research showed a potential gap in the range of products available. Together with the market analyzes, user studies were made in form of both a survey and interviews in order to understand needs and desires of outdoor kitchen users. The gained insights and conclusions were condensed into personas and a requirement specification. During the second phase, various idea generation methods were used together with an in-depth material

analysis which led to three developed concepts being presented to the company. In the third phase, the final outdoor kitchen concept design was developed including its construction and choices of material and hardware.

The final product is a modular outdoor kitchen design drawing inspiration from both Scandinavian and Japanese design. The kitchen was realized through CAD models, renderings and technical drawings.

Keywords: outdoor kitchen concept, modular, Scandinavian design, Japanese design

Prephase

This have been a very fun and interesting project where we have had the opportunity to utilize our knowledge gained from our education. From this, we have widen our skills as product developers and engineers. Firstly, we would like thank Petter Andersson from Livsyta AB who gave us the opportunity to work with this project and helped us on the way. It has been a pleasure. Secondly, we would also like to thank our supervisor, Martin Sjöman, and extra supervisor, Stefan Ståhlgren, for the entertaining meetings. You gave us valuable input and great guidance during the project work. And finally, a big thank you to all the people who shared their opinions and experiences with cooking and outdoor kitchens.

Filippa Skog
Maximilian Tallbom
Summer, 2022

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1 Background

The company Livsyta specializes in designing and building customized outdoor kitchens and cutting boards. It was founded in 2016 by the business developer Petter Andersson and started as a hobby project where Petter found a gap in the outdoor kitchen market. He felt that many of the current outdoor kitchens were compensated in favor of functionality instead of design and due to this wanted to produce more sophisticated and inviting kitchens. Petter has built two kitchen concepts and produced several cutting boards and in *Figure 1.1* examples of these can be seen.



Figure 1.1: Images of Livsyta's existing kitchens and cutting boards.

Livsyta is a company that also focuses on selling holistic cooking experiences. In addition to selling outdoor kitchens and kitchen utensils, Livsyta also attaches great importance to the experience of these products. Livsyta's kitchens are something that friends and family will gather around and together create memories. Therefore, the kitchens are imbued with a sense of inclusiveness, warmth and luxury. Furthermore, Petter also attaches great importance to well designed and long-lasting outdoor kitchens. During the year 2022, Petter wants to scale his company and start to produce more outdoor kitchens. In order to scale the production a basis from user studies, materials analyzes and construction need to be analyzed and created.

Therefore, this project will cover the development and redesign of an outdoor kitchen for Livsyta.

1.1 Introduction

The purpose of this project is to develop and design an outdoor kitchen concept for Livsyta. The focus will be on developing a modular outdoor kitchen in a Scandinavian and Japanese style. Furthermore, the design of the kitchen will also be based on literature and user studies. Among other things, standards in kitchens will be investigated as well as the needs and desires of today's users of outdoor kitchens. In addition, a deeper market analysis will be done in order to define where Livsyta as a company should position itself. To ensure that the outdoor kitchen lives up to a certain quality and durability, several material, manufacturing and production analyzes will be done. Lastly, the final design of the outdoor kitchen will be visualized in CAD-models and technical drawings which Livsyta can use for future production.

1.2 Deliverables

The deliverables in the project are the following:

- A finished design concept of the outdoor kitchen presented in CAD-models and renderings in form of product images.
- Technical drawings of the outdoor kitchen which can be used for manufacturing and production.

1.3 Delimitations

One limitation in the project was the study of standards within kitchens. The project was limited to Swedish indoor kitchen standards and recommendations regarding dimensions, hygienic and safety aspects. Should the final outdoor kitchen concept be manufactured and sold in other countries, it is therefore important to investigate whether other standards and regulations apply.

The project was limited to the overall construction and design of the outdoor kitchen and not the associated hardware. Chosen hardware, such as a sink, tap, grill and gas stove will be bought from a retailer and not be designed from scratch.

1.4 Methods

The project work follows the double diamond process and it has been divided into three different phases. The double diamond process usually has four phases: discover, define, develop and deliver (Ball, 2019). However, due to this project's limited time frame, the two last phases were combined in order to be able to generate a final concept.

In these three phases, various methods were used in order to create and develop understandings and insights that were crucial for the project and ongoing work. The three phases' methods, findings and conclusions will be presented in descending order, where one phase sets the foundation for the next one. In *Figure 1.2* the adapted double diamond process with the different methods that was used during this project as well as the divided phases is presented. Initial planning for the project was made in a GANTT-schedule and project work was gathered in Miro and Google Drive.

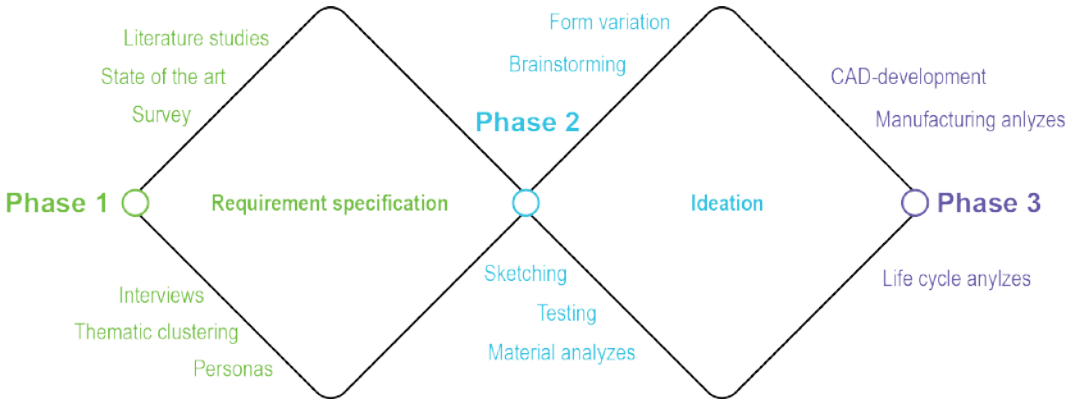


Figure 1.2: The adapted double diamond process in relation to phases and methods used in the project.

The first phase consisted of literature studies, market analyzes and user studies. These were used to establish an extensive understanding of the design guidelines within Scandinavian and Japanese design, the current market and users' behaviour and needs. In the first phase, the methods used were *State of the art*, *Perceptual mapping*, quantitative and qualitative methods such as *Surveys* and *Interviews*. From the user studies, the data was analyzed using *Thematic clustering* which emanated into *Personas*. Finally, all insights and conclusions from Phase I were summarized in a requirements specification which was used for further work.

During the second phase, the idea generation process began, where different outdoor kitchen concepts were developed using the methods *Brainstorming*, *Sketching* and *Storyboarding*. Furthermore, deeper material analyzes were done using the *Granta EduPack* software. In combination with the material analyzes, visits were paid to the Materials Library and different interior stores. At the end of Phase II an evaluation

of suitable materials for the final outdoor kitchen concept was made.

The third and last phase consisted of finalizing the chosen concept in CAD-models and renderings. The software used were the CAD software *Solid Edge 2021* and the rendering engine *Keyshot 10*.

2 Frame of reference

The first phase began with literature studies searching for relevant information regarding the layout and design of the finalized outdoor kitchen concept. The following chapter will describe the fundamental recommendations and guidelines that will set the overall formgiving for the finalized outdoor kitchen. The frame of reference is divided into three parts: standardization within kitchen design, Scandinavian and Japanese design. These parts will cover both the practical and aesthetical aspects which will affect the upcoming concept development work.

2.1 Standardization within kitchen design

When building kitchens certain standards and recommendations need to be taken into consideration. The are several recommendations regarding the different dimensions of the kitchen and how these are adapted depending on different arrangements. Furthermore, the kitchen design has also to be made concerning hygienic reasons, to create a cooking environment with a minimized risk of both food contamination and poisoning. Finally, there are also safety recommendations regarding handling fire and gas, and these will be followed to minimize the risk of any accidents.

2.1.1 Dimensions of kitchen modules

The Swedish Institute for Standards, shortened SIS, specializes in standards both nationally and internationally (Swedish Institute for Standards, 2022). SIS is in close connection with both public and private stakeholders to promote competitiveness and ensure products made are smart and sustainable.

SIS separates commercially used kitchens such as restaurant kitchens with kitchens used in general homes. According to the standard SS-EN 1116:2018, the overall dimensions of kitchens used in general homes differ depending on their arrangement (Swedish Institute for Standards, 2018). For example, if the kitchen is against a wall or displayed as a kitchen island. In the standard SS-EN 1116:2018, the different possible dimensions of a kitchen are summarized and clustered in their respective arrangement. The standard recommendations that have been taken into consideration are the dimensions of kitchen islands. For example the height, depth and width of each module.

The standardized dimensions of these three are the following:

- Height: Between 850-950 mm
- Depth: Minimum 600 mm
- Width: Between 450-900 mm

2.1.2 The work triangle

When designing a kitchen the overall layout should also be taken into consideration in order to create the most optimized and natural workflow. The work triangle is an established method that is used when planning kitchens. The method describes the best placements of the sink, oven, refrigerator, etc in relation to each other (Kvik, 2022) and how a triangle can be drawn between them. The triangle represents the movement between each kitchen module. In linear kitchens it is impossible to draw a triangle, however, efficiency and natural workflow can still be achieved through careful configuration of the kitchen modules.

The workspace where cutting boards and similar equipment can be used should be placed on the right side of the sink. Furthermore, tools connected to what is used in the workspace should be placed as close to the workspace as possible. In addition, the stove should be placed near the workspace area so it is easy to access the prepared food. Spices and oils should also be close to the stove so they are easily added when cooking the food. Finally, storage should be placed in the area where it is used in order to make the workflow more efficient.

2.1.3 Food hygienics

In kitchen design, it is important to take food hygienics into regard. This can easily be forgotten due to prioritizing efficiency, functionality, and practicality (De-Alwis, 2017). There are various things that affect the overall hygienic environment of a kitchen. Smaller gaps and other unreachable areas can often be overlooked when cooking and cleaning in the kitchen. In indoor environments, gaps between counters and walls could be a breeding ground for pests and microbes. In combination with having organic materials, such as wood and concrete, this can also create irreversible damage to the overall kitchen. In outdoor kitchens, these are even more common problems. Small animals and birds could open up to a potentially unhygienic environment, as feces and other animal secretions could create an unsafe environment for cooking. This can be avoided by reducing the number of gaps in the kitchen or using sealants to close these areas. Another option would be to have

certain removable parts, enabling easier cleaning or creating easily accessible covers for the entire kitchen.

The overall layout of the kitchen in regard to hardware and work areas needs to be considered as well (Mihalache et al., 2022). Cooking food should be convenient and have a naturally hygienic flow when handling ingredients. Certain ingredients could cause food poisonings, such as raw chicken and pork and the kitchen design should make separating these easy. Smaller kitchen designs are often associated with a one-wall kitchen, or single row kitchen, which leads to a linear flow of cooking food (Wallender, 2021). These kitchen designs are compact and space-efficient as the cooking area is focused on one area of the kitchen. This enables the user to be in close proximity to utensils and appliances. These types of indoor kitchens vary in size but are usually between 2.1 to 3.8 meters in order to include everything needed for a minimum kitchen (Dimensions.com, n.d.). Sharing similarities with an assembly line, the food is prepared in steps moving from preparation, to cooking and finally to dishes (Alcequiez, n.d.). Regular cleaning of hands and tools should be seamless and overall kitchen cleaning should follow the natural flow, from clean to dirty areas. Making cleaning easy and quick will also, not only increase safety but increase productivity as well.

2.1.4 Safety regulations

Safety regulations will be needed to adhere to when designing the final concept. This is in relation to the fire hazard of a potential built-in grill and/or stove. According to Räddningstjänsten Väst, there are a few guidelines that need to be followed when using a grill and will be followed during the design process (Räddningstjänsten Väst, 2018). These are:

- The gas canister should be located on the side of the gas grill as the heat from the grill can cause damage to the gas equipment.
- The gas canister should be able to stand up right during use, storage and transport.
- As a private person, a limit of 2 P11 gas bottles can be owned without applying for a license.

A common problem with gas canisters is the occurrence of leaking equipment (Myndigheten för samhällsskydd och beredskap, 2012). Liquefied petroleum gas, commonly used in Swedish households, is 1.5 times as heavy as air and will therefore sink towards the ground or floor. Leaking gas can be checked with a leak indicator attached to the gas valve of the canister. However, in order to ensure proper use,

ventilation will have to be incorporated in the outdoor kitchen if a gas stove or grill will be used (Abhaya, 2014).

Applied for both coal and gas grills when incorporated into an outdoor kitchen is the need for a heat-dissipating material surrounding the warm areas. This will depend on the final design and will be revised later in the process.

2.2 Scandinavian design

In the following section, Scandinavian design will be described to understand the characteristics and apply them to the final outdoor kitchen concept. This chapter is divided into the history of Scandinavian design and its characteristics.

2.2.1 History

Modernism is heavily associated with Nordic design and arouse in the early 1920s after the first world war and was a cultural movement characterized by self-conscience and deviation from social norms. It was a response to what was considered conservative and outdated and did not meet the needs of a new and industrialized society (Juliao, 2022). In the footsteps of industrialization, modernism was heavily tied to standardization and mass production and emphasized production rather than consumption (Vihma, 2003).

The characteristics of modernism reflect its fundamental principles as there is a big focus on clean lines and abstract shapes, moving away from the realism that was prevalent pre-war (RIBA, 2022). This was visible in the design, as ornamentation and decoration were heavily minimized, and focus shifted to comfort together with functionality (Designing Buildings, 2021). Functionalism is closely related to modernism as the focus was moved from ornaments and unnecessary details to deliberate form that reflects the purpose, functionality, and practicality.

”Scandinavian design” is a term used to describe a certain type of interior design originating from the Nordic countries Sweden, Norway, Denmark, and Finland and emerged alongside the modernist movement sharing many similarities. The low economic growth reflected the lifestyle of many as interior and home decoration was characterized by minimalism and functionality. Materials used were those that were nearby one’s home, such as spruce and pine alongside metal and stone (Sommar, 2004). Internationally, the style of Scandinavian modern was considered modern and functional with a subtle addition of traditional values, more so than other styles of design such as Central European modernism (Skou & Munch, 2016).

After the 1950s, there was a steady decline in furniture made in Scandinavian design. This was due to many reasons, mainly the increased production of cheaper products in bad quality materials. However, Scandinavian design got an upswing around the 1990s and has remained relevant since. Finally, there are many people who, through the decades, have made a big impact on the Scandinavian style. To mention a few, there are Alvar Aalto (1898-1976), Arne Jacobsen (1902-1971), Bruno Mathsson (1907-1988) and Gert Wingårdh (1951-).

2.2.2 Characteristics

Skou et al (2016) mention Nordic design as opposed to superficiality. This is done by a slight modernization in design while preserving Nordic traditions and craftsmanship. Elements commonly found in Scandinavian design is the use of a neutral color palette due to the long winters north of Europe (Invaluable, 2019). Colors often used are different shades of white and grey as well as tan colors to create a subtle and simple space. A recurring theme within Scandinavian design is the mantra of ”less is more”. This correlates to the lack of extravagant decor and overall simplicity of furniture, shifting focus towards functionality and craftsmanship. The minimalistic nature of Scandinavian design is categorized by the fine balance between clean formgiving, often using clean and straight lines, material, color, and function. In *Figure 2.1* characteristics representing Scandinavian design and formgiving is presented.

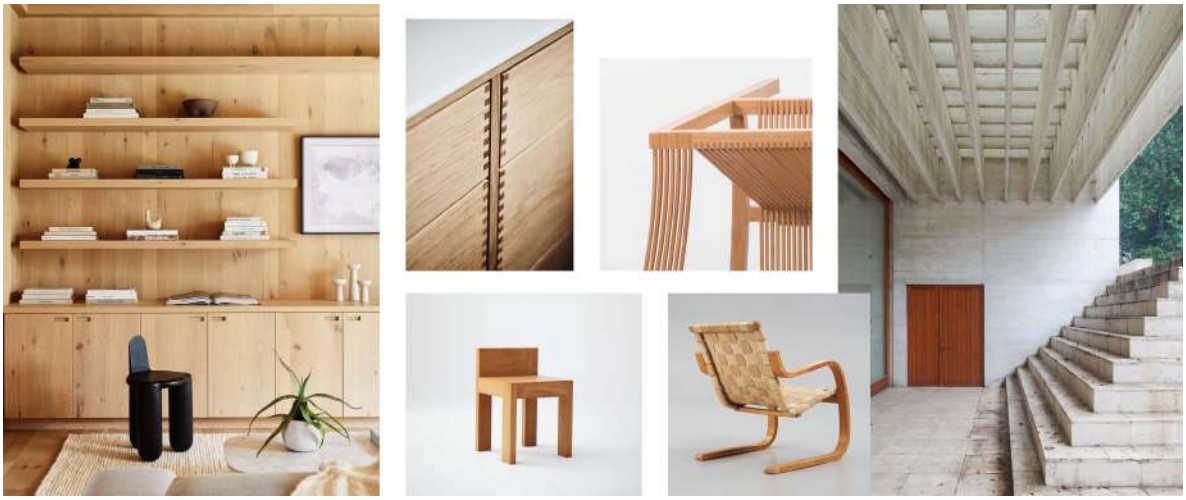


Figure 2.1: *Characteristics of Scandinavian design and formgiving.*

2.3 Japanese design

The following section describes the history and characteristics of Japanese design. The knowledge found within this section will also be used further on during the development process.

2.3.1 History

In the early twentieth century, much of the design used within Japan was heavily inspired by foreign countries (Nagata et al., 2006). Knowledge was drawn from many of the design movements present in Europe, which was later married into Japanese design. One of the earlier inspirations was the German ideas of Deutsche Werkbund and Bauhaus. Alongside the teachings of European design, emerged the philosophy of *Mingei*, which derived its design language from practicality and craftsmanship. Common themes included and were not limited to embracing tradition and anonymity of the craftsman, celebrating the simplicity of everyday items while letting go of luxurious dishonesty. *Mingei* connected the bridge between arts and crafts, something the modern European design philosophies at the time could not.

The societal damage brought onto Japan after World war II created a dire need to restore the economy and the normality of people's lives. To increase exports, an attempt to mass-produce products at low prices was made (Katsuhei & Bailey, 2016). This, however, quickly transformed into creating a large counterfeit market, causing problems internationally. This led to the creation of the Export Product Design Act in the 1950s. After the war, the focus shifted from the design philosophies of Europe to those of America. Japan started to gather information on international trends, inviting design experts from all over the world to promote new design ideologies while sending their researchers overseas (Boutard, 2021).

After the explosive economic growth in the 1960s, consumption among the people of Japan grew. It was at this time that modern Scandinavian design became increasingly associated with modern Japanese design. A relatively recent term combining elements of both Scandinavian and Japanese design is *Japandi* through similar aesthetics that are minimalism and comfort (Knierim, 2022). This merges neutral colors and clean lines with imperfection and carefully picked details.

2.3.2 Characteristics

Similar to Scandinavian design, Japanese design is characterized by its minimalism and sophistication (Swain, 2017). The Japanese style is deeply rooted in spirituality

and philosophy where each object has to be treated carefully and respectfully. There are several aesthetics within Japanese design and one that is often mentioned is *Wabi-Sabi* (Envato, 2020). *Wabi*, meaning rustic simplicity refers to the kind of beauty found in asymmetrical, uneven, or unbalanced objects (Heal's, 2019). *Sabi* means taking pleasure in the imperfect and refers to the beauty in aged objects that are impermanent. *Wabi-Sabi* inspires us to peacefully embrace the natural cycle of growth and decay where one should find beauty in the imperfect.

Furthermore, Japanese style is represented by simple, calm, and relaxed expressions. There is a balance between minimalism and complexion in every detail. Objects that may appear simple often have rich craftsmanship work behind them. For example, different sorts of joints are often used when building wooden furniture. Few and carefully selected materials often represent Japanese interior design. Natural materials in earthy colors like wood and paper are frequently used. In addition, the color white is essential within the Japanese style. White represents purity and emptiness and is often used within Japanese homes. In addition, creating natural light in Japanese homes is also very common since it brings the feeling of life and joy. Japanese design is also characterized by smart modularity. Japanese homes are often tiny and therefore spaces need to be planned efficiently. For example, dining areas can transform into workspaces or libraries with just a few steps. Finally, the overall shape and formgiving is often characterized by organic and smooth shapes with a low center of mass. In *Figure 2.2* characteristics of Japanese design and formgiving is presented.

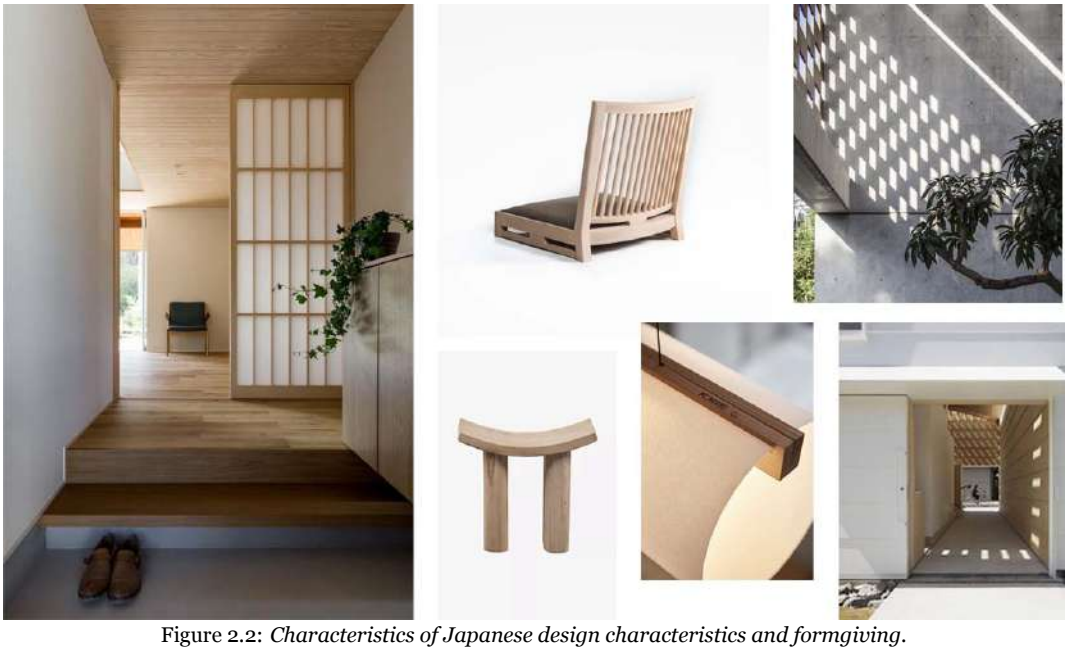


Figure 2.2: *Characteristics of Japanese design characteristics and formgiving.*

3 Phase I

After the frame of reference, the work during Phase I continued with a performed market analysis and several user studies. This laid the foundation for the upcoming concept development of the outdoor kitchen. The user studies were divided into quantitative and qualitative research in form of a survey and multiple interviews. These were summarized into insights and then personas. Finally, a requirements specification was done based on the knowledge and insights gained from the first phase.

3.1 State of the art

State of the art is a method of extensive market analysis where relevant products and potential competitors are researched (Nilsson et al., 2015). This was done in order to understand the current market of products and to decide where Livsyta as a brand and their new product should be placed in relation to similar products. In the state of the art, knowledge was gathered in relation to existing kitchen designs, brands, and prices.

At the beginning of the state of the art, research was made in order to find what defines as an outdoor kitchen. It was noted that the definition could vary a lot. For example, meanwhile, some retailers defined a simple charcoal grill as an outdoor kitchen, others called fully equipped with a grill, sink, fridge, etc. for an outdoor kitchen.

When different sorts of outdoor kitchens were found, they were also analyzed by their overall design, impressions and functionality. The found outdoor kitchens were categorized into six contrasting words: *bulky*, *bare*, *permanent*, *mobile*, *complete* and *spartan*. A *bulky* kitchen would for example comprise of multiple materials in varying colors and shapes while a *bare* would reference a more minimalistic and hygienic kitchen often in the material stainless steel, see *Figure 3.1*. A *permanent* kitchen displays robustness with a low center of gravity which is unable to move after placement. Meanwhile, a *mobile* kitchen is a lighter kitchen with a thinner frame which is possible to move because of the attached wheels, see *Figure 3.2*. Lastly, a *complete* kitchen would incorporate more functionality where everything that is needed to cook a complete meal is permanently placed within the kitchen. Meanwhile a *spartan* kitchen is reduced to the essentials and are often more open and spacious, see *Figure 3.3*. Dividing these kitchens into several categories helped gain an understanding of how different they could look in overall formgiving and design but also the functionality. This was later used during the upcoming concept

development.

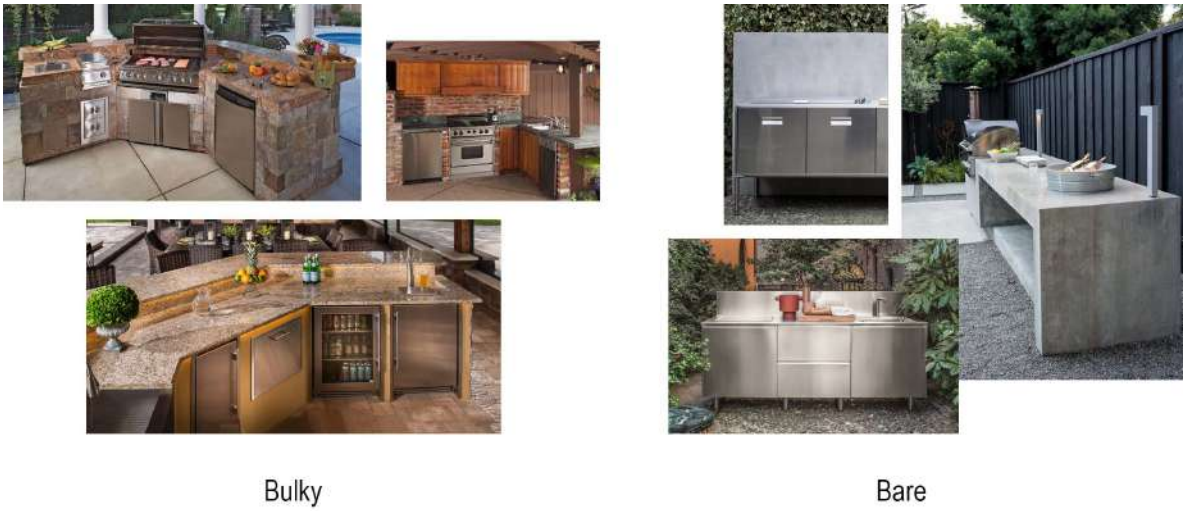


Figure 3.1: *Examples of Bulky and Bare kitchens.*



Figure 3.2: *Examples of Permanent and Mobile kitchens.*



Figure 3.3: *Examples of Complete and Spartan kitchens.*

Furthermore, different existing brands and their prices were analyzed and compared in a perceptual map. The method is based on exploring the user perception of certain brands and how they compare on a two-dimensional graph (Milton & Rodgers,

2013). The perceptual map was created to get an idea of where Livsyta should position itself concerning other competitive brands on the market. In *Figure 3.4* a perceptual map comparing several brands is presented. The axis in the maps have the variables expensive/cheap and conservative/radical. Conservative refers to a more traditional design of outdoor kitchen while radical being a more progressive design. A potential placement for Livsyta could therefore be in the gap where conservative meets expensive, see the green circle in *Figure 3.4*.

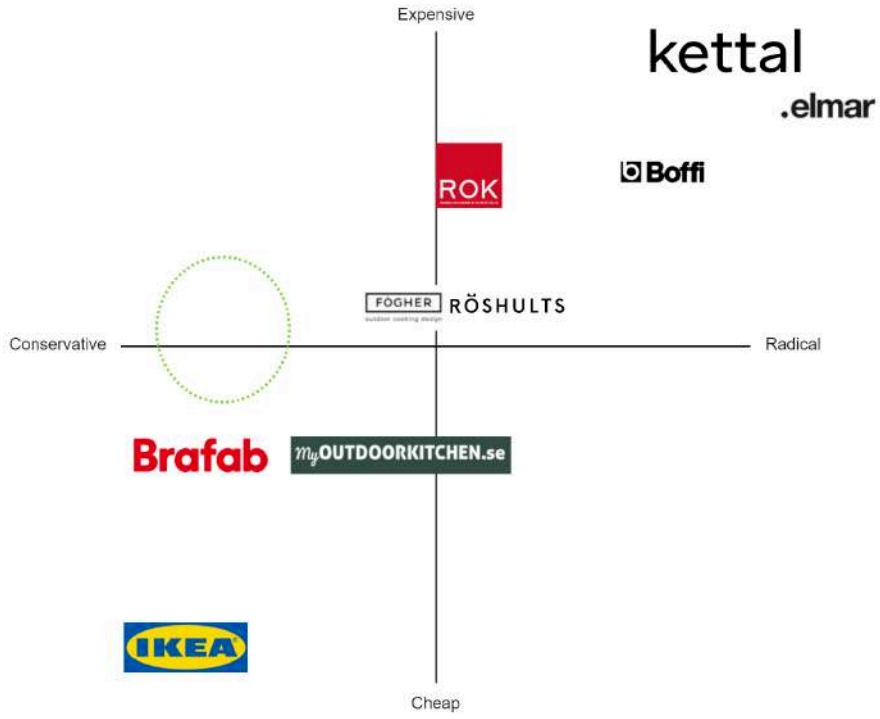


Figure 3.4: *A perceptual map of outdoor kitchen brands with the variables expensive/cheap and conservative/radical.*

3.2 Quantitative research

Following the market analysis, quantitative research was made in order to understand and define the potential target group for Livsyta.

3.2.1 Survey

The quantitative research was made as a survey, see *Appendix A*, and was sent to a dedicated outdoor kitchen and area group on Facebook. A total of 371 people answered the survey anonymously. The questions asked in the survey focused on identifying user behaviors, experiences, and needs within outdoor kitchens. Depending on how the survey was answered, different questions were asked. For example, a person who already owned an outdoor kitchen answered different

questions than a person who did not. A person who already owned an outdoor kitchen could for example answer questions regarding maintenance and cleaning of their outdoor kitchen since they have experience from it. After the collected data, the survey was cleaned by removing data that was deemed fake or not serious. Furthermore, the data were coded to easier make reasonable analyzes and assumptions regarding users.

The majority of the participants in the survey were between the ages of 31-60 years old, where 72 percent were men and the rest were women. Half of the participants owned an outdoor kitchen and the majority who did not wish they had one. It varied between how much the participants had paid for their outdoor kitchens, but most of them had not paid more than 30 000 SEK. Furthermore, a majority of the outdoor kitchen owners had built their kitchens from scratch. The most common hardware in the existing outdoor kitchens was a grill, workspace/unloading area, a sink with tap, storage, and a gas stove. Those who did not currently own an outdoor kitchen requested similar types of hardware as those who had kitchens. Furthermore, the participants answered a question regarding the best thing about their outdoor kitchens, and the answers were summarized into two categories: practical and experience. The practical category refers to the physical aspects such as the amount of storage, the modularization, and completion of the outdoor kitchen. Meanwhile, the experience category refers to the emotions that their kitchen brings such as connections made during social and common activities and feelings of freedom when being outside cooking.

The most common material used within the outdoor kitchens were wood, steel, and concrete. Other materials not as common were different types of stone, bricks, and tiles. Regardless of the materials used, winter storage was not common as around 62 percent left their kitchens out during winter and kept using them throughout the year. The rest of the participants either had their kitchen under a roof or covered it during the winter.

Except for oiling wooden parts, overall maintenance was not especially common among owners. However, those who maintain their kitchen seem to enjoy the task and usually do not spend more than a few hours per season. Those who work on their kitchen longer still seemed to enjoy doing it.

3.3 Qualitative research

The qualitative research was made in the form of semi-structured interviews. The interviews were conducted to gain deeper understandings and knowledge about

users' experiences and insights and also potential users' desires and wishes. The interviews were further summarized into personas which describe potential user needs and attitudes towards outdoor kitchens.

3.3.1 Interviews

In total, 12 semi-structured interviews were conducted with a variety of participants. There were 5 women and 7 men between the ages of 25 to 71. The interviewees' occupations ranged anywhere from content-creator, and consult to professional cook. They were people who are passionate about cooking food and enjoy doing it during their spare time. The interviewees were divided into two groups; non-outdoor kitchen owners and outdoor kitchen owners. The non-outdoor kitchen owners were chosen to understand what is required of a kitchen and what problems/irritations could arise when cooking food. The second group chosen were people who own one or several outdoor kitchens. From these people, experiences and knowledge could be gathered concerning the usage of outdoor kitchens as well as desired properties. All of the interviewees' age, sex, occupation, and ownership of an outdoor kitchen are shown in *Table* 3.1

Table 3.1: *Data of the 12 interviewed persons.*

Person	Gender	Age	Occupation	Kitchen owner
Person B	Woman	57 yr	Cook	Yes
Person C	Man	40 yr	Entrepreneur	Yes
Person D	Man	49 yr	Sommalier	Yes
Person F	Man	71 yr	Sales associate	Yes
Person G	Man	49 yr	Catering	Yes
Person H	Man	34 yr	Consult	No
Person J	Man	27 yr	Creative director	No
Person I	Woman	25 yr	Student	No
Person K	Woman	26 yr	Tech-talent recruiter	No
Person E	Woman	29 yr	Event coordinator	No
Person A	Woman	29 yr	Content Creator	No

After defining the two target groups for interviews, interview guides were created, see *Appendix B*. Here, two different interview guides were made in order to adapt the questions in regard to the two groups. The guides were then tested on people before actual interviews in order to refine it further, removing redundancy or adding

relevant questions. Each interview guide was used as a template but did not have to be followed exactly, meaning follow-up questions were asked when interesting knowledge was shared by the interviewees.

The insights from conducted interviews were clustered together according to relevant themes found, using the method thematic clustering (Braun & Clarke, 2006). Thematic clustering enables a way of finding correlations between certain statements and common problems and knowledge. These can, later on, be grouped according to themes found and how often they are mentioned. These were divided into Functionality, Design, and Experience and are presented in *Appendix C*. From the thematic clustering, it was possible to summarize the insights in categories and they are presented in the following chapters.

Hardware requirements, workspace and storage

A majority of the interviewees said that their kitchen must include a grill and a sink with a tap. Having a heat source like a grill creates both a different taste and also a feeling when cooking food than a stove/oven from an indoor kitchen. Furthermore, access to water is necessary however not mainly for cleaning dishes, but more to rinse vegetables and being able to wash hands and surfaces. In addition, some of the interviewees desired a more complete kitchen, meaning they wanted to have access to even more storage, a stove, and a pizza oven. More storage would reduce the need to go in and out to fetch tools, kitchenware, or similar. Some of the people interviewed saw no problem in leaving certain utensils and kitchenware outside while others kept only the practicalities, such as coal for the grill in their outdoor kitchens. Mentioned was having closed storage which would make storing utensils and kitchenware protected from water, dirt, or animals. Also mentioned during the interviews was that equipment common in indoor kitchens such as ovens, refrigerators and dishwashers was not necessarily something they wanted to have included in an outdoor kitchen.

”You can never have too much workspace in your kitchen.” - Person A

A consistent theme was the need for plenty of workspace and the interviewees often found themselves not having enough. This was especially a problem if there were friends and family involved in the cooking. Having more people over created a need for more space and the possibility of adapting the kitchen for this. A kitchen island was a recurring need from the interviewees as this created a sense of inclusion when having people over. A kitchen island also makes it possible for people to socialize

when the food is being cooked. Finally, being able to invite friends and family to cook food with, was one of the main reasons to get an outdoor kitchen.

Experience and social aspects

As mentioned earlier the interviewees could not emphasize more the social aspects of cooking food together. One of the main reasons to get an outdoor kitchen is to be able to invite friends and family to dine. It is a place where people meet, make memories and enjoy good-tasting food. Another aspect mentioned during the interviews was the feeling of being outside and cooking food. Being outside cooking food is an enhanced experience and also a lifestyle.

”Cooking outside is a lifestyle.” - Person G

Therefore, the owners of outdoor kitchens stressed the importance of decorating the outdoor kitchen to be as inviting and cozy as possible. Designing and decorating the outdoor kitchen as desired, leads to people using their kitchens more often and also through all seasons.

Mobility, maintenance and repairability

Also, another reason why the kitchen owners use their outdoor kitchen through all seasons was that it is too heavy to move inside. Their kitchens were built permanently in their gardens but some sensitive parts were possible to detach and move inside. In addition, being able to store sensitive parts inside could minimize the risk of theft or breakage. Furthermore, some of the kitchen owners had mobile tables that could be moved depending on how many people that are working in the kitchen. The possibility of having a mobile kitchen is applied to rearranging the layout of the kitchen more than having the opportunity to move it inside.

The kitchen owners also mentioned during the interviews that the kitchens should be easy to clean. Meaning, avoiding unnecessary seams and crevasses that could make cleaning more difficult. During spring and summer, pollen is a major issue when it comes to cleaning. Therefore, being able to rinse the outdoor kitchen makes the process easier and faster. In general, the interviewees had no problem with the maintenance of the outdoor kitchens such as oiling and sanding.

> *”I often think that everything has to be as convenient as possible. I don’t want to worry about pouring a bucket of water over it.”*

Lastly, another insight mentioned by person F was the repairability of the kitchen. Being able to remove and exchange parts that break, for example, a gas stove is important since a broken outdoor kitchen will become unusable and take up unnecessary space.

Function meets design

The people from the conducted interviews wanted their outdoor kitchens to be an addition to their overall home decor. Having a functional kitchen was important but having a good-looking kitchen added further value. The outdoor kitchen is a decoration detail. The materials within the outdoor kitchen should not only be attractive but also adapted to the environment they are in. For example, it should be resistant to water and sun but also able to manage for example heat from pans and pots. As mentioned by person B, who had bought his kitchen for its design, was very disappointed by how bad it performed.

> *”It should look like a kitchen where someone enjoys cooking food” - Person E*

Finally, also mentioned during the interviews was the aspect of the overall layout. The layout should be well planned so that the workflow becomes natural and efficient. People work differently in their kitchens and being able to adapt them accordingly is key.

3.3.2 Personas

From the completed interviews, surveys and thematic clustering, it was possible to summarize the findings into personas (Milton & Rodgers, 2013). Creating the personas made it possible to show the thoughts that overlap between interviewees and instead of showing the differences in their answers, it would highlight their similarities. In the project, four personas were created and they reflect the needs, desires and attitudes that were observed and generalized from the survey and interviews.

The first persona, Mats, is a man who sees cooking as a social activity and a cozy space. He has few needs outside of the basics and sees maintenance as a fun past-time. The second persona, Felicia, likes to experiment in the kitchen with her home grown vegetables and hunted animals. She do not care about the design and focuses more on the functionality. The third persona is Jesper who uses his kitchen as a social space, regardless of how little he likes to cook. He likes to spend money on branded

things and wants to minimize maintenance as much as possible. Finally, the fourth persona is Maria who enjoys high quality ingredients and likes to make every dinner to an experience. She considers cleaning to be very important but would like to reduce maintenance. The four personas are presented in *Appendix D*. These four personas were later considered and used during the concept development process where their different desires, needs and attitudes were taken into account and balanced.

3.4 Requirement specification

At the end of Phase I a requirement specification was made for the concept kitchen design, see *Appendix E*. This was made by summarizing the aspects and properties mentioned in the analysis of the qualitative and quantitative research. The requirement specification was categorized according to; partial functions, fundamental user function, material and design, kitchen standards, production, ergonomics and safety, quality, and market and target group. Within these categories were properties ranked according to Main function (MF), Requirement (R), and Desired (D) depending on how important they were to incorporate in the final concept design. The requirement property represents an attribute that has to be included in the final concept while the desired property is good to include but not necessary.

4 Phase II

The gathered insights and knowledge from Phase I laid the foundation for the idea generation process during Phase II. In the following phase, the definition of an outdoor kitchen was rephrased. Furthermore, the overall shape, components, and modularity spectrum of the outdoor kitchen were also investigated. To facilitate the idea development process, deeper material analyzes were made in combination with store visits narrowing it down to suitable materials. All of this allowed three outdoor kitchen concepts to be developed and the work is presented in the following sections.

4.1 Redefinition of an outdoor kitchen

From the state of the art, conducted surveys and interviews it was possible to make new decisions about what defines as an outdoor kitchen for this project. Therefore, a redefinition regarding which components an outdoor kitchen consists of was made. The final outdoor kitchen design should include modules that have the following components:

- A sink with tap
- Workspace/unloading areas
- A grill, either charcoal or gas
- A gas stove

A fully loaded kitchen by Livsyta should therefore include all of these components but can be mixed and matched according to the customers' needs.

Furthermore, it was decided that the outdoor kitchen should be divided into three parts: the countertop, framework, and cladding see *Figure 4.1*. The countertop is defined as the overall work and unloading area that is attached to the framework. Where the tap, sink, grill, and gas stove will be installed upon. The framework is the main structure of the kitchen and where not only the countertop is attached but the cladding as well. The cladding is the rest of the kitchen, as in the shelves, cabinet doors, and inner and outer walls. Excluded from this are the fittings, hinges, and brackets as well as the potential means of the attachment, such as screws and glue.

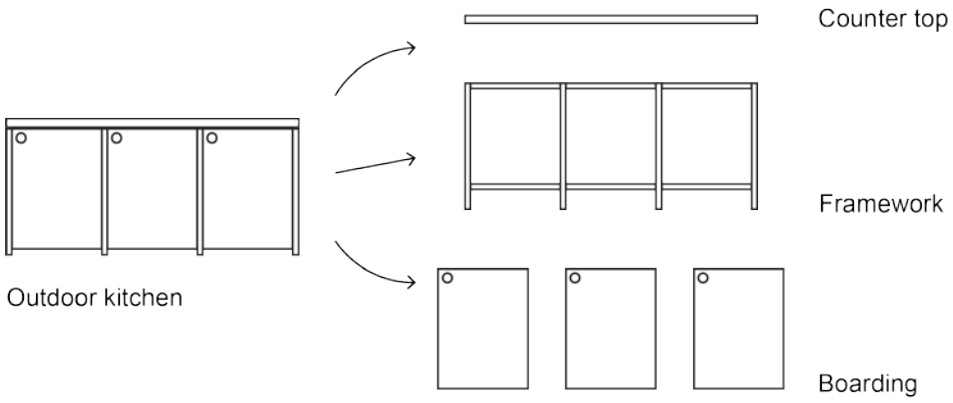


Figure 4.1: Simpler figures showcasing the basics of each part.

Dividing the outdoor kitchen into three parts made it easier to investigate the modularity of the outdoor kitchen. Furthermore, it facilitated the material analysis of the three parts since these were evaluated differently.

4.2 Kitchen shape

Having a modular kitchen design would enable the customers to create a kitchen that would be more tailored to their outdoor area than a standard conventional kitchen. However, some placements could cause problems such as hygiene issues and/or unnecessary dead spaces. To understand and tackle which setups would provide reasonable placement, simpler figures were drawn and placed in three common kitchen setups. These were; L-shape, U-shape and I-shape, see *Figure 4.2*. Various modules were created based on previous decisions and were; a sink with a tap, workspace/unloading area, grill, and gas stove. These were later placed in different setups to visualize advantages or problems with each setup. According to the work triangle, some designs were better than others while others provided too much dead space, usually in the corners of the L- and U-shaped designs.

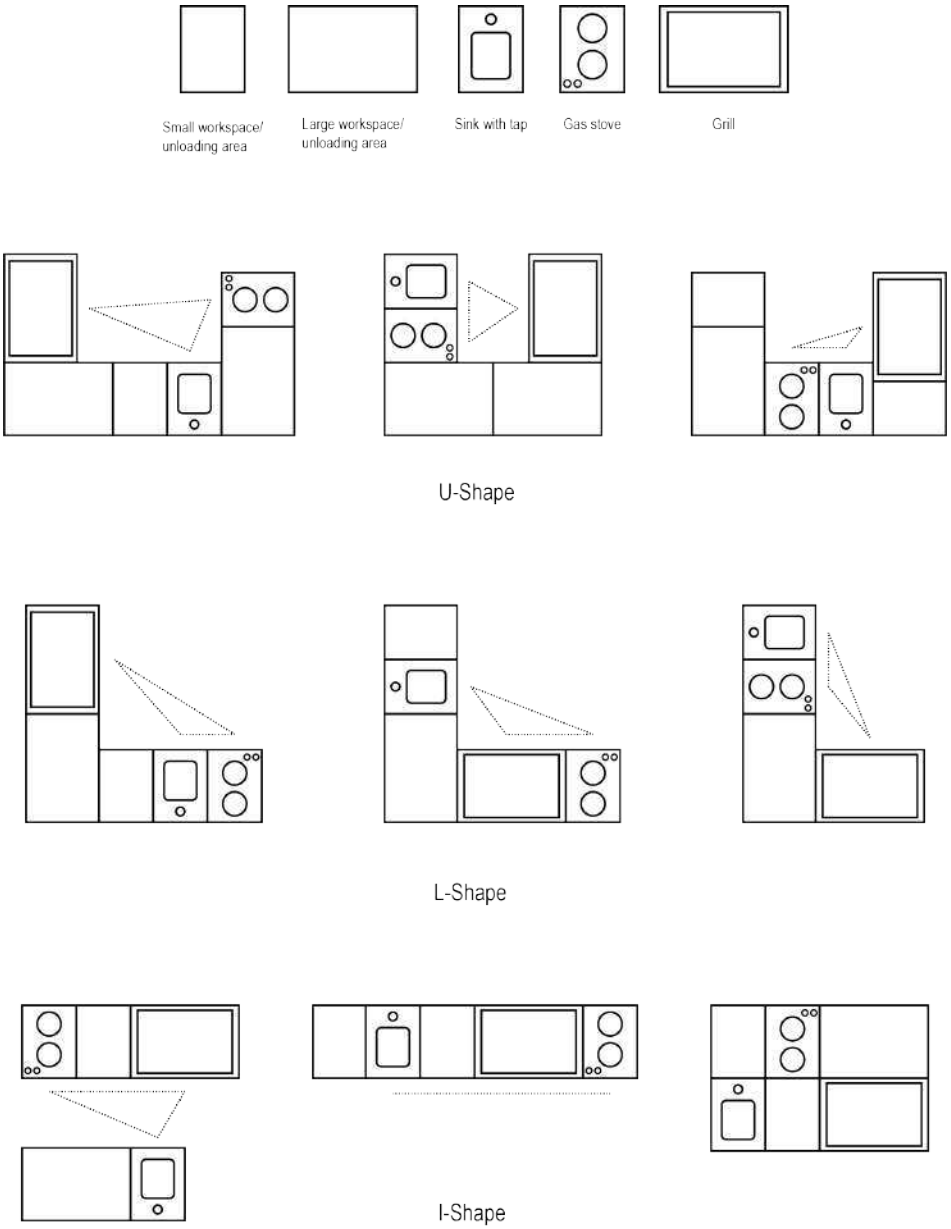


Figure 4.2: The three different setups that were analyzed.

In addition to this, real life prototyping was made with cardboard boxes, see *Figures 4.3-4.5*. This was done in order to get an understanding and feeling of the different setups.



Figure 4.3: Testing the L-shaped kitchen with placed out modules for gas stov, grill and sink.



Figure 4.4: Testing the U-shaped kitchen in similar fashion.


Figure 4.5: Final testing of the linear shaped kitchen with a kitchen island.

From this it was decided that the outdoor kitchen should be made in an I-shape, offering the best optimization of modularity. The biggest advantage of the I-shaped format was that it prevents cabinets to be blocked by other modules, preventing an unnecessary design challenge. Furthermore, a linear-shaped kitchen also creates the least amount of dead space as it offers the possibility of having a kitchen island, which was found to be essential by the interviewees in Phase I. Lastly, the format also prevents potential hygiene issues such as cross-contamination and bacterial growth in crevices.

4.3 Modularity spectrum

The spectrum of a modularized kitchen can also vary a lot and it is restricted by many factors such as cost, manufacturing, time, etc. Therefore, during this phase, it was also important to discuss where Livsyta should position itself in order to avoid problems during the concept development phase. Thus, the grade of modularity and pre-manufactured parts had to be defined first. This is also the basis for which an upscale of the company can be made. It was possible to define the spectrum from the gained insights from Phase I. The spectrum was then narrowed down into three categories: Tailored, Mix-and-match and Standardized. These three categories were then compared with each other regarding advantages and disadvantages and then a decision of which one to use was made.

4.3.1 Tailored

A tailored outdoor kitchen means that Livsyta would not have any pre-designed kitchens of its own, but would only take orders on demand. Here the customer would design together with Livsyta their outdoor kitchen and choose which parts, components, and materials it should be made of. The kitchen's parts would then be made by a carpenter and later delivered to the customer where it would be built and finished by Livsyta's employees.

The benefit of having a tailored solution is that the customer gets to decide completely how they want their kitchen. This makes it easier for the customer to make the right choices but also to feel more satisfied with their product. Furthermore, it also gives the customer a unique kitchen that adds to the feeling of exclusivity and luxury.

A disadvantage of having a tailored solution is that it becomes more expensive for both Livsyta and the customer. It also takes longer time to produce an outdoor kitchen which means more work for Livsyta. In addition, Livsyta would have to decide whether they should employ carpenters who will build the outdoor kitchens in-house or if they should outsource it to other manufacturers. Lastly, it will also make Livsyta's brand more difficult to define as all outdoor kitchens will look different.

4.3.2 Mix-and-match

Compared to the tailored solution, the mix-and-match solution would be more limited. Meaning it would provide the customer with only some options and possibilities to customize their outdoor kitchen. The customers would be able to choose between different modules that include the different components defined

earlier, such as a sink with a tap, grill, etc, and combine these as they wish. However, the overall design and its materials would already be decided. The modules would be made beforehand and then delivered to the customer and installed by Livsyta's employees.

An advantage of the mix-and-match solution is that pre-defined modules would make the manufacturing of the outdoor kitchen faster. Leading to minimization of costs and time. Furthermore, it would remain the customers' ability to choose and customize their kitchen regarding the overall shape and its components included. However, this solution is not as free as the tailored one which could make the customers less satisfied since they have less influence regarding the materials and components of the kitchen. Lastly, it could also make the transportation of the outdoor kitchen harder since the modules will be built beforehand.

4.3.3 Standardized

On the opposite side of the spectrum to the tailored solution, is the standardized solution. Here, Livsyta will design and sell an outdoor kitchen with predefined modules, components and materials leading to no customization options for the customer. The outdoor kitchen would have a pre-defined dimension and layout. The outdoor kitchen would then be transported to the customer where the customer itself would build and install the outdoor kitchen.

An advantage of this solution is that it minimizes manufacturing and production costs marginally as all parts will be standardized in the outdoor kitchen. Furthermore, it also makes it easier to produce replacement parts in case something breaks after use. It would also make the transportation of the outdoor kitchens easier as it is sent in packages that are built up by the customer itself, which minimizes the work for Livsyta's employees. A disadvantage of this solution is that it contributes to no choice or customization for the customer. It may also contribute to a less luxurious feeling since the customers will have to build and install the outdoor kitchen themselves. Lastly, it may also be difficult for the customer to assemble it because it requires the customer to have all the tools needed.

4.3.4 Chosen modularity

After comparing the advantages and disadvantages of the different solutions, it was decided that the mix-and-match solution would be chosen for this project. The mix-and-match solution suits Livsyta's brand the best as it preserves the feeling of exclusivity while having standardized modules. The solution also makes it easier for

Livsyta to scale their business since they will balance standardized parts with some customization for the customers. Finally, customers will also be happier with this solution since they will be able to customize their kitchens according to their needs and prerequisites.

4.4 Idea generation methods

When the redefinition of the outdoor kitchen was made in combination with decisions regarding the shape and modular spectrum, it was possible to start the idea generation phase. The idea generation phase consisted of several iterations over Phase II and multiple methods were utilized. These methods consisted of form variation and brainstorming that was manifested into sketches, see *Figure 4.6*. It was important to maintain a wide mindset to gather as many ideas as possible in the beginning and then slowly narrow down, locking in the various aspects that were favorable. Certain ideas were remade as certain needs were taken into consideration such as materials and structure. It was also important to keep the Scandinavian and Japanese design and formgiving in mind.

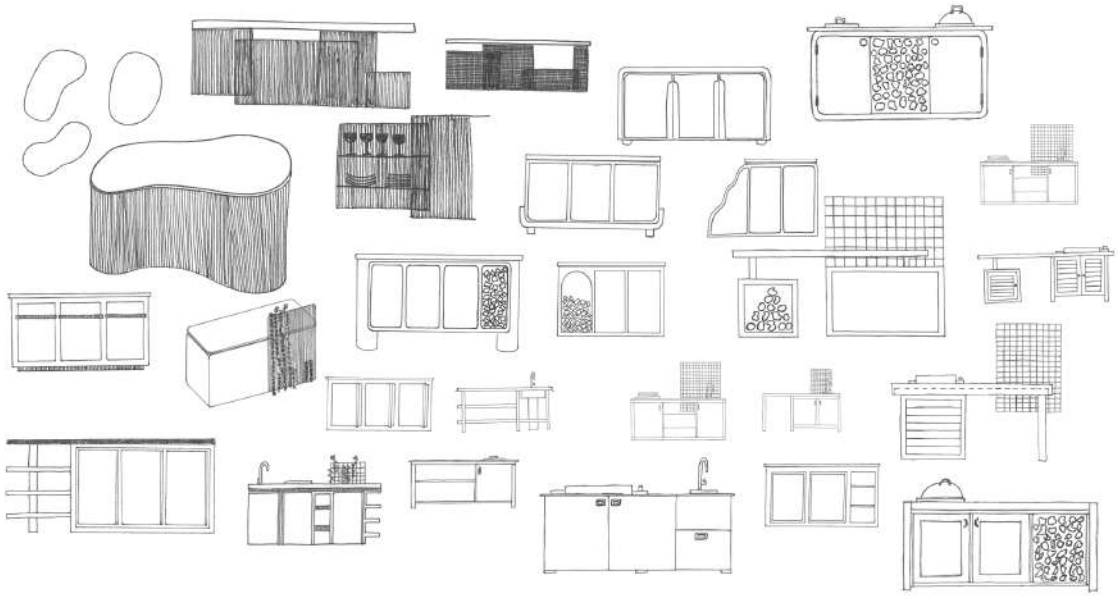


Figure 4.6: Different sketches from the idea development.

4.4.1 Storyboard

A storyboard was created in order to visualize the process of using an outdoor kitchen and problems commonly mentioned in the qualitative and quantitative research. It was drawn based on how the users from interviews and surveys used their outdoor kitchen and common problems associated with certain steps, see *Figure 4.7*.

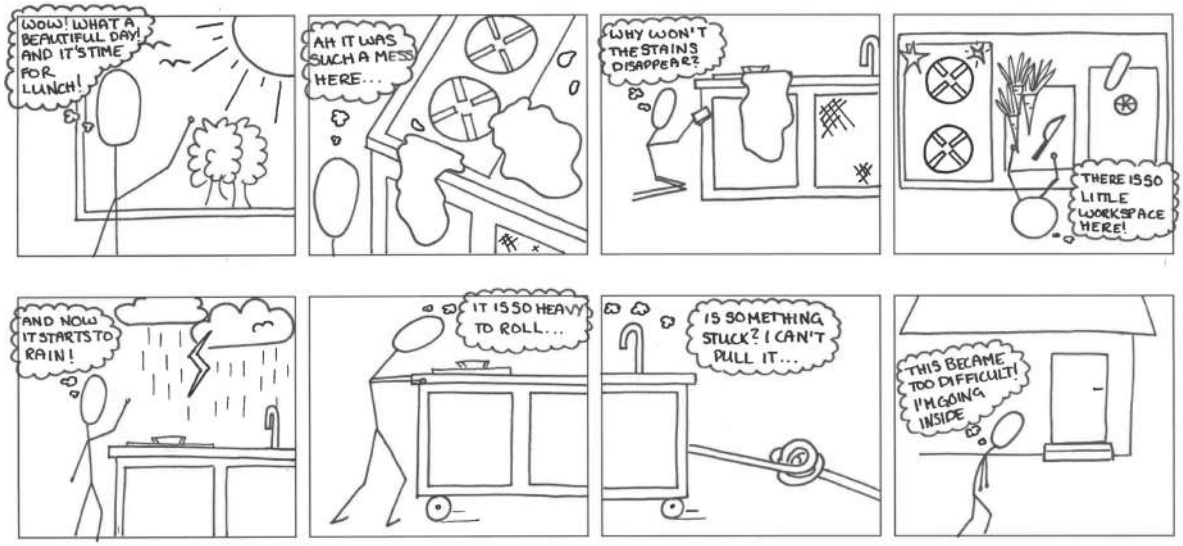


Figure 4.7: A storyboard highlighting potential problems associated with an outdoor kitchen.

Described problems within the storyboard were the maintenance and cleaning process of the outdoor kitchen. Highlighting that the process should be easy and time-efficient. Factors that could help facilitate the process are choosing suitable materials, minimizing extra details that make it harder for dirt to be removed, or instead creating a cover that encloses the kitchen when not in use. Furthermore, another identified problem was the mobile function of the outdoor kitchen showing that if the function is available it should be easy to move or roll the kitchen into another area. All of these highlighted problems were kept in mind during the idea generation development.

4.5 Materials suited for outdoor use

In order to facilitate the idea generation phase, several material analyzes were made. Materials used within Scandinavian and Japanese interior design are many. However, only a few are suitable for outdoor use and furniture nonetheless. Especially since the outdoor kitchen is meant to be used outdoor all year round and harsh weather will occur. Furthermore, the product needs to have materials suited for kitchen environments meaning it will not take any damage from example food

stains. During the research for suitable materials for the outdoor kitchen, several new criteria and guidelines were listed. These were the following:

- *Weather resistant*, the materials should be able to cope with different types of weather conditions such as sun, rain, etc.
- *Stains and dirt*, the materials should be resistant to various types of stains and easy to clean and maintain.
- *Physical damage*, the materials should be resistant to physical damage such as strikes and hits in order to prevent scratch marks or breakage.
- *Cost minimization*, manufacturing and production costs should be minimized as much as possible.
- *Sustainability*, the materials environmental impact should be lowered as much as possible.

4.5.1 Material categories and GRANTA Edupack

Different materials are better suited for the three defined parts of the outdoor kitchen in relation to manufacturing, costs, durability, and also the overall design. This was also decided according to materials commonly used for such purposes in relation to outdoor kitchens. There were several interesting materials but in relation to the mentioned aspects, a selection was made. Therefore, the investigated materials were clustered differently with regard to the three parts and the following were:

- *Framework*: wood and steel
- *Cladding*: wood and steel
- *Countertop*: steel, stone and composites

In order to know which sorts of wood, steel, stone, and composite to use deeper analyzes were made for each material category. During the studies of each material category, the software *GRANTA Edupack* was used. The software is useful when it comes to analyzing different material properties and qualities. It can also be used to compare several materials with each other with different chosen variables (Ansys Inc, 2022) In order to determine the material criteria in *GRANTA Edupack*, several variables in relation to material properties were associated with each criterion. For weather resistance, the durability against fresh water and UV radiation were analyzed. The effects of dirt and stains, for example, lemon juice or wine stains, were analyzed by examining durability against weak acids and alkalis. Further,

withstanding shocks and hits were analyzed through the properties of flexural and compressive strength. These variables were summarized in tables on a scale of unacceptable, limited use, acceptable and excellent. The summarized tables and graphs will be presented in each material category section. Lastly, the analyzed variables are in relation to materials that are untreated, meaning the data presented are raw materials, the exception being galvanized steel.

Considering that not all materials exist within the database of *GRANTA EduPack*, certain materials had to be investigated from other sources. Furthermore, the materials' effects of treatments were not analyzed in *GRANTA EduPack* as there was no data available. This was examined through other sources. In the following sections the general information regarding each material will be presented as its usage and suitability for outdoor furniture.

4.5.2 Wood

Wood is an organic and natural material used within many different areas (Svenskt Trä, 2022). Wood has been used for centuries as a building material and is a self-evident choice of material for many home purposes, ranging from construction, flooring, trellis and paneling. Wood differs widely in structure and properties, meaning it is important to choose the right wood for the right purpose (SkogsSverige, 2021). The common properties of wood in comparison to other material groups is the overall low density, providing easy handling and transport. Another is the flexibility, as it is easy to work with using basic tools.

Certain woods are more often used for outdoor furniture, as they have properties suitable for such environments. Naturally, since wood is an organic material, bacteria and rot are common problems associated with wood in outdoor environments (Taylor & Vila, 2020). However, certain families of wood are naturally protected from these types of infestations. Some of these are cedar, oak, teak, and pine. There are treatments that can be applied to wood as a way of creating artificial protection against these threats. Furthermore, different treatments can also be used in order to maintain the durability and appearance of the wood. By applying wood oil or wax, a protective surface will make the material more durable. In addition, newer ways of treatment such as applying silicon to the wood under pressure have also become more popular (OrganoWood, n.d.). Another common treatment of otherwise non-protected wood is using heat (Moelven, 2022). By applying high heat to the material, increased protection against rot and form stability is added. The most common wood to treat with heat is pine. Pressure treatment is another option that forces chemicals into the wood which makes it unsusceptible to rot and increases the strength of the

wood (Curtis Lumber and Plywood, 2020).

Analysis of cedar, pine, teak and oak was made according to necessary properties and can be seen in *Figure 4.8* and *Table 4.1*.

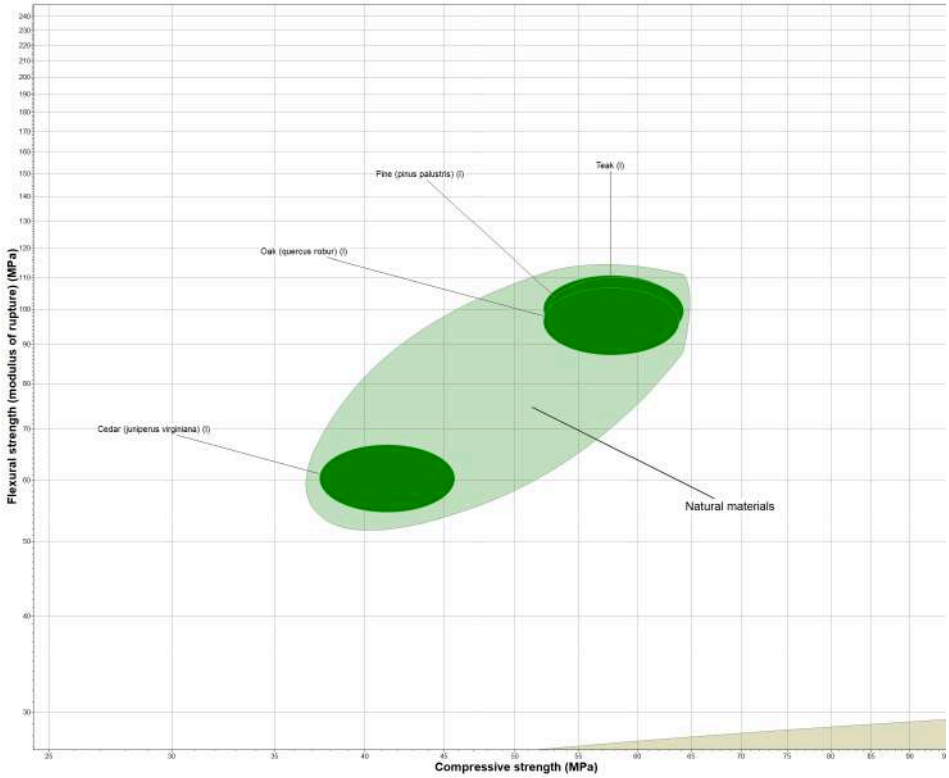


Figure 4.8: A graph showing flexural strength in relation to compressive strength in unit MPa.

Table 4.1: *Wood analysed in relation to weak acids and alkalis, UV-radiation and freshwater.*

Material	Weak acids	Weak alkalis	UV-radiation	Freshwater
Cedar	Limited use	Acceptable	Acceptable	Limited Use
Oak	Limited use	Acceptable	Acceptable	Limited Use
Pine	Limited use	Acceptable	Acceptable	Limited Use
Teak	Limited use	Acceptable	Acceptable	Limited Use

From *Figure 4.8*, cedar is the overall weaker wood material while oak, pine, and teak are all similar in flexural and compressive strength. From *Table 4.1*, it shows that all wood materials share the same properties in durability against weak acids and alkalis, UV radiation, and freshwater. However, the properties are something that surface treatments could affect and improve if applied. UV radiation could also be something that is favorable and desired, as the natural aging of wood is an effect of sun exposure.

Wood prices can be measured in board feet which is 1 foot x 1 foot x 1 inch in

dimension as a reference between wood types. Price-wise, teak is the most expensive at approximately 410 SEK per board feet, European oak at 156 SEK, Spanish cedar at 78 SEK and untreated pine at 29 SEK (Hearne Hardwoods, 2022).

4.5.3 Metal

Metals are inorganic compounds with metallic characteristics, such as malleability and high electrical and thermal conductivity (University of Maryland, 2022). As a material, metals are rarely used in their pure forms, such as iron and titanium, but are compounded into alloys, such as different types of steel (Minay & Boccaccini, 2005). Metals are categorized according to two main groups: ferrous and non-ferrous metals. There are many different types of metals, but depending on the application and the requirements placed on the metals, there are more or less suitable ones.

Commonly used metals in structures in outdoor environments include stainless steel, galvanized, and powder-coated steel. Steel is often used in outdoor high-end, larger types of furniture or constructions due to its high weight, load-bearing properties, and durability against physical damage (Ferris, 2022).

Steel, like most metals, is highly prone to corrosion when coming in contact with air or water. This usually means requiring some sort of surface treatment to keep the integrity of the material protected. Some common steel alloys have natural protection against this, such as stainless steel. Where the surface is protected by a passive layer of chromium and ”self heals” when damaged by reacting with air (Encyclopaedia Britannica, 2021). Other surface treatments include galvanizing which creates a protective layer of zinc, or powder coating, which is a layer of resin or polyester sprayed on the surface (Keller Technology Corporation, 2021).

Analysis of steel was made according to necessary properties and can be seen in *Figure 4.9* and *Table 4.2*. Powder coated steel was not possible to examine in *GRANTA EduPack* but from further research it was found as a suitable material for outdoor use, sharing many similarities with galvanization in regard to durability (Weaver, 2022), (Mchone Industries, 2021). In the *Figure 4.9* and *Table 4.2* the steel type carbon steel (AISI 1010) was used, as it is a common steel two either galvanize or powder coat.

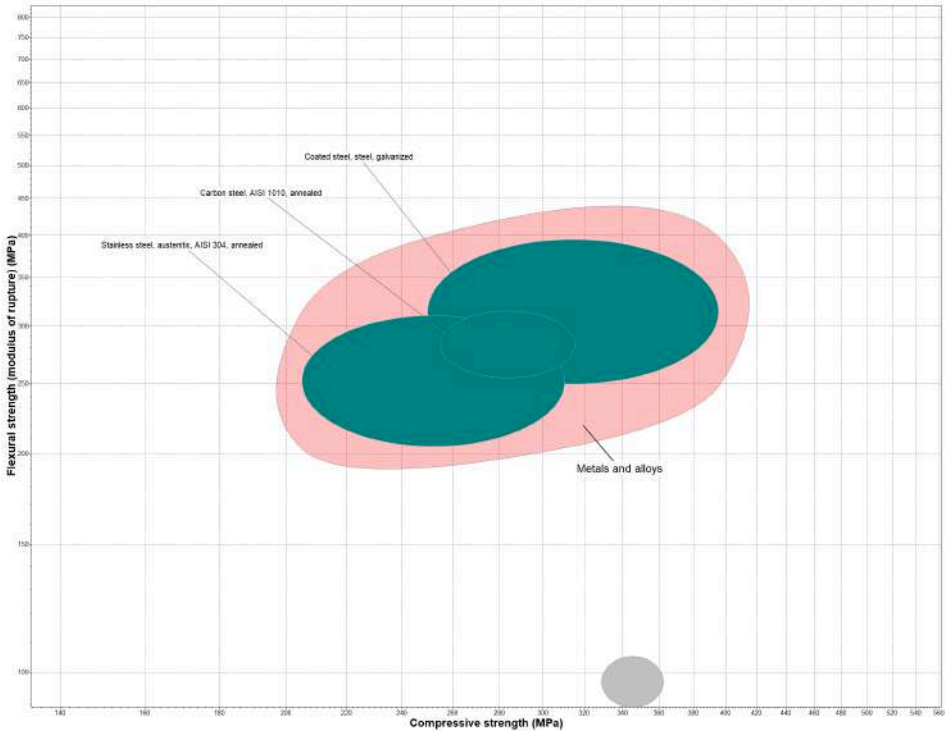


Figure 4.9: A graph showing flexural strength in relation to compressive strength in MPa.

Table 4.2: *Metal analysed in relation to weak acids and alkalis, UV-radiation and freshwater.*

Material	Weak acids	Weak alkalis	UV-radiation	Freshwater
Carbon steel (AISI 1010)	Limited use	Acceptable	Excellent	Acceptable
Coated steel (galvanized)	Limited use	Acceptable	Excellent	Excellet
Stainless steel (AISI 304)	Excellent	Excellent	Excellent	Excellent

From *Figure 4.9* it was possible to see that galvanized steel scored higher than both stainless steel and carbon steel. However, from *Table 4.2* stainless steel was the metal who scored the highest. Price is dependent on use, whether it will be used for the frame, countertop, or cladding but stainless steel is overall more expensive than powder coating or galvanization. In addition, powder-coated and galvanized countertops are not commonly used. This could be the hygienic nature of stainless steel over other surface treatments (Clinical Furniture, n.d.).

4.5.4 Stone

Stone and rock are naturally occurring masses that are composed of various minerals merged together (National Geographic, n.d.). There are three types of rock, categorized by the type of minerals and how they are composed: igneous, sedimentary, and metamorphic. The type of rock affects the mechanical properties such as porosity, strength, and hardness and differs widely.

Various types of rocks are used within outdoor furniture and interior design, often as detailing due to the natural and unique texture. Stone is often used as countertops, both indoors and outdoors, due to this. Common outdoor uses include tiling and construction because of the durable and highly resistant nature of certain rocks. For this project, relevant and commonly used stone materials were examined and included: granite, slate, limestone, and sandstone.

Stone materials can be surface-treated by multiple methods. Machine treatment is the act of manually polishing the surface in order to get a new look or other properties (Lundmark, 2020). Numerous surfaces can be retained from this for example brushed, sawn, honed, leathered and polished. Along with this, impregnation of stone is possible in order to retain a more protected surface against example water and oils (ABC Stone Inc, n.d.).

Analysis of granite, slate, lime- and sandstone was made according to necessary properties and can be seen in *Figure 4.10* and *Table 4.3*.

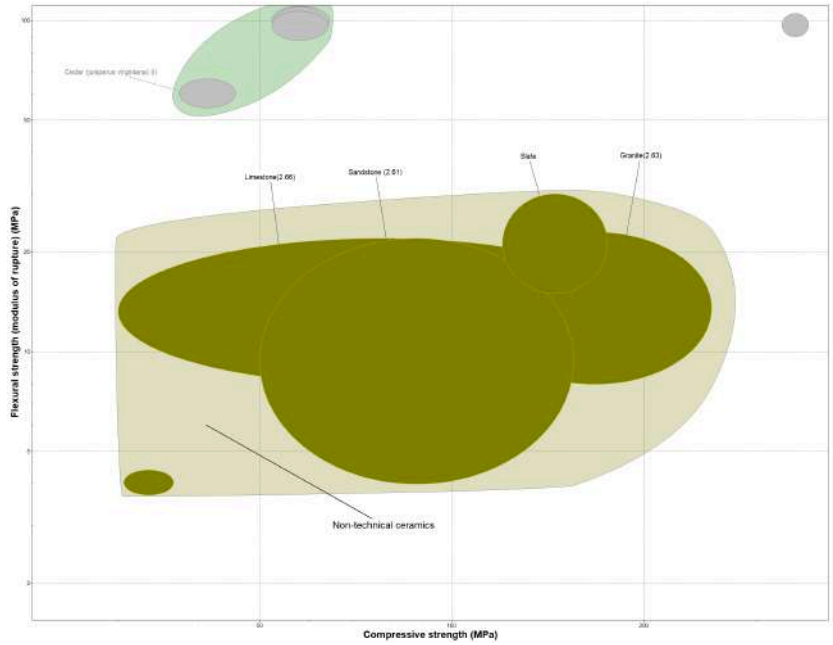


Figure 4.10: A graph showing flexural strength in relation to compressive strength in MPa.

Table 4.3: *Stone analysed in relation to weak acids and alkalis, UV-radiation and freshwater.*

Material	Weak acids	Weak alkalis	UV-radiation	Freshwater
Granite	Excellent	Excellent	Excellent	Excellent
Limestone	Unacceptable	Excellent	Excellent	Excellent
Sandstone	Limited use	Excellent	Excellent	Excellent
Slate	Excellent	Acceptable	Excellent	Excellent

From the results seen in *Figure 4.10* both slate and granite provide better flexural and compressive strength in comparison to lime- and sandstone. Furthermore, from *Table 4.3* all of the different stone types have excellent resistance against UV-radiation and freshwater. Granite is also the most durable material against weak acids and alkalis, which untreated lime- and sandstone struggle with due to high porosity. Sandstone was not chosen to continue with due to it not being commonly used for the purpose that applies to the final design (Stenskivor.se, n.d.).

Price-wise, limestone is considered more expensive at approximately 4500 SEK per running meter, in comparison to granite at 2500 SEK (Westergren, 2019). Slate costs around approximately 3500 SEK (Marble.com, 2022) .

4.5.5 Composites

Composite materials are engineered combinations of two or more materials with different properties and offer a wide range of functionality (Sabhadiya, n.d.). The created material can be more specialized for specific usage, for example reducing weight while remaining strong, creating a conductive or waterproof material. Some notable composites are reinforced concrete and ceramics which are interesting materials for the concept development.

Composites can easily be created into many different shapes, whether it is beforehand or after production (Knight & Curliss, 2003). Concrete and ceramics can easily be molded into many forms, often used as details and countertops due to this. Place-built furniture in the form of tables and chairs are common with concrete.

Composites can be widely different due to the possibility of specializing in each material. However, previously mentioned composites can be treated further to enhance durability. Concrete and ceramics can be polished in order to change the texture of the surface (Convergent Concrete Technology, n.d.). Concrete is a porous material and can be surface treated with example wax or epoxy to make the material protected from water and oils.

Analysis of concrete and ceramic was made according to necessary properties can be seen in *Figure 4.11* and *Table 4.4*.

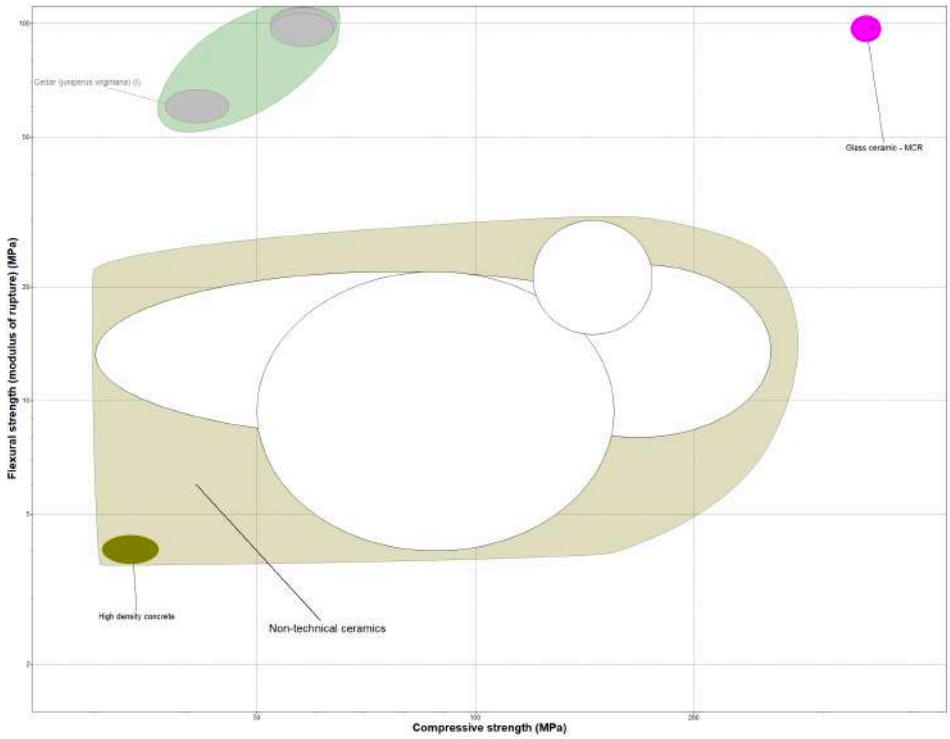


Figure 4.11: A graph showing flexural strength in relation to compressive strength in MPa.

Table 4.4: Composites analyzed in relation to weak acids and alkalis, UV-radiation and freshwater.

Material	Weak acids	Weak alkalis	UV-radiation	Freshwater
Ceramic (MCR)	Excellent	Excellent	Excellent	Excellent
High density concrete	Unacceptable	Excellent	Excellent	Excellent

As can be seen in *Figure 4.11* and *Table 4.4*, ceramic is the best material based on the chosen criteria, scoring the highest in every property. However, the chosen concrete was an untreated one, meaning the properties can be improved heavily with various surface treatments or by utilizing concrete enforcement.

Prices here differ widely as well as concrete as a material is cheap and if in-house manufacturing is an option, prices can be lowered. Outsourcing manufacturing would cost around 4000-5000 SEK per running meter (depending on thickness and size). Ceramic has an overall higher price, at around 4000-8000 SEK per running meter.

4.5.6 Ecological sustainability within the materials

Adhering to sustainable product development is important and something that was possible to affect by choosing sustainable materials and production methods. Except for the previously decided criteria in regard to design, the ecological effects had to

be considered as well. For this, the materials chosen were analyzed in *GRANTA EduPack* by carbon dioxide (CO_2) footprint and energy consumption (MJ) during primary production, see *Figure 4.12*. This was done by examining virgin grade materials in order to get an estimate for the ”worst-case scenario”.

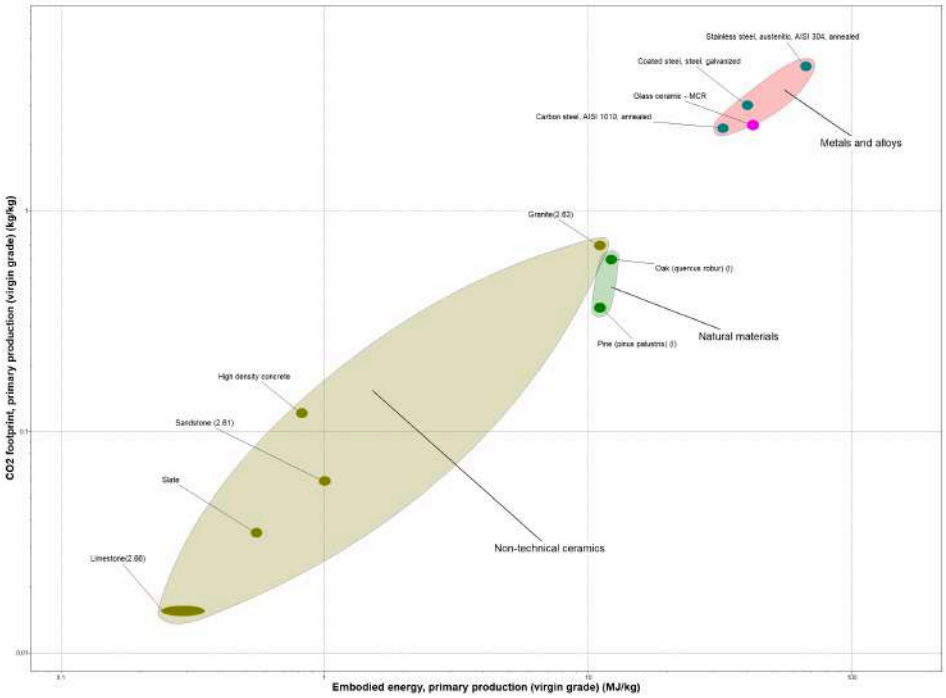


Figure 4.12: The potential materials graphed according to CO_2 footprint and energy consumption.

From *Figure 4.12* it is possible to see that the metal category of materials is the least sustainable with both a high CO_2 footprint and high energy requirement, together with ceramic. However, based on information provided by *GRANTA EduPack*, the chosen metals are 100 percent recyclable in comparison to stone and composites which are usually down-cycled. Wood is another material group showing relatively high outputs of CO_2 and high energy consumption. Similar to metal, wood is not only recyclable and reusable but also renewable. Among the chosen stone materials, granite is one of the least environmentally friendly but is also the most durable, meaning a long life span could compensate for the lack of sustainability.

4.6 Visit to The Materials Library

After narrowing down the suitable materials for each component into material categories, a visit to the Materials Library in Älvsjö was made, see *Figure 4.13*. The Materials Library is a permanent showroom displaying over 3000 material samples as well as showcasing new ways of manufacturing. The visit was conducted in order to get a feeling of the various materials’ finishes as well as to get a better understanding of weight and durability. Suitable materials for the final design were also discussed

with the project leader of the Materials Library.



Figure 4.13: Images from the payed visit to the Materials Library.

From the paid visit, it was possible to draw the conclusion that most materials discussed in the previous chapters are suitable for an outdoor kitchen. However, a tip from the project leader was to choose carefully where the material is used. For example, if the framework is hidden by a wooden cladding, the framework could be made in steel or another material, in order to minimize costs and maintenance and to optimize the lifespan.

The visit to the Materials Library also gave a feeling of which finishes to use for the final design of the outdoor kitchen. For example, several samples of granite were studied where varying grades of finishes were made. A too polished finish made the impression of the granite feel tacky and cheap meanwhile a too unpolished finish made the granite feel dirty and rough. Therefore, a finish in between could be best suited for the final concept design. Another example was the finishing of stainless steel, where a brushed finish was found to be the best one. The outdoor kitchen may attract some scratches and marks during its usage and therefore a brushed finish could help hide these better than a fully polished finish. These insights were important for the upcoming concept development and were kept in mind since they will affect the feeling and impression of the final outdoor kitchen design.

4.7 Store visits

In connection to the visit to the Materials Library, several store visits were also made, see *Figure 4.14*. The purpose of the store visits was to get a further feeling of different materials and details in existing kitchens and furniture. During the store visits, the overall formgiving of different kitchens and furniture were studied as well as smaller details such as fittings and components. The visits were paid to indoor kitchens and furniture at Ballingslöv, Kvik, Nordiska Galleriet, Norrgavel, IKEA and Svenssons i Lammhult in Stockholm. Here it was interesting to see how different qualities of the same materials could feel and look. This knowledge will have to be applied to choosing materials later on in the final design. Furthermore, it was rewarding to physically see how transitions between shapes contributed to the overall impression. The visits gave good insights and ideas for the ongoing idea generation.

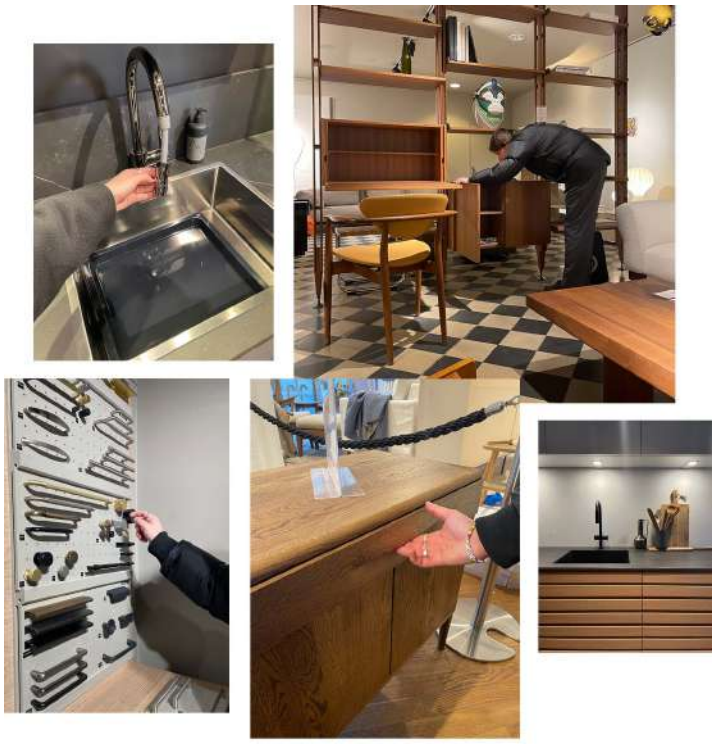


Figure 4.14: Images from the paid store visits.

4.8 Conclusion of materials

From the different material analyses, it was possible to narrow down which materials were of interest. The following materials in each material category that were of interest are:

- *Wood*: Teak, oak, cedar and pine
- *Metal*: Stainless steel, galvanized steel and powder coated steel
- *Stone*: Granite
- *Composites*: Concrete

The materials used are dependent on which part of the kitchen they are used for. This is related to price and intended use. Powder-coated and galvanized steel was decided not to be used as countertops but could potentially be used for the framework and cladding. Due to price, teak and oak will not be used for construction as pine can be treated, giving it similar properties. However, it might be a potential material for the cladding depending on the chosen concept. Granite is a material durable enough to be used as a countertop within the stone materials while being relatively cheap. Finally, due to the steep price of ceramic, concrete would be the only composite used, provided in-house manufacturing is possible.

4.9 Concept development

From the idea generation development, creating of four personas, material analyzes and paid visits to the Materials Library and furniture stores, three different concepts could be developed. The focus during the concept development was mostly on the overall formgiving, expressions and chosen materials. However, smaller details such as fittings and manufacturing processes were also considered since it affects the overall impression of the outdoor kitchen. For each concept, a moodboard, overall design and modular design were made and will be presented in the following sections. Further development and iterations of the chosen concept will be made when the decision on which one to continue with has been taken.

4.10 Concept I

Concept one is an evolution of Livsyta's previous kitchens. The concept represents the minimalistic shapes of Scandinavian and Japanese design, focusing on smaller manufacturing details that express craftsmanship and luxury. The concept expresses weight, stability, and durability. A moodboard representing the overall shapes, details, and materials can be seen in *Figure 4.15*.

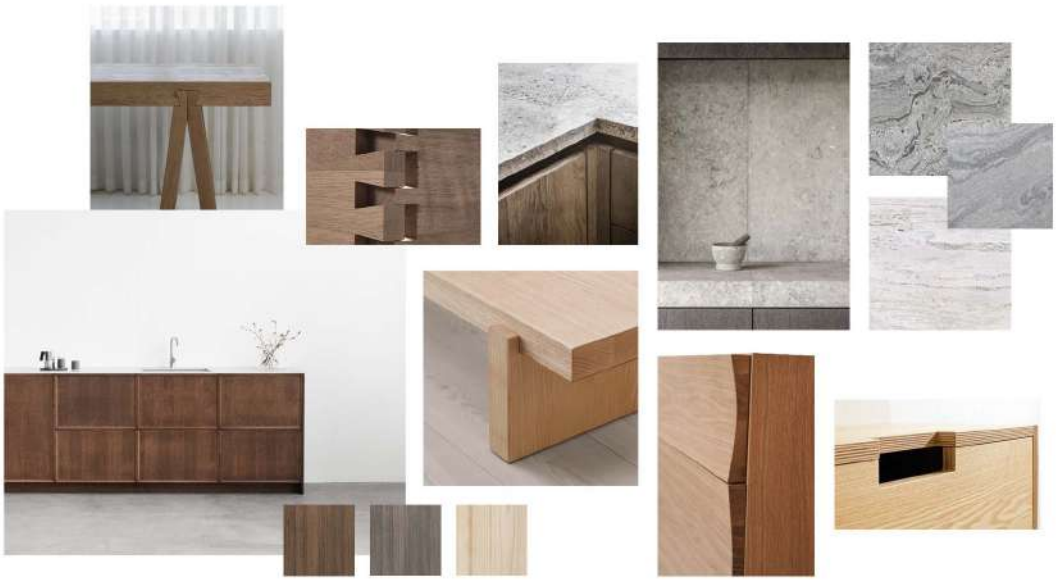


Figure 4.15: A moodboard of concept 1 representing its overall formgiving and materials.

An overall shape of the concept with its chosen materials can be seen in *Figure 4.16*. The chosen materials for concept one are: granite for the countertop, oak or pine for the cladding and framework, and lastly stainless steel for the components like the sink, tap, grill, and gas stove. As mentioned earlier the concept has a minimalistic style with a focus on material transitions and smaller craftsmanship details such as dovetail joints. Due to the intricate nature of the craftmanship and varying weather conditions, materials that are dimensionally stable will have to be chosen. To create a more sophisticated style, the framework is covered by a cladding which hides everything that is stored within the kitchen. In *Figure 4.17* suggestions of how the shelving system is installed is shown. The idea is that the shelf system is made in the same material as the cladding and is a cut-out from the cladding which highlights the focus on small details.

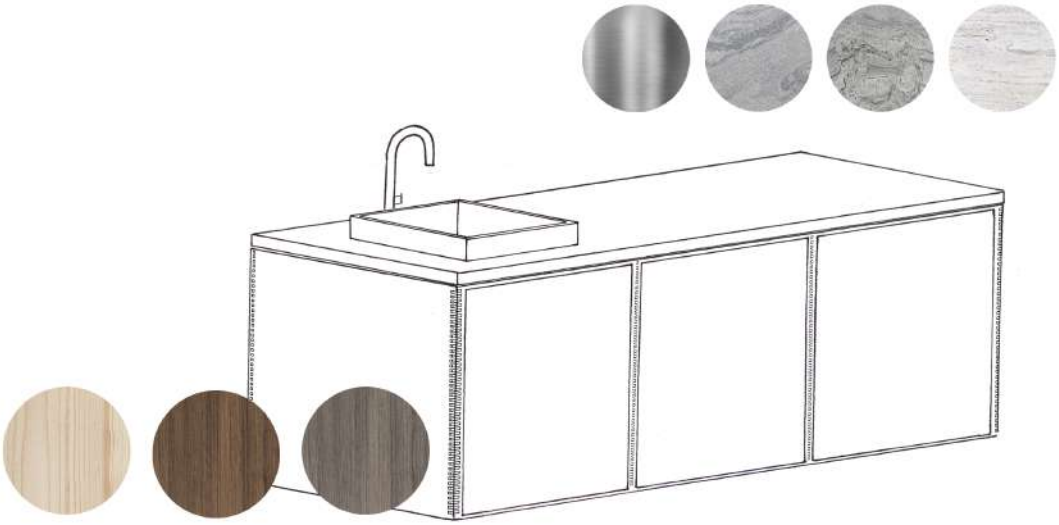


Figure 4.16: An overall design of concept 1 with its chosen materials.

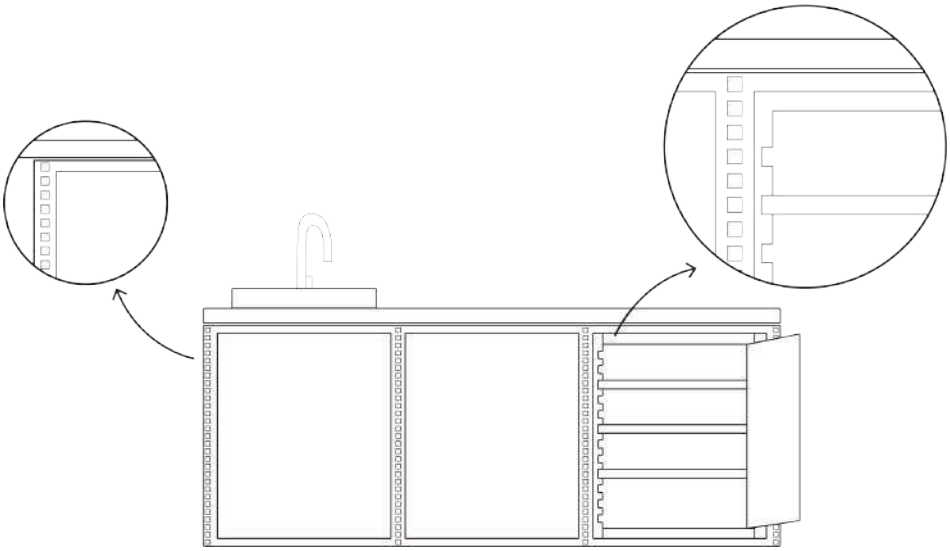


Figure 4.17: Smaller details in concept 1 such as dovetail joints.

To create a more dynamic impression of the kitchen different fittings were ideated. These can be seen in *Figure 4.18* and *Figure 4.19*.

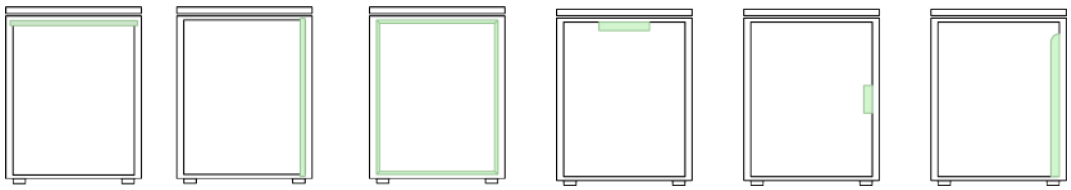


Figure 4.18: Different fittings on the modules of concept 1.

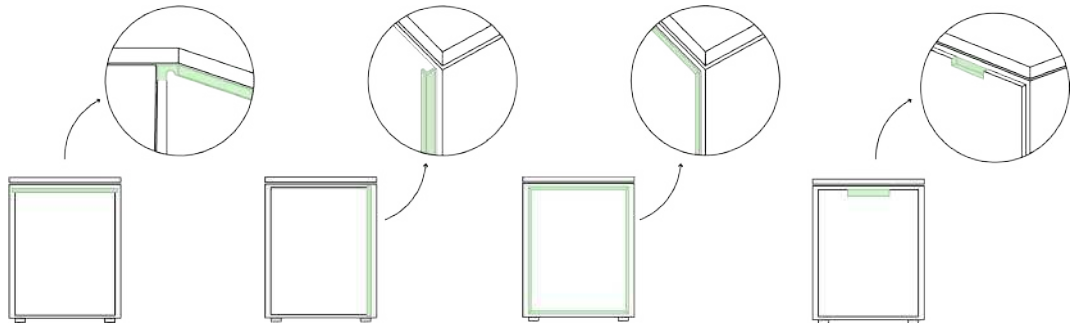


Figure 4.19: Different fittings on the modules of concept 1.

Concept one comes in five different modules which can be seen in *Figure 4.20*. These modules include all of the previously mentioned components such as a sink with tap, workspace/unloading areas, a charcoal grill/gas grill, and a gas stove. These modules can be combined in various versions following the I-format shape.

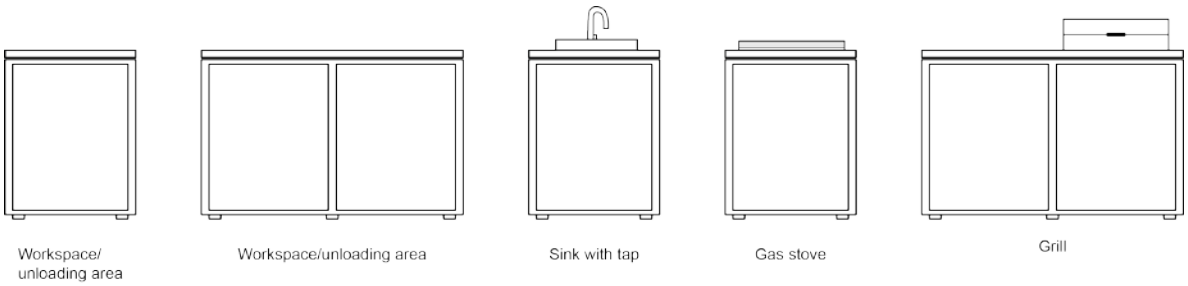


Figure 4.20: The five different modules that concept 1 comes in.

4.11 Concept II

Concept 2 is also an evolvement of Livsyta's previous kitchens and it expresses a more airy, spacious, and inviting feeling. The idea of the concept is to communicate the beauty of aging in materials, as in the aesthetics of *Wabi-Sabi*. Here, a thin but dense countertop in brushed stainless steel meets a thick aging pine wooden cladding, maintaining the feeling of hygiene as the work area is in steel. A moodboard expressing the overall shape, materials and details can be seen in *Figure 4.21*.



Figure 4.21: A moodboard of concept 2 representing its overall formgiving and materials.

The overall design of concept 2 is more open and transparent than the previous concept and can be seen in *Figure 4.22*. Concept 2 has a visible framework with open shelving systems expressing a more inviting feeling since it is possible to see what is stored inside the kitchen. The overall formgiving of the concept is smooth with curved edges meeting straight framework beams. The idea is to have several material meetings between steel and wood, creating multiple transitions between the materials. For example, fittings and hinges that will be fixed onto the pine wooden cladding will be made of stainless steel.

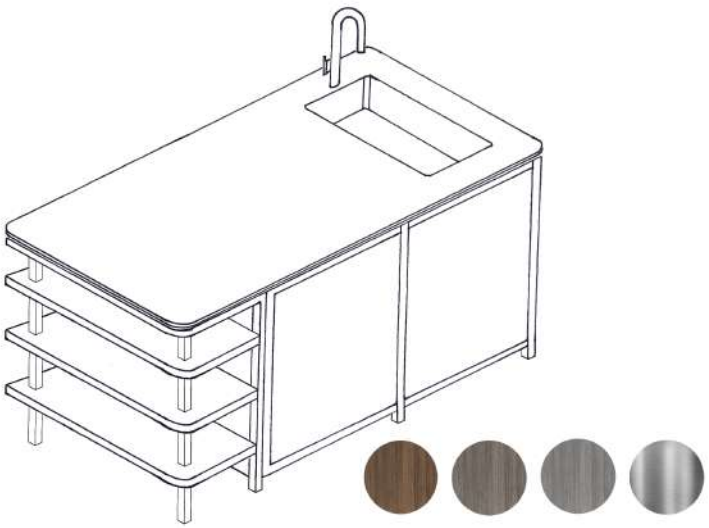


Figure 4.22: An overall design of concept 2.

The concept comes in six different modules where two of them have open shelves and one of them can only be places at the end of the kitchen, see *Figure 4.23*. The kitchen modules include all of the necessary components such as sink with tap, work space/unloading space, charcoal/gas grill and gas stove. The modules can be combined differently in order to create different expressions of the outdoor kitchen. In *Figure 4.24* two examples of how the modules can be composed is shown.

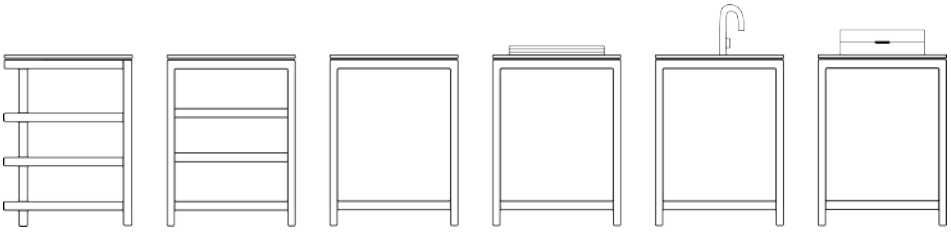


Figure 4.23: The different modules of concept 2.

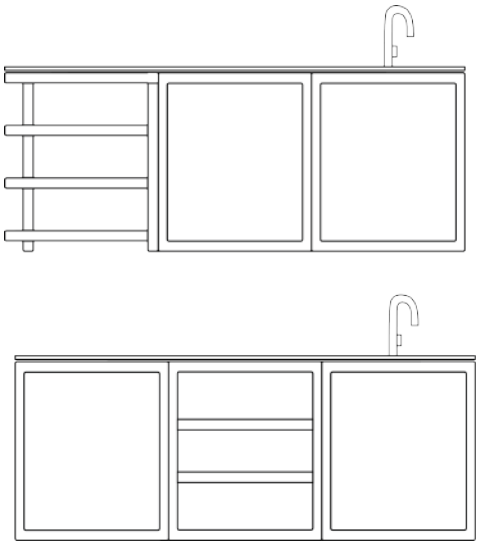


Figure 4.24: Different combinations of the modules in concept 2.

4.12 Concept III

Concept 3 has a more evolutionary design and takes a larger step than the previous presented concepts. The key focus in this concept was to create an eye-catching outdoor kitchen meanwhile preserving the functionality. Focus was also put on choosing materials that are non-aging and requires little maintenance. The concept expresses robustness and playfulness. A moodboard of concept 3 is presented in *Figure 4.25*.



Figure 4.25: Moodboard of concept 3 representing its overall formgiving and materials.

The idea of concept 3 was to create an extra dimension of the kitchen, playing with light and shadow. The materials chosen are concrete, perforated stainless steel and powder coated steel. The idea is that the materials go in the same color scheme but the material transitions creates the dynamic and depth of the outdoor kitchen. Adding the perforated steel to the outdoor kitchen also gives the possibility of integrating the surroundings such as plants and vegetables to it. An overall design of the outdoor kitchen with its materials can be seen in *Figure 4.26*.

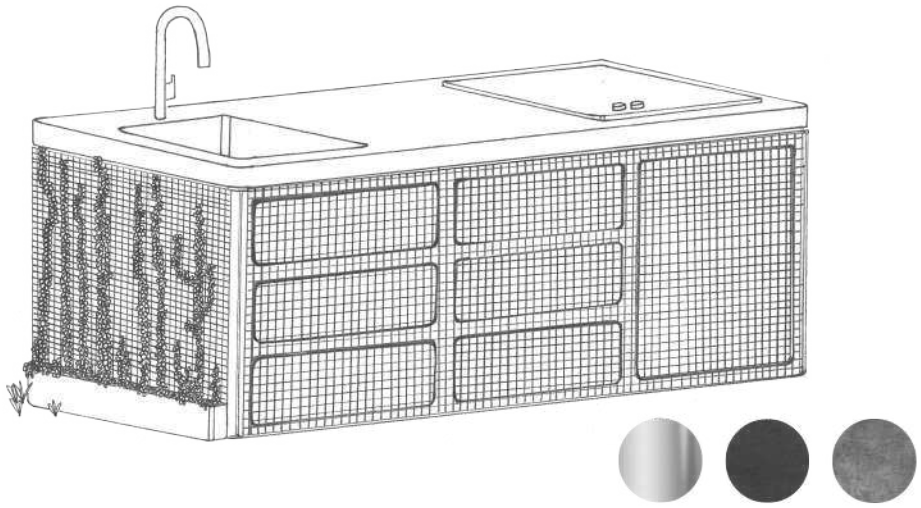


Figure 4.26: An overall design of concept 3.

The countertop is made out of concrete, the framework is in stainless steel or powder coated steel and the cladding is made of perforated steel. The perforated cladding creates a playful shadow when struck by light, where the boxes and cabinets will become visual. The cladding encloses the kitchen but the patterns create a see through view of the kitchen. The cladding can also be used as a climbing wall for plants. The idea of the concept was to use materials that express roughness and coldness but adding patterns and soft edges will contribute to a more inviting and soft feeling. In *Figure 4.27* a more detailed picture of how the outdoor kitchen is built can be seen.

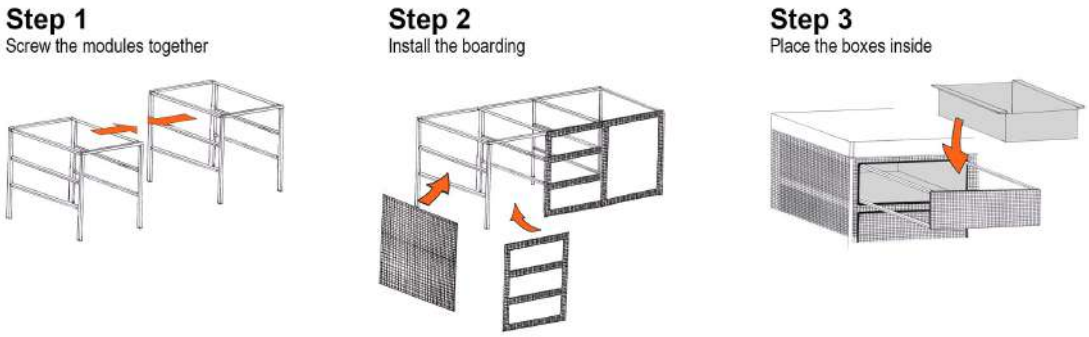


Figure 4.27: A step by step instruction of how concept 3 is built.

In order to create different expressions of the outdoor kitchen, different patterns of the perforated steel were ideated. In *Figure 4.28* some of the patterns are presented. The two patterns at the bottom right plays with the ”L” and ”li” as in Livsyta, in a way to integrate the name in the overall design of the concept.

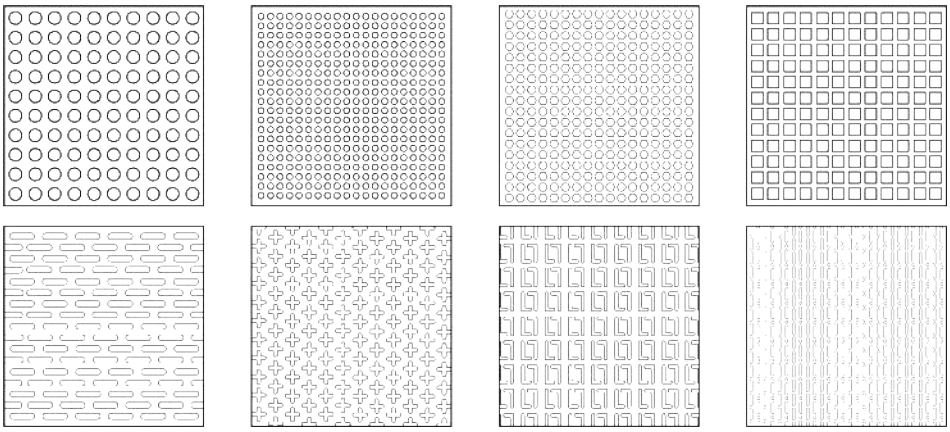


Figure 4.28: Different patterns of the perforated steel.

Concept 3 comes in five different modules, including all of the components that are necessary for the kitchen, see *Figure 4.29*. Something that makes concept 3 differentiate from the previous ones is that it comes with both doors and boxes. The idea of the boxes is that they can easily be lifted out of the kitchen in order to facilitate the maintenance of the outdoor kitchen. The boxes can also be used as storage inside the house when not in use.

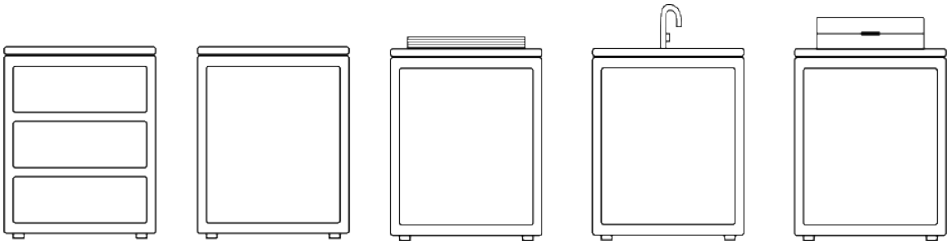


Figure 4.29: The five different modules that concept 3 comes in.

4.13 Chosen concept

After a presentation and discussion with the owner of Livsyta it was decided to continue with a combination of concept 1 and 2. The two concepts correlates heavily with the brand and where Livsyta is today, meanwhile concept 3 was more of a re-branding of Livsyta, as this takes away from the fundamental competences of the company in the form of carpentry. The smaller details and closed modules from concept 1 will be combined with the open shelving system from concept 2. Furthermore, after evaluation it was decided to make the framework and cladding in cedar. Cedar is adapted for outdoor use and is a very form stable material, meaning moisture and temperature will not change the dimensions of the material compared to other wooden material. Therefore fulfills the different criteria set on the materials earlier during the analyzes. Cedar also ages beautifully which goes in line with Scandinavian and Japanese design and adds an extra touch of luxury. After the evaluation it was also decided that the countertop should be made in stainless steel. The combination of these two materials creates a balance between hygiene and natural aging of materials. The chosen materials were also decided in regard to maintenance, manufacturing, costs and accessibility.

5 Phase 3

After the presentation for the owner of Livsyta and deciding on the chosen concept, a new iteration had to be made in order to include further requirements and design aspects. In Phase III the concept was developed into the final design. The final phase included fundamental CAD-development split into the three parts: framework, cladding and counter top including technical drawings and renders. Following this, market research into the required white goods was made, proposing which grill, gas stove, sink and tap could be used from which manufacturer. Lastly, the final design concept was also analyzed in regard to price and sustainability.

5.1 Further concept development decisions

Taking into account the chosen mix-and-match modularity as well as the chosen concepts, the development of the three different parts of the outdoor kitchen will vary. The framework and counter top will be made as standardized as possible. Meaning, they will have pre-defined standard dimensions for the different parts which will facilitate the manufacturing and storage before the assembly of the outdoor kitchen. The cladding, on the other hand, will be the ”tailored” part of the outdoor kitchen. The focus will be put on smaller details and embellishments that add to the feeling of solid craftsmanship. It will also require higher demands on the manufacturing process as more special measurements will be added. More about each part and the development of these will be presented in the following chapters.

5.2 Framework

The development of the framework was divided into several parts and were the following: defined dimensions, chosen material and potential retailers for the materials. All of these mentioned parts made it possible to finalize the construction of the framework and to ensure that it is durable.

5.2.1 Dimensions of the framework

In the begging of the third phase, the dimensions of the overall outdoor kitchen had to be determined. Therefore, the dimensions of the framework were first investigated. The outdoor kitchen will be developed in regard to the components that must be included, these are a sink with tap, working/unloading areas, a grill and a gas stove. The modules for each component will have the same dimensions except for the module with the grill, which will have a two times larger width than the others.

From the chapter *1.0 Frame of reference* standards and recommendations regarding height, depth and width of a kitchen were presented. These dimensions were used in order to set the dimensions of one module and the following are:

- *Height*: Between 850-950 mm
- *Depth*: Minimum 600 mm
- *Width*: Between 450-900 mm

The framework was created as simple as possible in order to optimize storage, ease of construction and preventing possible problems with the modularity. Since the framework will be hidden by the cladding, focus could be put purely on durability and also modularity of the construction within the framework. In order to get an idea and inspiration of how the framework could be built, different retailers and DIY (Do-It-Yourself) websites were investigated, one of them being from the company Byggmax (Byggmax, 2022). The final construction was also verified by a person working professionally with kitchen drawings. Recommendations found for similar sorts of constructions were 45x45 mm studs and therefore the framework will be built by these. In *Figure 5.1* the overall dimensions of one framework module are shown. The height of the feet will be adapted to the total height of the outdoor kitchen, more about this comes in the following chapter.

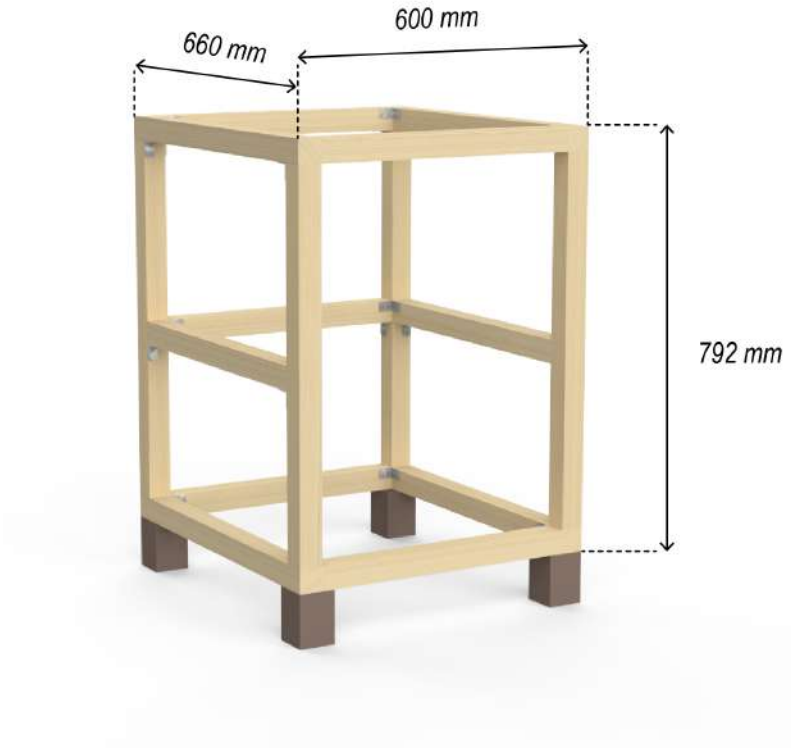


Figure 5.1: The different dimensions of one framework module.

In order to avoid double legs when combining the different modules, two legs of one module will be excluded. In *Figure 5.2-5.3* an example of three modules and how these are combined is shown. Combining the modules like this facilitate the modularity of the outdoor kitchen because it will make the upcoming design of the cladding easier.

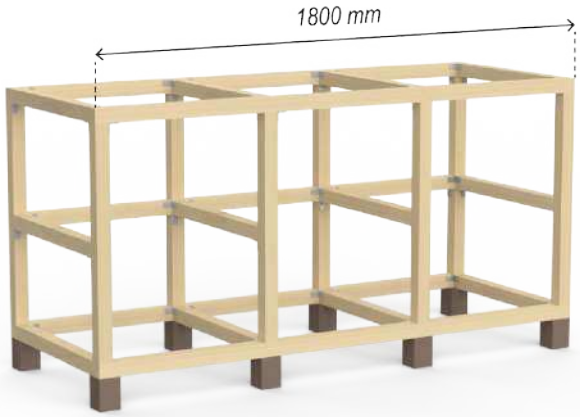


Figure 5.2: An example of three framework modules and how these are combined.

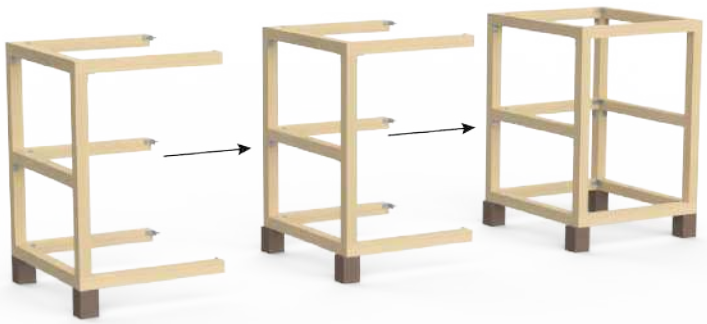


Figure 5.3: An exploded view of three framework modules.

The technical drawings of the framework modules can be found in *Appendix F*. The framework modules will be assembled with galvanized brackets and wooden screws, which can be found at several retailers. The holes for the brackets and feet can be pre-drilled in order to minimize potential cracks in the studs. The framework modules' studs are adapted for each module, but when several are assembled together, these dimensions can be changed in order to minimize the number of studs which will also optimize the assembly.

5.2.2 Material and retailers

The material chosen for the framework is pressure treated pine. This material was chosen because of the earlier material analyzes and also because of its accessibility. Pressure treated pine will ensure protection against rot, bacterial growth and weakening of the material over time. Further, it is a commonly used material for similar constructions. The previously mentioned dimension of the studs, 45x45 mm, will be ordered by meter from a manufacturer. There are multiple retailers that offer pressure treated wood and it is available throughout Sweden. Depending on the retailer, the price varies between 15-24 SEk/meter (Bygghemma Sverige AB, 2022). Part of the construction is the option of having wheels in order to easily move the built kitchen. There is a whole range of available wheels that could be possible options for the final design but has to follow the requirement of a max diameter of 50 mm. Limiting the diameter to 50 mm is due to the total height of the kitchen, preventing it from becoming to tall. Furthermore, the wheels should preferably be constructed in steel to match the rest of the outdoor kitchen and be able to carry at least 40 kg in dynamic load.

5.3 Cladding

When the construction and dimensions were set for the framework modules it was possible to develop the cladding for the outdoor kitchen. The process for the cladding was iterative and many different designs were explored. The cladding will hide the framework modules and it was divided into two parts: the inner cladding and the outer cladding. The inner cladding are the ones that cover the outdoor kitchen from the inside and includes inner walls, floor and shelves. The outer cladding are the walls that cover the outdoor kitchen from the outside and includes doors for the modules.

5.3.1 The outer cladding

When iterating on the outer cladding, several wooden joint types were investigated. The idea was to keep the joints minimalistic. Meaning, they would have a discreet but also good craftsmanship expression. Therefore, it was decided to make the wooden joints in square shapes which draws inspiration from Scandinavian furniture. A picture of the outer cladding can be seen in *Figure 5.4*.



Figure 5.4: The corner of the kitchen showing the wooden joints of the external cladding.

It was also decided to make the wooden joints only at the corners of the outdoor kitchen, creating a more seamless impression. It also helped for the manufacturing of the outer cladding, since the joints require different walls in order to fit together. Requiring more walls, contributes to an increase of the production costs and also more steps during the assembly. The seamless cladding also prevents infestation, water and dirt from getting inside of the outdoor kitchen. In *Figure 5.5* the seamless walls of the back of the outdoor kitchen can be seen. Here, three walls are put together in order to hide the framework modules.



Figure 5.5: The back view of the cladding put onto the outdoor kitchen.

Included in the development of the outer cladding was also the design of the fittings of the outdoor kitchen. From the decision making of which concept to continue with, it was decided to use wooden fittings that match the rest of the outdoor kitchen. Several different types of wooden fittings were tested, based on the ones that were ideated in concept one, in order to investigate the different expressions they would make. Here, it was looked at fittings that are cut out from the doors or added to the doors. It was decided to use a fitting list that is added onto the door and that goes from the top to bottom. Choosing an external type of fitting prevents dirt from getting inside of the outdoor kitchen. Furthermore, making it go in the vertical direction also prevents water from gathering in the fittings. In *Figure 5.6* the fittings of the kitchen are shown.

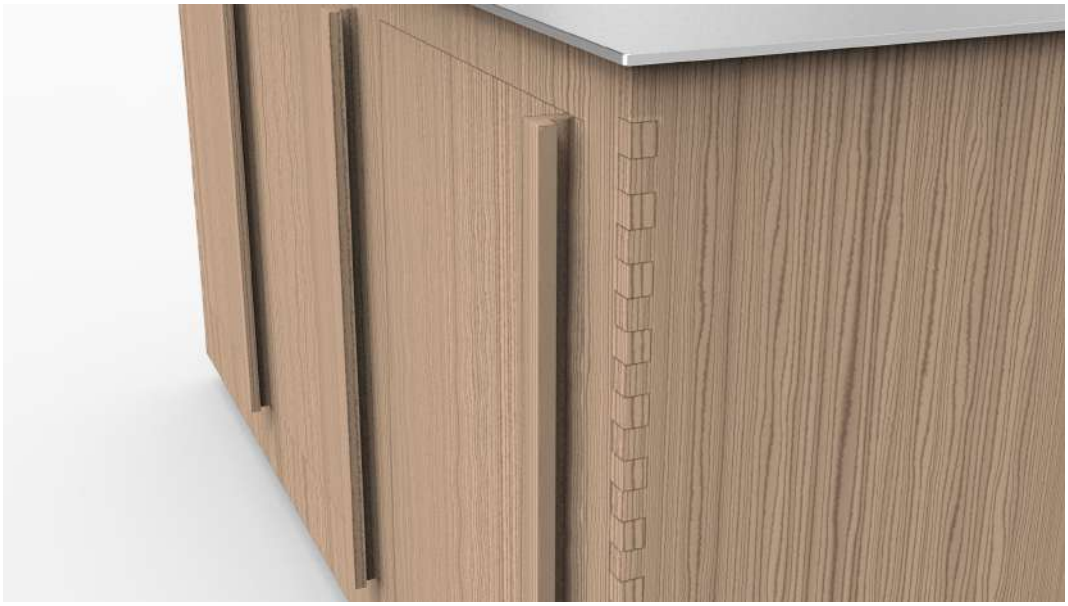


Figure 5.6: A view of the fittings of the outdoor kitchen.

Furthermore, the fittings are attached to the doors similar to the external wooden joints and in *Figure 5.7* this can be seen. The fittings will be fastened to the doors by using wood glue.

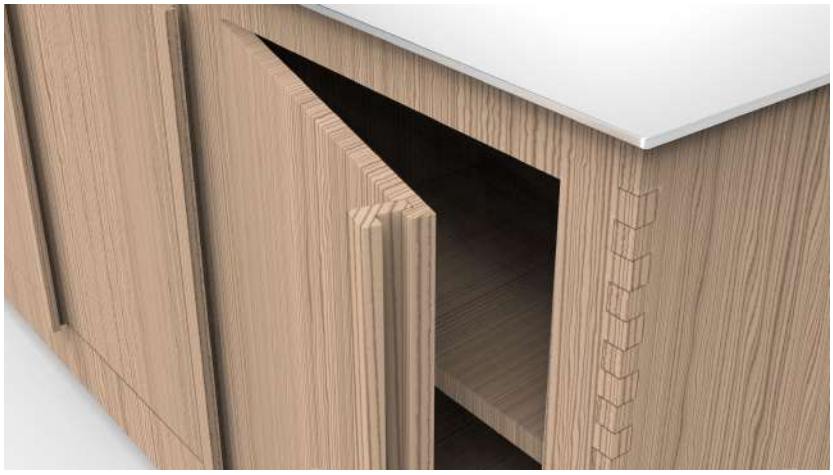


Figure 5.7: A zoomed in view of the fittings.

The dimensions of the outer cladding are adapted to the framework modules. The walls will have pre-defined dimensions where these can be combined differently depending on how many framework modules that are put together. In *Figure 5.8* the dimension of the wooden joints are shown as well as the shelves thickness. For further measurements see *Appendix F*.

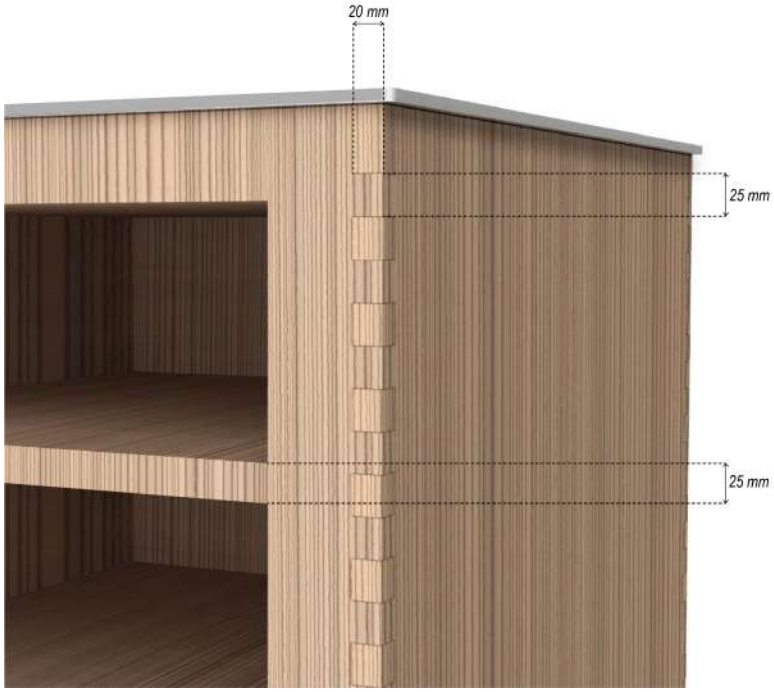


Figure 5.8: The dimensions of the outer claddings' wooden joints.

During the presentation where the concept was chosen, it was also decided that the outer cladding should cover the feet or wheels of the outdoor kitchen. However, some space was left between the ground and the outer cladding in order to facilitate cleaning around the outdoor kitchen. The space also helps to facilitate the locking of the wheels, see *Figure 5.9* for the dimension of the space.

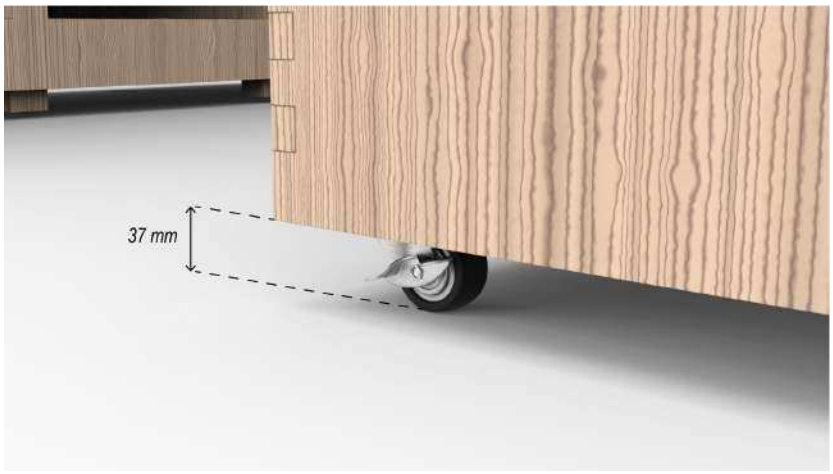


Figure 5.9: The dimension of the space left between the external cladding and ground.

The dimension of the space between the ground and the cladding is the same for both the wheels and the feet. When designing the feet, focus was put on creating feet that gives the impression of stability and mass. Different feet in both stainless steel and cedar wood with varying shapes were tested. It was decided to have square shaped feet made out of cedar wood. It goes in line with the rest of the kitchen and expresses a discreet yet stable impression. The dimensions of the feet are 70x70 mm and in *Figure 5.10* they are presented.



Figure 5.10: The feet of the outdoor kitchen.

5.3.2 The internal cladding

When the outer cladding was designed the inner cladding could be developed. Here, focus was put on enclosing the outdoor kitchen in order to prevent infestation and protection from water and dirt. The inner cladding is composed of the inner walls, floor and inner roof. The inner walls have movable support which makes it easy to adjust the shelves depending on need. The shelves placement will be in line with the outer claddings' wooden joints, in order to create symmetry within the outdoor kitchen. There are two different types of inner walls, one that can be used on the left side and another for the right side. Furthermore, there are two types of back inner walls used for the modules, where one of them is used for the two times larger grill module and the other one is used for the rest of the modules. The inner roof is dependent on the white goods as there will be a specific cutout for each. The floor's length is dependent on the amount of modules combined as well as which modules. For example, the grill and gas stove will have cutouts for ventilation in their modules meanwhile the sink and tap will have a cutout for the water hose. In *Figure 5.11* the internal cladding of one framework module is presented. In *Figure 5.12* it is shown how the shelves are hanged in line with the external claddings' wooden joints.



Figure 5.11: The internal cladding of one framework module.



Figure 5.12: The shelves placement can only be hanged in line with the external claddings' wooden joints.

In order to have proper ventilation of both the charcoal grill and the gas stove, square shaped cuts was placed in the bottom of respective modules. This will create airflow to the charcoal grill as well as let leaked gas from the tank escape. Squared shapes was chosen in order to follow the rest of the outdoor kitchens formgiving. The cutouts was also placed closed to the inner back wall in order to prevent blocking the holes if something is stored inside. See *Appendix F* for the placement and dimensions of the cutouts.

The cutout for the hose in the sink and tap module was made as a circle. This cutout was also placed close to the inner back wall and vertically in line with the sink and tap. In *Appendix F* the placement and dimension of the cutout can be seen.

5.3.3 Material, components and dimensions

The cladding itself is made from glued together planks of Eastern red cedar wood. The material choice is based on price, suitability and design. Generally, Eastern red cedar is cheaper than other suitable types of oak and it is cheaper than other exotic woods such teak. It is naturally resistant to weather conditions and infestations due to a high sap content. Also, for being a softer wood, it is very durable for use such as furniture due to its form stability. When cedar is exposed to UV light from the sun it tends to become more gray in color. The species itself is not vulnerable and is considered invasive in many areas. It is ranked by ”Least Concern” in relation to its conservation

status (Farjon, 2011) and is therefore a suitable material for production.

The dimensions of the planks will vary depending on which retailer is chosen and the designed construction is a proposal. The set dimensions of the inner and outer cladding will determine the necessary plank quantity and dimensions. Depending on the retailer, this will also change the overall impression of the kitchen due to variations of the wood. The thickness of the outer cladding is 20 mm meanwhile the inner cladding has a thickness of 10 mm. In *Appendix F* the different dimensions of the cladding is presented. The price is based on rough estimates due to the varying wood price over time and where it is purchased from. For the inner cladding, a 11x92 mm plank is around 69 SEK per meter while the thicker outer cladding is made from planks of 18x140 which is 236 per meter (Calexico Wood, 2022).

The cladding will be attached to the framework with hot-dipped galvanized screws and brackets. First the inner cladding will be installed in order to avoid visible screws from the inside. Then the outer cladding will be attached to the framework from the top in order to hide the screws and brackets. The cabinet doors will attach via overlay hinges in order for a flush fitted and adjustable cabinet door.

5.4 Countertop

The countertop is composed of a massive stainless steel sheet, being 8 mm in thickness and 720 mm in depth, and is adapted to each module. The total length depends on the amount of modules that the customer decides on. Since the cutout dimension for each white goods is varying it has to be adapted for each module. However, each cutout for the white goods are in line with each other creating symmetry within the outdoor kitchen. The cutouts begin at 75 mm from the front of the countertop. The sink is attached by gluing the bottom of the countertop and attaching the lip of the sink onto it. This creates a more seamless transition from the countertop to the sink. The stove and grill are placed in the cutouts of the countertop and then sealed with silicone in order to create waterproof modules. The countertop itself will be attached by glue to the framework and then a sealant will be applied around the countertop. As to avoid leakage from harsh weather and unnecessary visible crevices, the countertop will be 10 mm longer on each side for every module. The smaller sized modules has a 660 mm wide countertop while the larger grill module has 1215 mm. The overall dimensions can be viewed in the *Appendix F*. The price of the countertop will vary depending on chosen retailer, however a price suggestion is presented in *Table 5.1*.

5.5 Choosing white goods

When choosing the white goods, research was done according to both function and design. The hardware had to match the rest of the outdoor kitchen in shape and texture as much as possible. Here, it was important to find products with a similar finish and hue of the material as well as overall formgiving, avoiding styles that do not match with the overall outdoor kitchen. Considering the chosen criteria for the outdoor kitchen and desired target group, quality had to be in focus as well, choosing hardware from known and reputable retailers. This is only a recommendation for hardware and these can be switched out if needed. However, none of the hardware found had standard dimensions, meaning simpler adjustments had to be made to match the chosen products.

5.5.1 Grill

It was decided for the final concept to have a charcoal grill instead of a gas grill. This was decided due to several reasons. Firstly, it was deemed easier for the customer and for the sake of modularity to only have one module connected to a gas canister, the gas stove. Since private persons can only have access to two canisters per household without a specific license, a reserve canister for the stove is therefore possible. Secondly, it will clear out space from storage underneath as well as remove the need for tube management as canisters can not be located directly under a gas grill. Finally, the feeling associated with a charcoal grill will further match Livsytas' brand better in comparison with a gas grill and was favored by survey and interview insights.

For this, market research was made in order to find a charcoal grill suitable for the kitchen. This was done by finding retailers, mostly within Sweden, but internationally was a possibility depending on assortment, transport and cost. Here a built-in charcoal grill was desired instead of a ready-to-use grill.

It was decided to go with a charcoal grill from Roths, seen in *Figure 5.14*, due to the overall aesthetic matching the rest of the kitchen. Furthermore, the functionality of the charcoal grill matches the expectation of Livsyta's target group. In connection with the decision-making of the charcoal grill, the company Roths was also contacted. A potential collaboration was discussed with one of the owners which were found very interesting. If a collaboration would happen, a re-designing and customizing of the charcoal grill could be possible in order to create a grill that suits Livsyta's outdoor kitchens even better.

In regard to heat and the main composition of the final design being wood, some



Figure 5.13: The chosen grill from Roths.



Figure 5.14: Open grill with visible grid.

sort of heat protection might be needed between the grill and the frame. One option would be to buy already manufactured grill jackets, which do not exist for this size but could be manufactured by Roths for the specific grill. Another option commonly used within outdoor kitchen setups is attaching fiber cement boards. These act as isolation around the grill and would protect potentially exposed wood within a specific distance from heat. Since the grill is not explicitly touching any parts of the framework, the heat will not be direct and will dissipate into the air.

5.5.2 Gas stove

For the gas stove there were no specific needs required except for limiting the size to one module. This was due to the need of a gas stove and how the users from Phase I utilized it, demanding not having too many burners. This meant to reduce the amount

of burners to only two. Here, focus on finding a gas stove was placed on the quality, overall looks matching the concept and brand. After some research for potential gas stoves, the PGF32G from Smeg gas stove was chosen. The gas stove matches the overall design of the charcoal grill from Roths and it also has an affordable price. In *Figure 5.15* the Smeg gas stove is presented.



Figure 5.15: The chosen gas stove, PGF32G from Smeg.

To protect the gas stove from rain and snow, a lid has been designed as well, which can be seen in *Figure 5.16*. It is composed of a stainless steel sheet being bent into a boxy shape that can easily be placed and removed before or after use and can be seen in *Appendix F*.



Figure 5.16: The added lid, protecting the stove from water

5.5.3 Sink and tap

The choice of sink was required to be at least 300 mm in width but smaller than 600 mm to fit in the module while remaining large enough for normal use. Furthermore, some basic functionality of the tap was desired such as having an extendable pip. This

would make it easy for users to quickly fill up larger sized pots with water, quickly rinse kitchenware and spray down the kitchen to remove dirt and pollen. When searching for suitable sinks and taps, it was also important to keep in mind that they should match the chosen charcoal grill from Roths and the gas stove from Smeg.

The chosen sink for the final concept design was the Smeg Mira, seen in *Figure 5.17* (Smeg, 2022). Further, the chosen tap for the final concept design was the Primy Steel Accenture Squared tap from Scandtap (Scandtap, 2022). These were chosen in regard to costs, overall design and customer reviews.



Figure 5.17: The chosen Smeg Mira sink and Primy tap for the outdoor kitchen.

5.5.4 Price analysis

Estimates were made in regard to final price in order to map the total costs in relation to material. The analysis was made through online research, finding retailers selling the necessary material and components. These are only estimates as these might differ, ordering in bulk might decrease the price further. The price calculated is for one module. Individual costs and length can be seen in *Table* 5.1 for raw materials and for white goods.

Table 5.1: Price estimates of the materials and hardware.

Part	Materials	Quantity	Price
Framework	Pressure treated pine	9,3 m	340 SEK
Cladding	Cedar	34,7 m	4800 SEK
Countertop	Stainless steel	0.6x0.7 sqm	6700 SEK
Roths grill	Stainless steel	1 piece	15000 SEK
Smeg gas stove	Stainless steel	1 piece	10000 SEK
Smeg sink	Stainless steel	1 piece	2700 SEK
Primy tap	Stainless steel	1 piece	4000 SEK

From this it is possible to determine the prices of each module. The cheapest module will be the workspace at around 11800 SEK while the most expensive will be the grill module at 32900 SEK. Between these are the sink and gas stove module at 18500 SEK and 21800 SEK.

From the analysis it is possible to determine that the final design is within the price range of Livsyta and matches their previous iterations. However, cost of manufacture and shipping has not been included as well as potential sales tax, meaning the price for the customer will differ. These prices are also dependent on the hardware chosen which could be replaced in order to reduce costs. Similar can be done for the cladding as cedar is a more luxurious wood and that is reflected in the price. Other potential material options for the cladding could be pine from ThermoWood or OrganoWood.

5.6 Final concept design

The final design is a new iteration of Livsytas' previous designs with added functionality, new aesthetic and modularity, giving the customer the option of choosing their own layout. This iteration also fits well in the gap of the current market

in regard to storage, hardware and design. Offering a more traditional outdoor kitchen than the thin stainless steel ones or bulky outdoor kitchens. The new outdoor kitchen includes four modules that can be combined in multiple ways and these are presented in *Figure 5.18*. The four modules consists of a sink with tap, charcoal grill, gas stove and workspace/unloading areas. These modules can come with or without cabinets if a more open outdoor kitchen style is desired. The materials chosen was pressure treated pine for the framework, Eastern red cedar for the cladding and massive stainless steel for the countertop. The materials chosen are durable enough to withstand not being covered when not in use, which was one of the deciding factors. Cedar has a natural patina where it gets a gray color when exposed to sun. The combination of a hygienic stainless steel countertop and an aging wooden cladding creates a balance between the materials in the outdoor kitchen. But, if not wanted it is possible to oil the cladding in order to remain the feeling of a new outdoor kitchen.



Figure 5.18: All modules included in the final design.

A potential layout of the outdoor kitchen is shown in *Figure 5.19* and its framework in *Figure 5.20*. Here, two kitchen islands are built including all of the four modules as well as one open module. The suggested layout offers great storage, large working and unloading areas as well as social areas where family and friends can easily be part of the cooking. Furthermore, having wheels on one of the kitchen islands also makes it easier to adapt the layout depending on the amount of people around. Based on the earlier price analysis, the price of the individual kitchen islands will be 66500 SEK for the larger setup while the smaller setup is 42100 SEK, leading to a total price of 108600 SEK.

Inside the modules there is enough space for storing anything from charcoal and gas cans to kitchenware and the adjustable shelves make storage convenient and provide order, seen in *Figure 5.21*.



Figure 5.19: A view of a setup with all the modules included.



Figure 5.20: A bare-bone view of the construction together with the hardware and countertop.

Finally, a step by step guide on how to build one of the kitchen islands is shown in *Appendix G*.



Figure 5.21: The dimensions of the storage space.

5.6.1 Further sustainability analysis

To further map the effects on the environment of the product, *Eco Audit* in *GRANTA EduPack* was used to get a rough approximation of environmental burden and which part of the outdoor kitchens life cycle had the biggest impact. This function makes it easy to calculate energy consumption and CO2 footprint from certain variables, such as material, manufacture, transport and end-of-life. The created report and its results can be seen in *Appendix H* and is done for the working space/unloading area module. In the report, pressure treatment of the wood is not calculated since there was no option for this. However, pressure treated wood is not recyclable and had to be deposited into a landfill as end-of-life.

For the analysis, pine from Sweden was used as the construction material for the framework. The cladding was in cedar from Canada and the stainless steel was from Sweden as well. The construction of the outdoor kitchen will be made in Sweden. End-of-life for the product follow the requirement specification of 10 years minimum. The sustainability analysis is not made in regard to potential surface treatments and other smaller components needed but is strictly done on the materials.

From the report of *Eco Audit*, it was noted that the biggest negative impact on both energy and CO2 emissions come from the material gathering. Stainless steel was

the material with the biggest impact on both aspects while cedar comes in at second. However, both materials have a better end-of-life potential as stainless steel can be recycled and cedar can be reused. Pressure treated pine had no end-of-life potential due to it being deposited into a landfill.

6 Discussion

During the project there have been several aspects to keep in mind when developing the final outdoor kitchen concept. Aspects such as, key insights from users, potential competitors on today's market, finding suitable materials, the overall design in regard to Scandinavian and Japanese design, the modularity and functions, maintaining a sustainable thinking and finally developing an outdoor kitchen in relation to the brand Livsyta. These aspects have been kept in mind and balanced in order to be able to implement them on the final design concept. However, there are some things that could have been done differently or should be investigated further.

For the survey, some questions could have been asked differently. Some of the answers from the respondents were misleading as people assumed things differently, example is the price of their current outdoor kitchen. It was difficult to understand if people were referring to only their kitchen or including other products as well, such as grills. Changing the question to be more clear would have given a better picture of price and how much they would have been willing to pay.

In order to be sure that the construction of the final concept can be manufactured and produced a prototype should be built. Here, it is possible to see if anything should be changed or re-designed in order to optimize the assembly in regard to time, materials and costs. A lot of research based on how kitchens are built and manufactured was found during the project. Thus, in order to ensure that it is physically possible to produce the kitchen in the design that it has, a prototype needs to be built. Building a prototype will facilitate later large-scale production of the outdoor kitchen. This will allow for an investigation on whether any minor corrections to the design have to be done. For example, in this project the two proposed kitchen islands are built based on their individual modules in order to present the modularity. However, when manufacturing these some changes could be done, for example minimizing the number of studs if several framework modules will be combined together. Furthermore, the possibility of repairability could be looked over. Since the final concept design has a cladding that encloses the whole outdoor kitchen it makes it harder to exchange parts that break or get damaged. This could also be investigated further in order to have a more sustainable mindset where the outdoor kitchen's life-span could increase if parts could be exchanged. However, it was chosen to only place the charcoal grill and gas stove onto the countertop, providing only waterproof sealant around the crevices because if the white goods break they could be exchanged. For hygenic reasons, the sink could be welded on the countertop to increase durability and reduce crevices. This would however, make

repairability harder if the sink would need to be replaced.

The materials chosen for this project and for analyzing in Phase II were common materials used within outdoor kitchen design. However, there were many other potential materials that could have been analyzed as well but were weeded out early. This due to price and availability. Price is also difficult to analyze since the market for raw materials fluctuates, this especially applies to wood as prices have been soaring as of writing this report. Utilizing *GRANTA EduPack* also made it difficult as some aspects could not be taken into consideration, such as material treatments. This meaning further research had to be made by other means.

From a price and sustainability standpoint, other wood materials could be more suitable than cedar. Cedar was chosen as it was form stable and would add a sense of luxury to the product and is a proven material to this purpose. However, this effects the price as it was higher than other materials filling the same purpose. Moreover, since cedar does not grow in Sweden, importing it from other countries is necessary, creating longer transport routes for the material. ThermoWood, as mentioned earlier, is a viable option and is also made in Sweden. However, not as much data was available on how well the material works in relation to other proven materials.

Further development of the outdoor kitchen is also the potential of designing more modules. Insights from both the survey and conducted interviews displayed a small need for a pizza oven. However, it was decided that priority should be placed on the chosen modules of a sink with tap, workspace, grill, and gas stove. A pizza oven could be added in further iterations of Livsyta with added modules. Further iterations should include more modules depending on what the user base wants. Such as more alternatives for materials and components. The option of having a smaller or bigger grill is relevant as well considering the grill in this project is only possible in the bigger module size.

7 Conclusion

The project resulted in a final design of a modular outdoor kitchen, inspired by both Scandinavian and Japanese design. Adhering to the needs of users, the necessary functionalities and also the requirements specification the purpose of the project has been fulfilled. The material analyzes that were made in regard to suitability and sustainability made it possible to develop an outdoor kitchen concept that is optimized for its purpose.

There are several outdoor kitchen products available on today's market but there is also a gap when it comes to their aesthetics and functionalities. With this modular outdoor kitchen concept, Livsyta could start scaling their business and fill this gap within the market. Providing the modular function of the outdoor kitchen concept, Livsyta also has the potential of iterating and creating new modules that could be offered in the future.

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Appendix A

Questions asked in the Facebook forum are presented in the following appendix.



Enkät om utomhuskök

Hej! Vi är två studenter från KTH som ska, under vårt examensarbete, designa och utveckla ett utomhuskök. Vi är därför intresserade av att veta vad du har för kunskaper och erfarenheter om just utomhuskök! Vi skulle verkligen uppskatta om du hade tid att svara på några frågor, det tar bara några minuter och enkäten är anonym! Tack på förhand!

Hur gammal är du?

- 20 år eller yngre
- 21-30 år
- 31-40 år
- 41-50 år
- 51-60 år
- 61-70 år
- 70+ år

Könsidentitet *

- Man
- Kvinna
- Annat
- Vill ej ange

Hur bor du? *

- Lägenhet
- Radhus
- Villa
- Gård
- Other...

Har du tillgång till något av följande? Flera alternativ är möjliga. *

- [ ] Trädgård
- [ ] Innergård
- [ ] Altan
- [ ] Brygga
- [ ] Balkong
- [ ] Växthus
- [ ] Inget av det nämnda
- [ ] Other...

Tycker du om att laga mat? *

	1	2	3	4	5	6	7	8	9	10	
Avskyr att laga mat	○	○	○	○	○	○	○	○	○	○	Älskar att laga mat

Har du ett utomhuskök idag? *

- ○ Ja
- ○ Nej

Vad ingår i ditt utomshuskök? Flera alternativ är möjliga. *

- [ ] Diskho
- [ ] Vattenkran
- [ ] Stekhäll
- [ ] Gashäll
- [ ] Grill
- [ ] Ugn
- [ ] Kylskåp
- [ ] Frysskåp
- [ ] Förvaring
- [ ] Avlastningsyta
- [ ] Arbetsyta
- [ ] Other...

Var är ditt utomhuskök placerat? *

- [ ] Altan
- [ ] Gräsmatta
- [ ] Asfalt
- [ ] Stenplattor
- [ ] Brygga
- [ ] Strand
- [ ] Other...

Hur ofta använder du ditt utomhuskök i snitt? *

- ( ) Aldrig
- ( ) Cirka en gång per år
- ( ) 2-5 gånger per år
- ( ) Varje månad
- ( ) Varje vecka
- ( ) Varje dag

Vad är det bästa med ditt utomhuskök? *

Long answer text

Är det något du saknar i ditt utomhuskök? *

Long answer text

Brukar det uppstå några problem eller irritationer när du använder ditt utomhuskök? *

Long answer text

Hur vinterförvarar du ditt kök? *

Long answer text

Vilket/vilka material är ditt utomhuskök gjort av, t.ex. rostfritt stål, betong, trä mm? Specificera gärna så noga som möjligt! *

Long answer text

Behöver ditt utomhuskök någon typ av underhåll och i så fall vad? T.ex. inoljning mm. *

Long answer text

Om ditt utomhuskök behöver underhållas, hur mycket tid tar det och hur upplever du det?

Long answer text

Hur mycket betalade du för ditt utomhuskök? *

- ○ Mindre än 5000 kr
- ○ 5000 - 10 000 kr
- ○ 10 000 - 30 000 kr
- ○ 30 000 - 50 000 kr
- ○ 50 000 - 100 000 kr
- ○ Mer än 100 000 kr

Vad är det för varumärke på ditt utomhuskök? *

Short answer text

Skulle du vilja ha ett utomhuskök? *

- ○ Ja
- ○ Nej

Varför vill du inte ha ett utomhuskök? *

Long answer text

Vad vill du ska ingå i ditt utomhuskök? Flera alternativ är möjliga. *

- [ ] Diskho
- [ ] Vattenkran
- [ ] Stekhäll
- [ ] Gashäll
- [ ] Grill
- [ ] Ugn
- [ ] Kylskåp
- [ ] Frysskåp
- [ ] Avlastningsyta
- [ ] Arbetsyta
- [ ] Other...

Var skulle du vilja placera ditt utomhuskök? Flera alternativ är möjliga. *

- [ ] Altan
- [ ] Gräsmatta
- [ ] Asfalt
- [ ] Stenplattor
- [ ] Brygga
- [ ] Strand
- [ ] Other...

Vilket/vilka material skulle du vilja att ditt utomhuskök var gjort av, t.ex. rostfritt stål, betong, trä mm? Specificera gärna så noga som möjligt! *

Long answer text

Hur mycket skulle du kunna betala för ett kök? *

- ○ Mindre än 5000 kr
- ○ 5000 - 10 000 kr
- ○ 10 000 - 30 000 kr
- ○ 30 000 - 50 000 kr
- ○ 50 000 - 100 000 kr
- ○ Mer än 100 000 kr

Om du skulle kunna tänka dig att delta i en mer detaljerad intervju som handlar om utomhuskök och användning av dem, får du gärna skriva din emailadress nedan!

Short answer text

Appendix B

Presented here is first the interview guide aimed for participants with outdoor kitchen and then participants without outdoor kitchens.

Intervjufrågor - Personer med utekök

Om oss

Vi är två studenter från KTH som ska, i vårt examensarbete, designa och utveckla ett komplett utomhusök. Syftet med intervjun är att få en bredare förståelse och kunskap om hur utomhusköket ska vara utformat och vilka funktioner som är viktiga när man lagar mat utomhus.

Grundläggande info

Kan du börja med att berätta lite om dig själv.
Vad heter du?
Hur gammal är du?
Könsidentitet?
Hur bor du?

Djupgående info

Tycker du om att laga mat? Varför?

- Vad brukar du laga?
- Hur ofta lagar du mat?

Kan du visa/beskriva ditt utomhusök? Placering? Material? Layout?

- Varför har du ett utekök?
- Var är ditt kök ifrån (varumärke)?
- Vad ingår i ditt kök?
  - olika funktioner
  - förvaring?
  - utrustning?
  - förvarar du mat i köket?
- Brukar ni äta mat vid utomhusköket?
- Brukar ni bjuda över kompisar och laga mat tillsammans här ute? Hur brukar det se ut?
- Vad är det bästa med ditt utomhusök och varför?
- Är det något som är mindre bra med ditt utomhusök och finns det något du saknar? Varför?
- Brukar det uppstå några irritationer/problem när du använder ditt utomhusök? Och isåfall vad?
- Använder du ditt kök året runt? Varför/Varför inte?
- Hur vinterförvarar du ditt kök? Hur gör du med grill, kastruller eller övrig utrustning (det som personen nämnt finns i köket) under vintern?
- Hur förvarar du ditt kök resten av året? (Tex täcker du för det under sommaren etc.)

- Hur rengör du/håller ditt kök rent?
- Är det något som gör det svårt att rengöra?
- Om du skulle jämföra ditt inne- och utekök, finns det några fördelar/nackdelar som du önskar att vardera kök hade/inte hade?
- Beskriv ditt drömutomhuskök.
- Har det stor betydelse hur köket ser ut? Hur ser funktion vs. estetik ut?
  - Gömda vitvaror?
- (Ifall personen byggt uteköket själv) Vad skulle du säga till en nybörjare som ska göra sitt första kök?
- Slutligen, finns det något mer du vill tillägga?

Intervjufrågor - Personer utan utekök

Om oss

Vi är två studenter från KTH som ska, i vårt examensarbete, designa och utveckla ett komplett utomhuskök. Syftet med intervjun är att få en bredare förståelse och kunskap om hur utomhusköket ska vara utformat och vilka funktioner som är viktiga när man lagar mat utomhus.

Grundläggande info

Kan du börja med att berätta lite om dig själv
Vad heter du?
Hur gammal är du?
Könsidentitet?
Hur bor du?

Djupgående info

- Gillar du att laga mat? Varför?
- Hur ofta lagar du mat?
- Vad brukar du laga?
- Brukar du bjuda över kompisar/familj och laga mat tillsammans?
- Beskriv ditt kök.
- Vad tycker du är bra med köket?
- Finns det något som är mindre bra med köket och isåfall vad?
- Finns det något du saknar?
- Brukar det uppstå några problem/irritationer när du lagar mat i köket? Isåfall vad?
- Vad är viktigt när du lagar mat?
- Beskriv ditt drömkök. Hur ser funktion vs. estetik ut?
  - material?
  - utrustning?
  - hur ser förvaringen ut?
  - dölja vitvaror?
- Hur rengör du/håller ditt kök rent?
- Något som gör det svårt att rengöra?

Frågor om utekök

- Vad tycker du om utomhuskök? Har du några erfarenheter av det?
- Har du ett utomhuskök eller skulle du vilja ha ett?
- Om du skulle vilja ha ett:
  - Vad vill du ska ingå i det? Varför?
  - Var hade du placerat ditt utomhuskök? Varför?
  - Vilka material skulle köket ha varit gjort av? Varför?

Appendix C

Presented here are the thematic clustering that was divided into three categories: Functionality, Design and Experience.



Design

Layout



Sustainability



Color and materials



Details



Experience

Social



Maintenance



Appendix D

Presented here are the four personas that were made based on summarized insights from interviews.

Mats 63 yr

Mats lives with his husband in a villa in Ekerö. Both their children moved out a few years ago but they visit regularly. After 30 years in the construction industry, he retired early and with the time off he has rediscovered life anew. In his spare time, Mats enjoys inviting family and friends over where they cook together.

- Mats sees cooking as a social activity where it's nice if everyone helps out.
- Doesn't need to be fully equipped, can manage with the basics.
- Sees maintenance as a fun pastime and is happy to do so.
- Mats' outdoor kitchen has become a cosy gathering place for socialising where guests feel at home.

"If you have people over, the kitchen becomes the 'heart of the house'. It's where people want to hang out and get fed."



Maria 42 yr

Maria lives with her husband in an apartment in Vasastan, Stockholm. Maria has been running her own successful business for a few years now and spends much of her time working. She has a great interest in cooking and eating good food and therefore spends a large part of her income on it. On weekends, she and her husband tend to spoil each other with dinners out on the town or cook three-course meals at home.

- The quality of raw materials and equipment is important and only the best is good enough.
- Likes to make dining an experience where every detail is carefully thought out for the overall impression.
- Would like the kitchen to be maintenance free but hygiene is paramount and is therefore good at cleaning.

"It should be neat and look like a kitchen where someone likes to cook."



Jesper 35 yr

Jesper lives with his partner and 3 year old daughter in a house in Nacka. He has spent several years abroad and especially in Senegal because his family lives there. Jesper works as a consultant and has had an intense career in his younger days. Nowadays a lot of time is spent with his family but he rarely turns down an afterwork with his boys.

- I'm not really interested in cooking but I like to invite friends over for dinner.
- Likes to spend a lot of money on nice branded stuff and is not shy about telling friends how much it cost.
- Wants and buys things that are usually not used to their fullest potential.
- Wants to minimise time spent on maintenance as well as cleaning and gets annoyed when things don't work as they should.

"When the first scratch is made it's already over."



Felicia 48 yr

Felicia is divorced and lives in a townhouse in Huddinge with her daughter. In her spare time she enjoys being out in nature where a big part of it includes hunting and fishing. On her small plot of land she also has several crops that she enjoys working with. On weekdays she works as a chef at a famous restaurant in Stockholm.

- She doesn't care about looks and design, the main thing is that it should be functional and durable.
- She likes to experiment and try out new cooking methods on what she has hunted and grown herself.
- Is prepared to spend time on maintenance and cleaning to prevent things from breaking or wearing out.

"This is my zen zone."



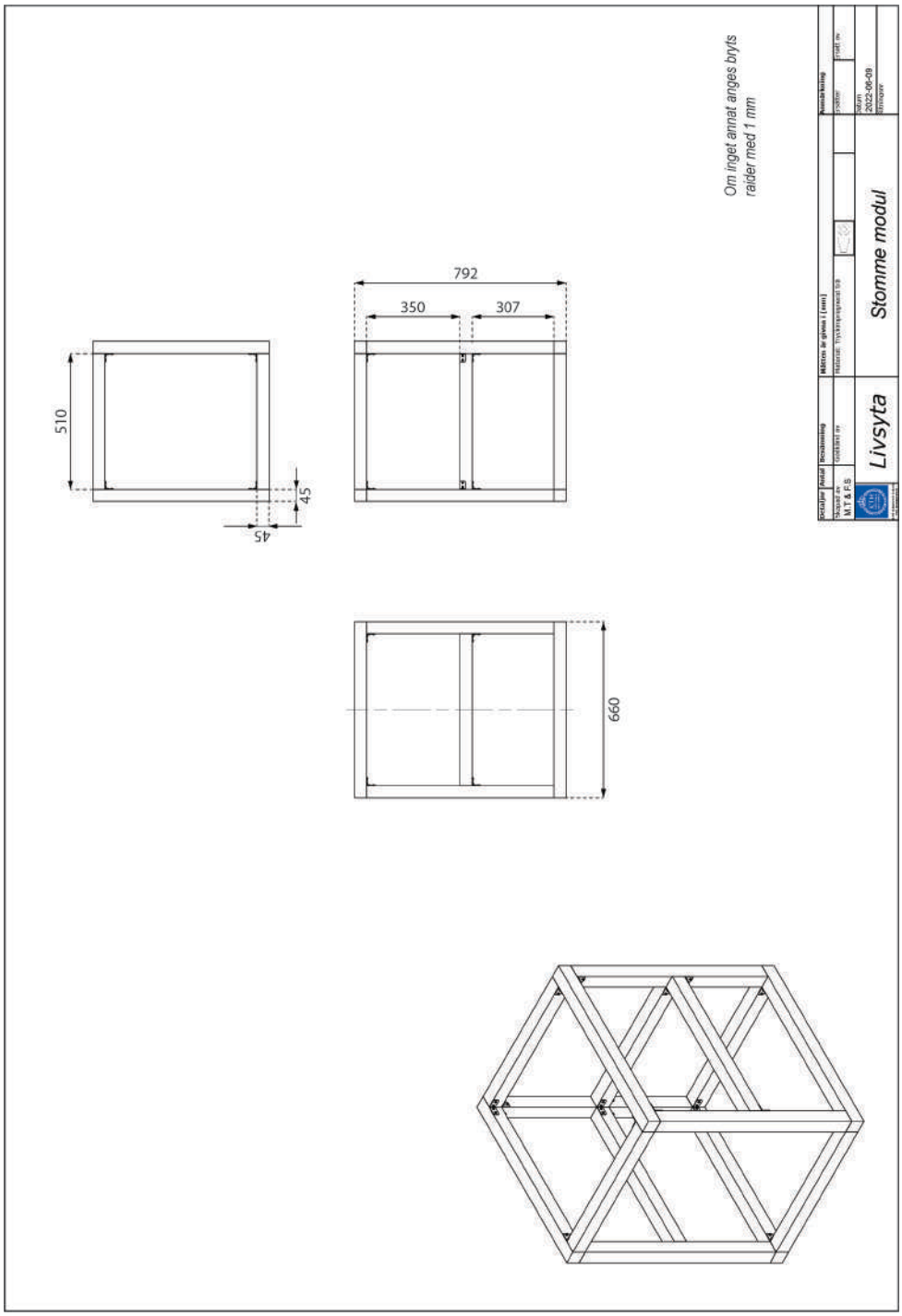
Appendix E

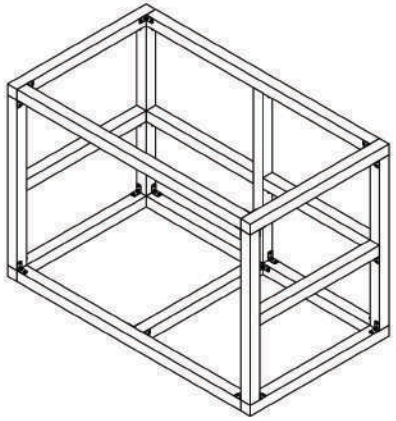
The formulated requirements specification based on knowledge and insights gained during the first phase.

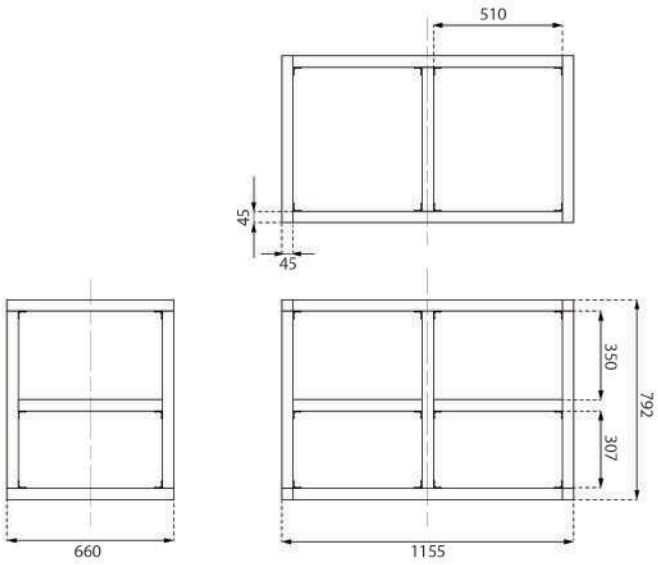
Main function		***Comments***
Offer cooked food	MF	
Subfunctions		
Offer work space	R	Minimum of 1 sqm
Offer unloading space	R	Minimum of 1 sqm
Offer sink	R	
Offer tap	R	Possibility of connecting both hot and cold water
Offer storage	R	Atleast 1 cbm
Provide one grill	R	Either gas or charcoal grill
Provide one gas stove	D	
Offer electrical outlets	D	
Offer storage of chilled food	D	
Offer storage of frozen food	D	
Offer mobility of modules	D	Module should be mobile and able to move
Basic user functions		
Minimize and optimize assembly	D	
Offer the assembly of different types of modules	D	
Offer tailor-made installation	D	
Minimize accumulation of dirt and water	D	
Possibilities for placement on different grounds	D	For example, should be able to place on asphalt, grass etc.
Material and design		
The solution must be modularised	R	
Durable materials	R	
Withstand different types of weather conditions	R	Must withstand water, snow, wind and sunlight etc.
Minimize maintenance	D	Maximum of 10 hours per year
Minimize cleaning	D	Maximum of 20min per use
The solution should be able to be covered over	D	For example covered by a tarpulin
Offer waterproof storage	D	
The design should follow the brand guidelines of Livsyta	D	
Kitchen standards		
Offer the possibility to repair broken/worn components	R	
Comply with kitchen safety standards	R	
Follow standards of hygiene in kitchen environments	R	
Follow standards for dimensioning of kitchens	D	For example SIS recommendations
Production		
Minimize manufacturing costs	D	
Minimize material consumption	D	
Technical drawings must comply with ISO standards	D	
Ergonomics and safety		
The solution should be easy to understand	D	
The solution must be user-friendly	D	
Minimize the risk of various injuries	D	Minimize pinching and burn damages
Quality		
Withstand long use	R	Atleast 10 hours of continues use
Withstand heat and cold	R	
Withstand shocks and strokes	R	
The kitchen should have a long life	D	Atleast 10 years
Market and target group		
The solution must appeal to the target group	D	
Allow high recognition	D	
Differentiate from competitors	D	
Express high quality	D	
Perceived as affordable in relation to quality	D	

Appendix F

The technical drawings of each module within the outdoor kitchen concept.

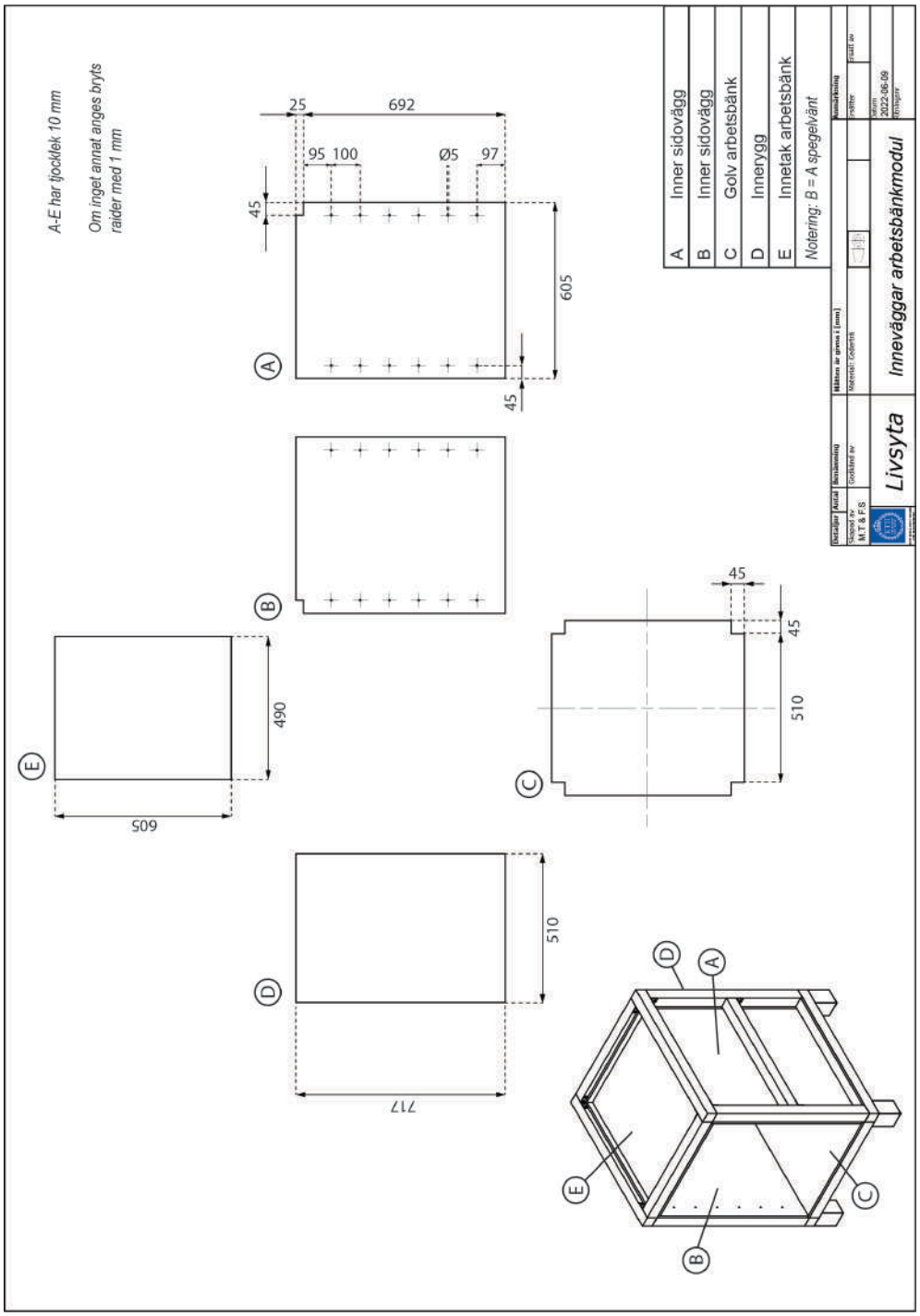


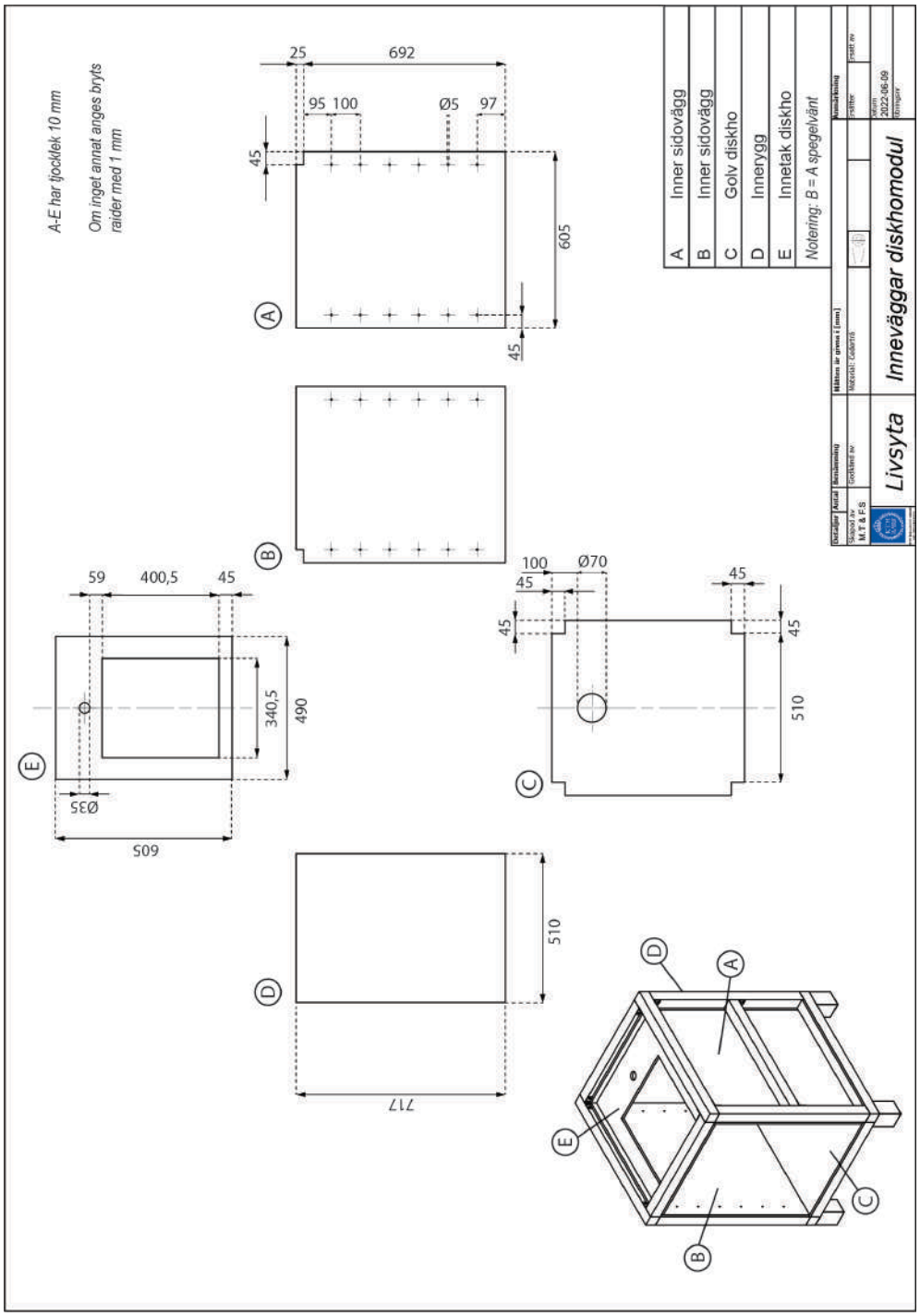


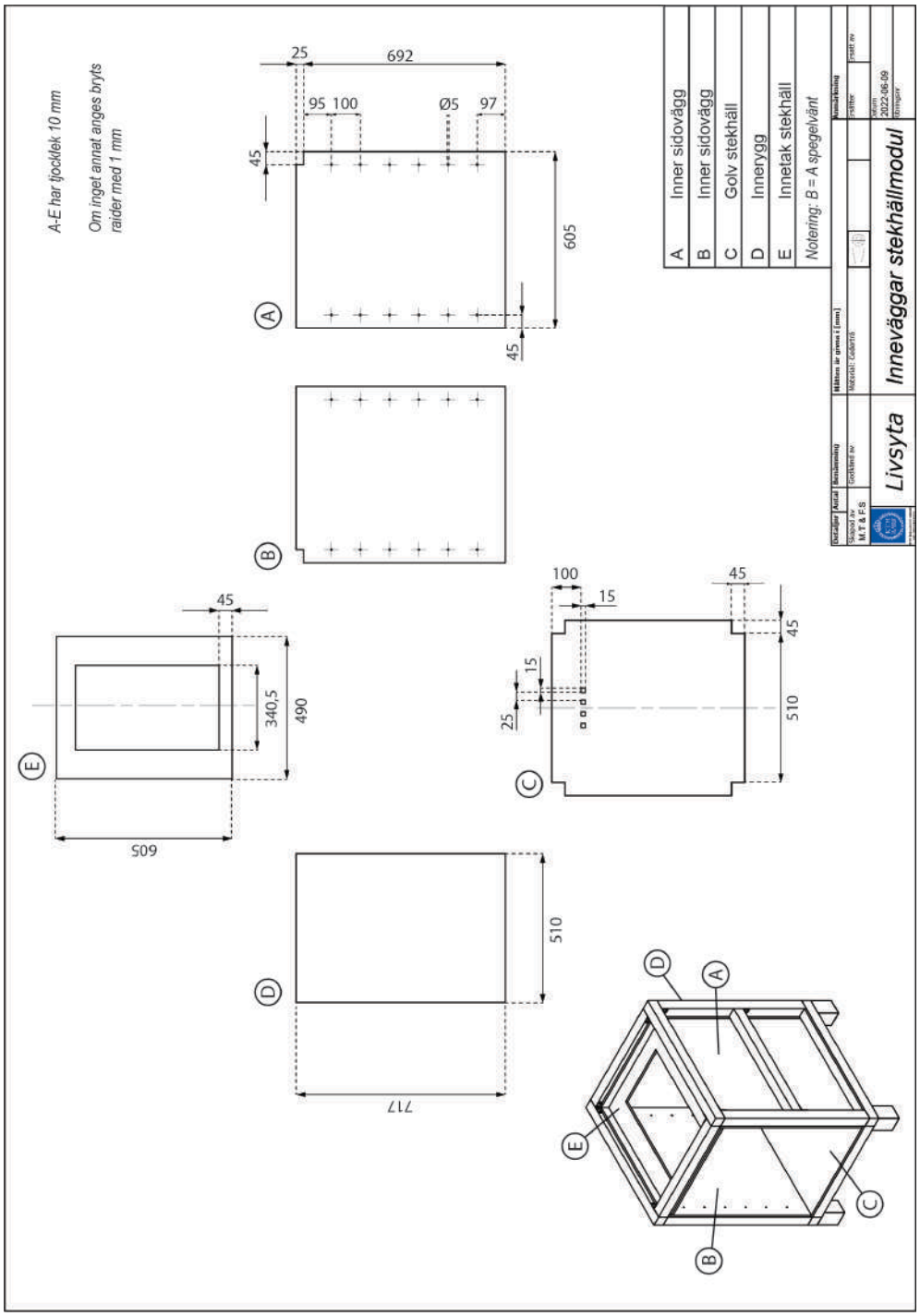


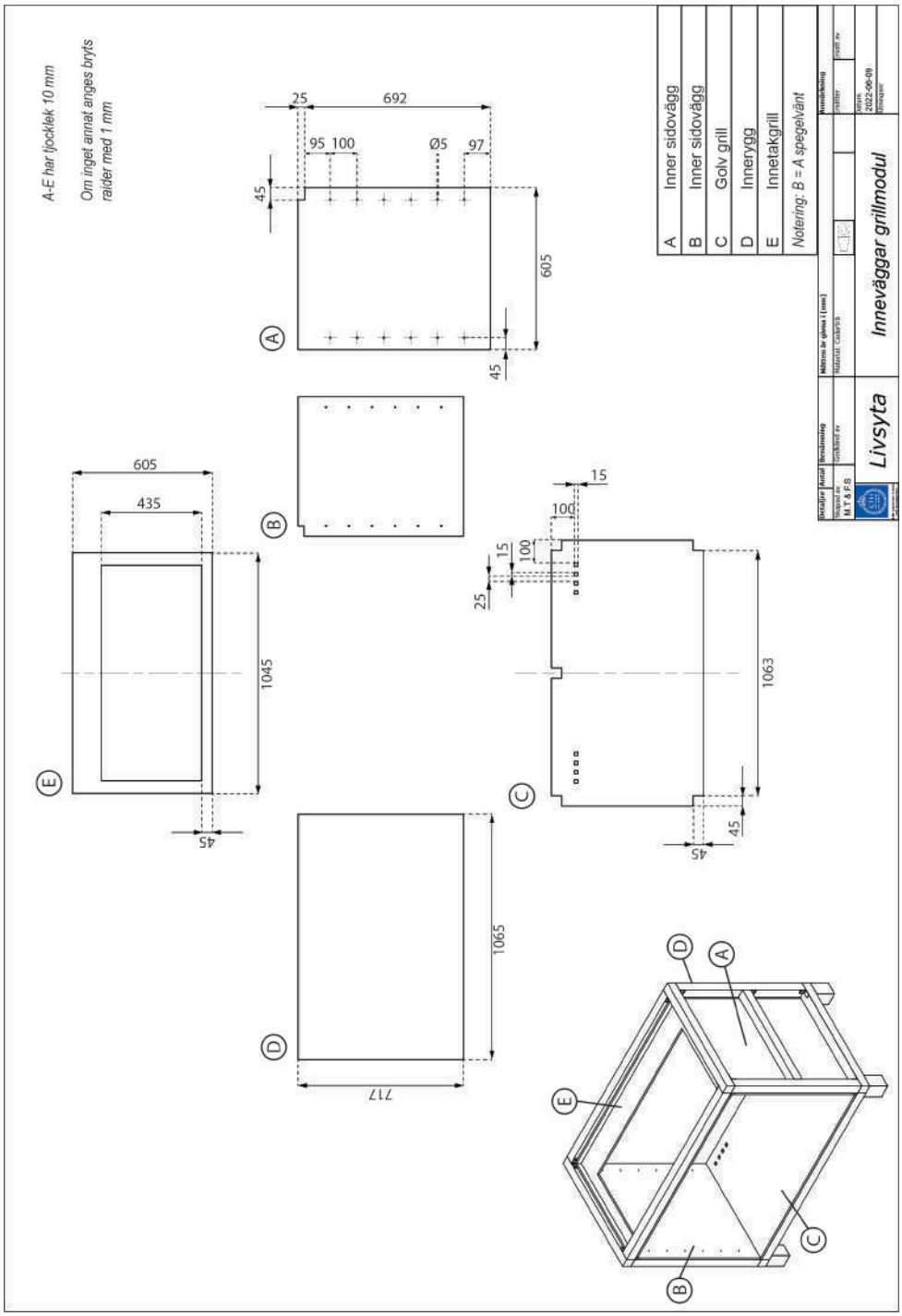
Om inget annat anges bryts raider med 1 mm

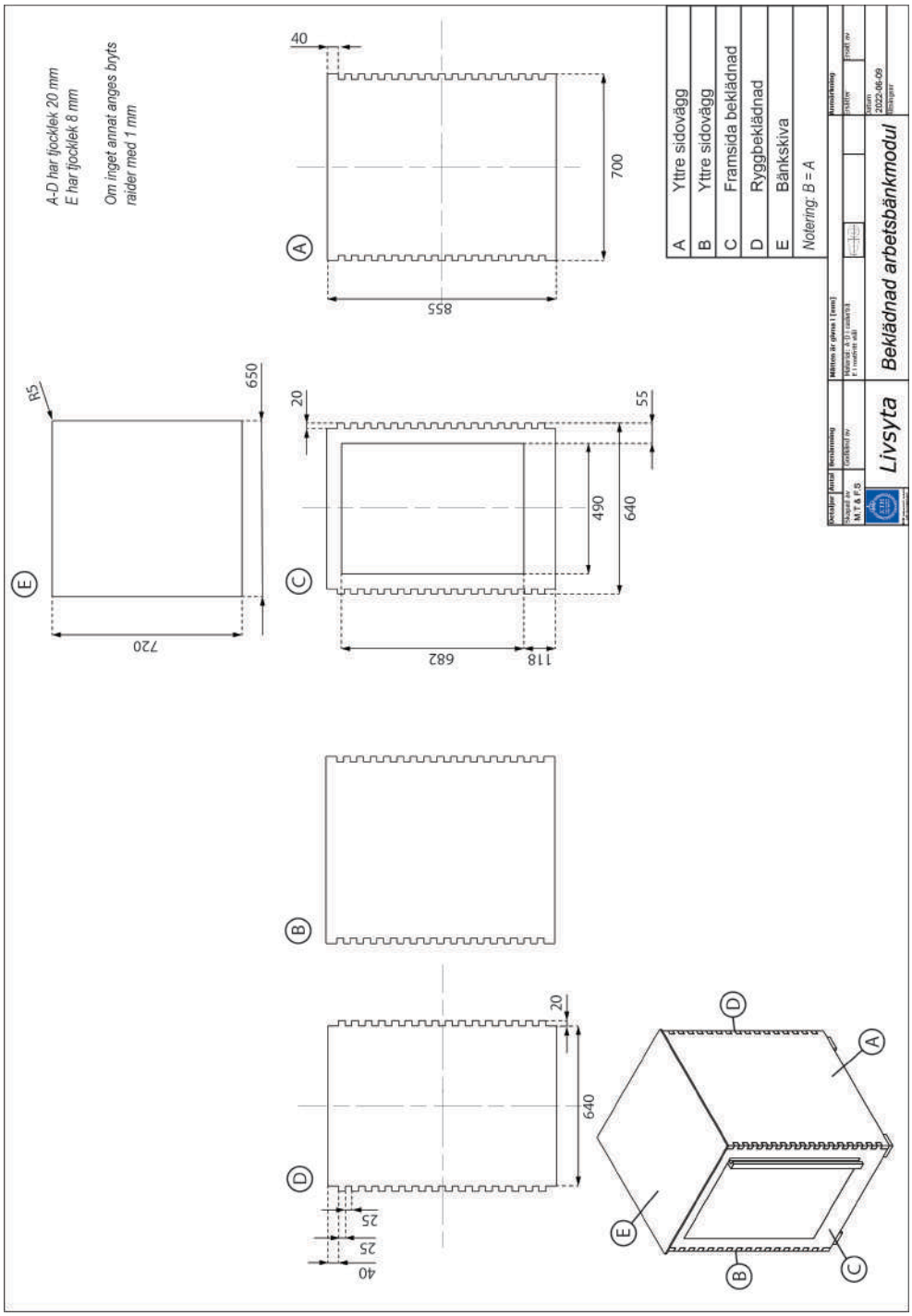
Detal[nr]	Antal	Benämning	Mått[en är givna i [mm]]	Anmärkning
Skapad av: M.T & F.S		Godkänd av:	Material: Tryckimpregnerat trä	Ersätter / Ersatt av:
		Livsyta	Stomme grillmodul	Datum: 2022-06-09

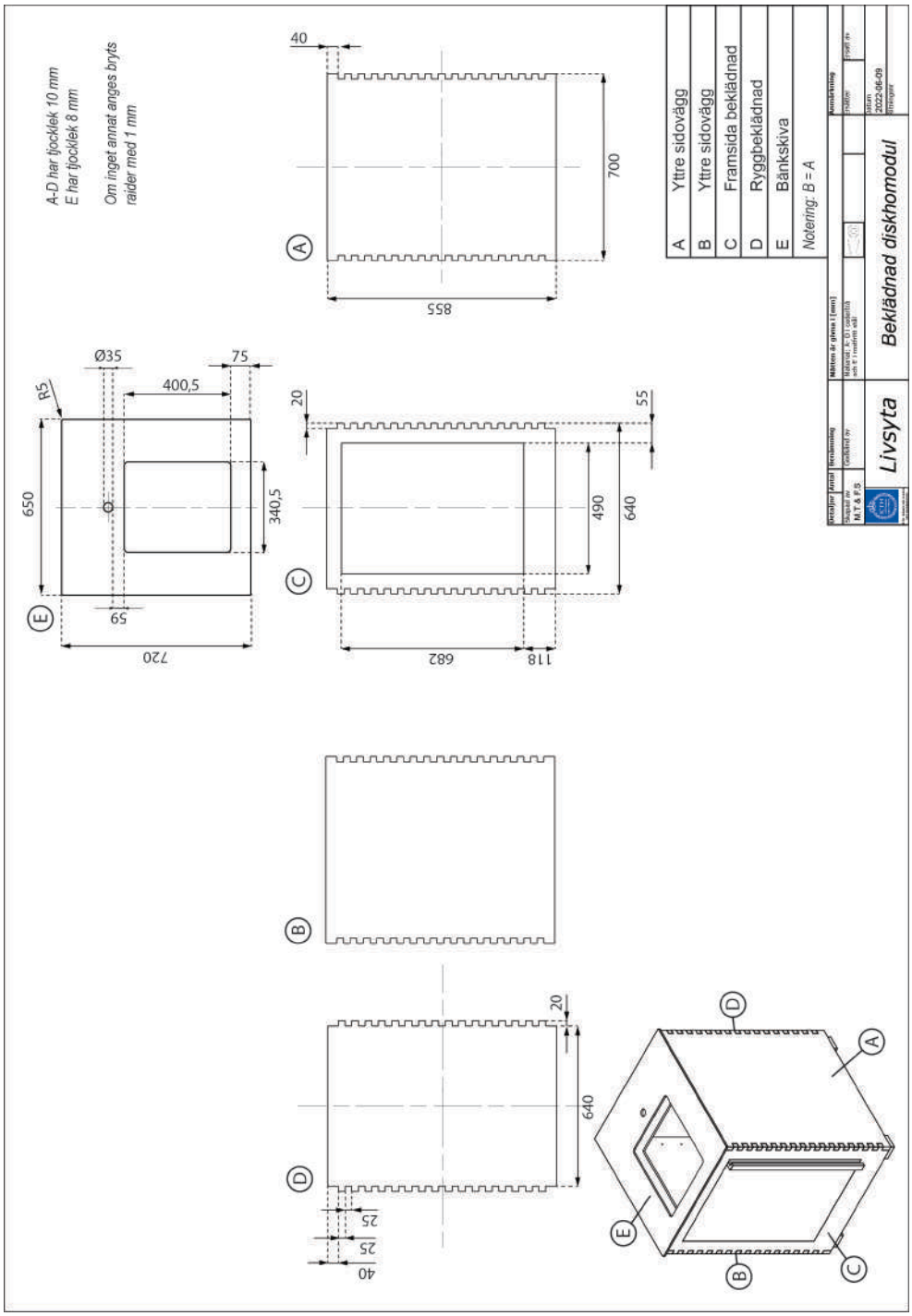

A-E har fjocklek 10 mm
Om inget annat anges bryts raider med 1 mm
A Inner sidovägg
B Inner sidovägg
C Golv arbetsbänk
D Innerygg
E Innetak arbetsbänk
Notering: B = A spegelvänt
Livsyta
Innevägger arbetsbänkmodul
2022-06-09
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692
95
100
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97
45
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717

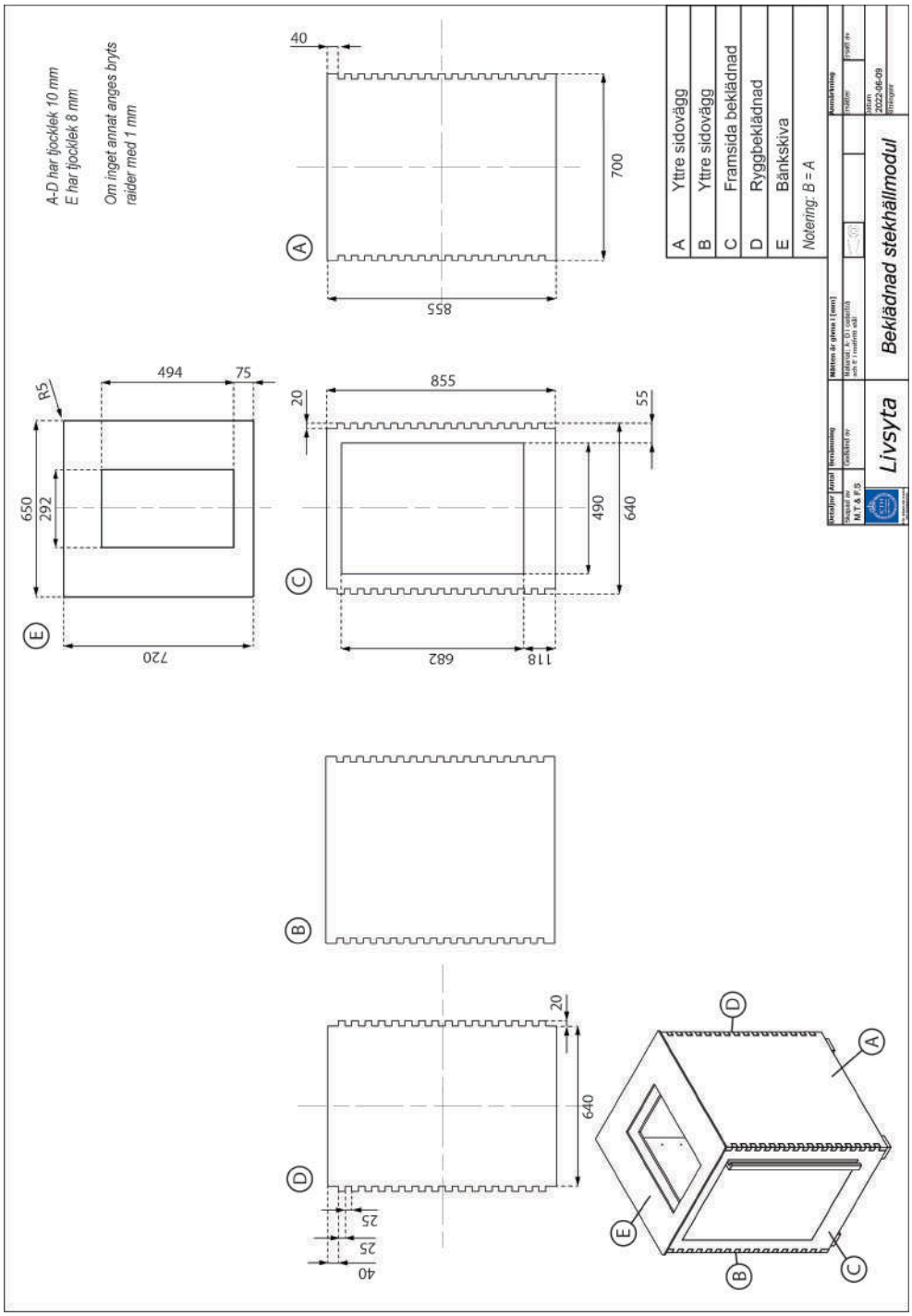

A-E har fjocklek 10 mm
Om inget annat anges bryts raider med 1 mm
A Inner sidovägg
B Inner sidovägg
C Golv diskho
D Innerygg
E Innetak diskho
Notering: B = A spegelvänt
Livsyta
Innevägger diskhomodul
25
692
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717

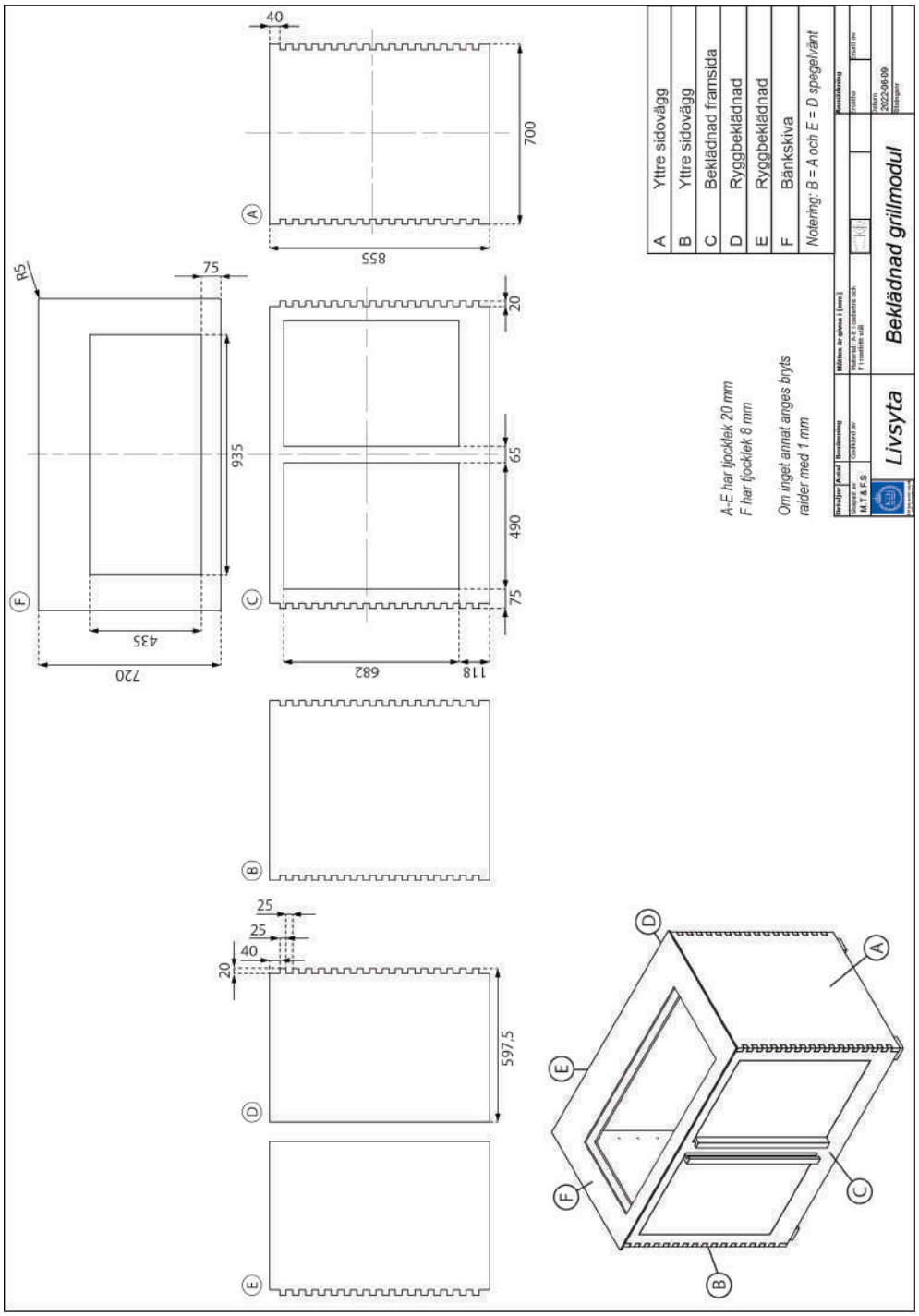

A-E har fjocklek 10 mm
Om inget annat anges bryts raider med 1 mm
A Inner sidovägg
B Inner sidovägg
C Golv stekhäll
D Innerygg
E Innetak stekhäll
Notering: B = A spegelvänt
Livsyta
Innerväggar stekhällmodul
2022-06-09
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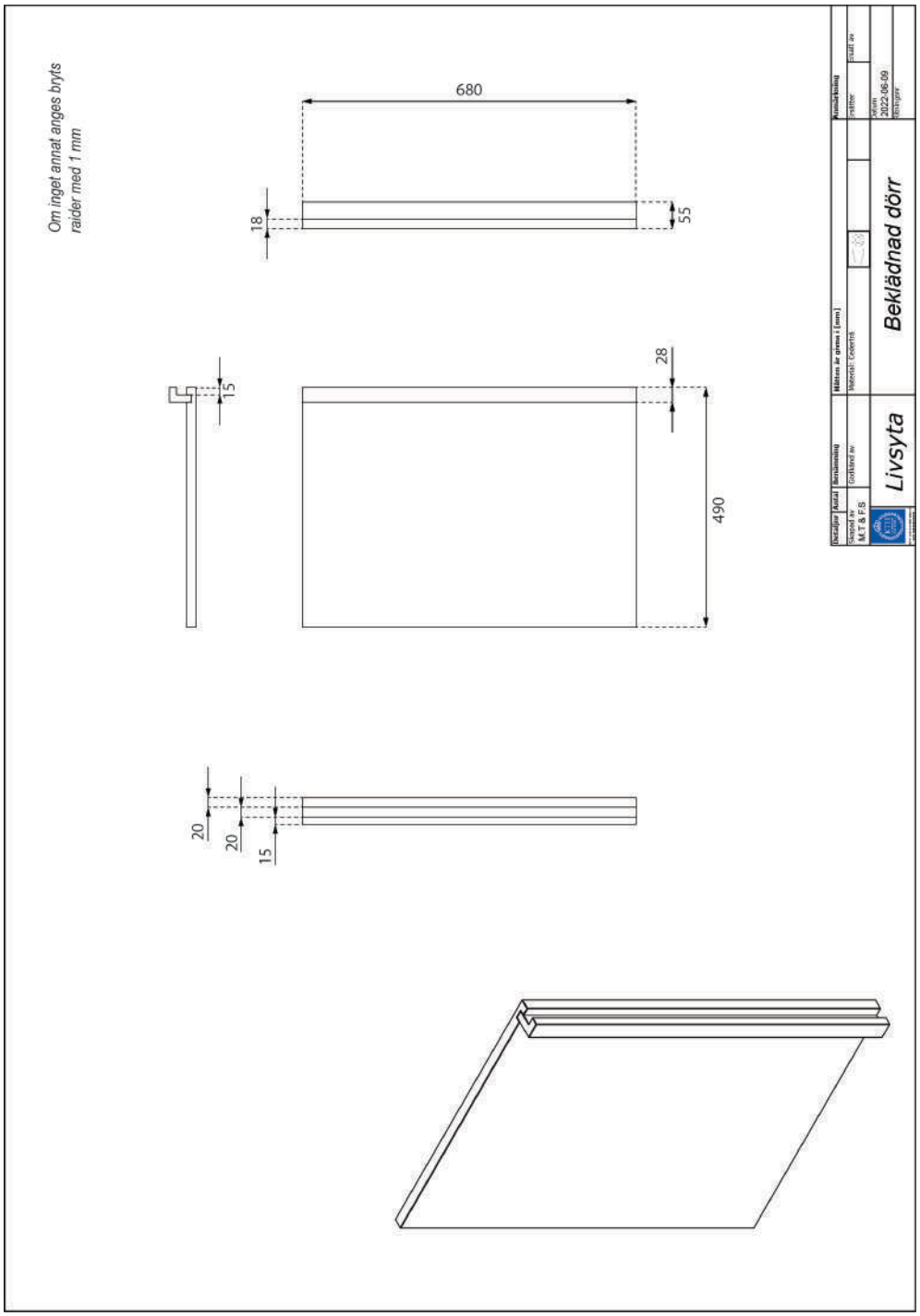

A-E har fjocklek 10 mm
Om inget annat anges bryts raider med 1 mm
A Inner sidovägg
B Inner sidovägg
C Golv grill
D Innergg
E Innetakgrill
Notering: B = A spegelvänt
Livsyta
Innerväggar grillmodul
2022-06-09
25
692
95 100
Ø5
97
45
605
435
1045
1063
1065
717
15
100

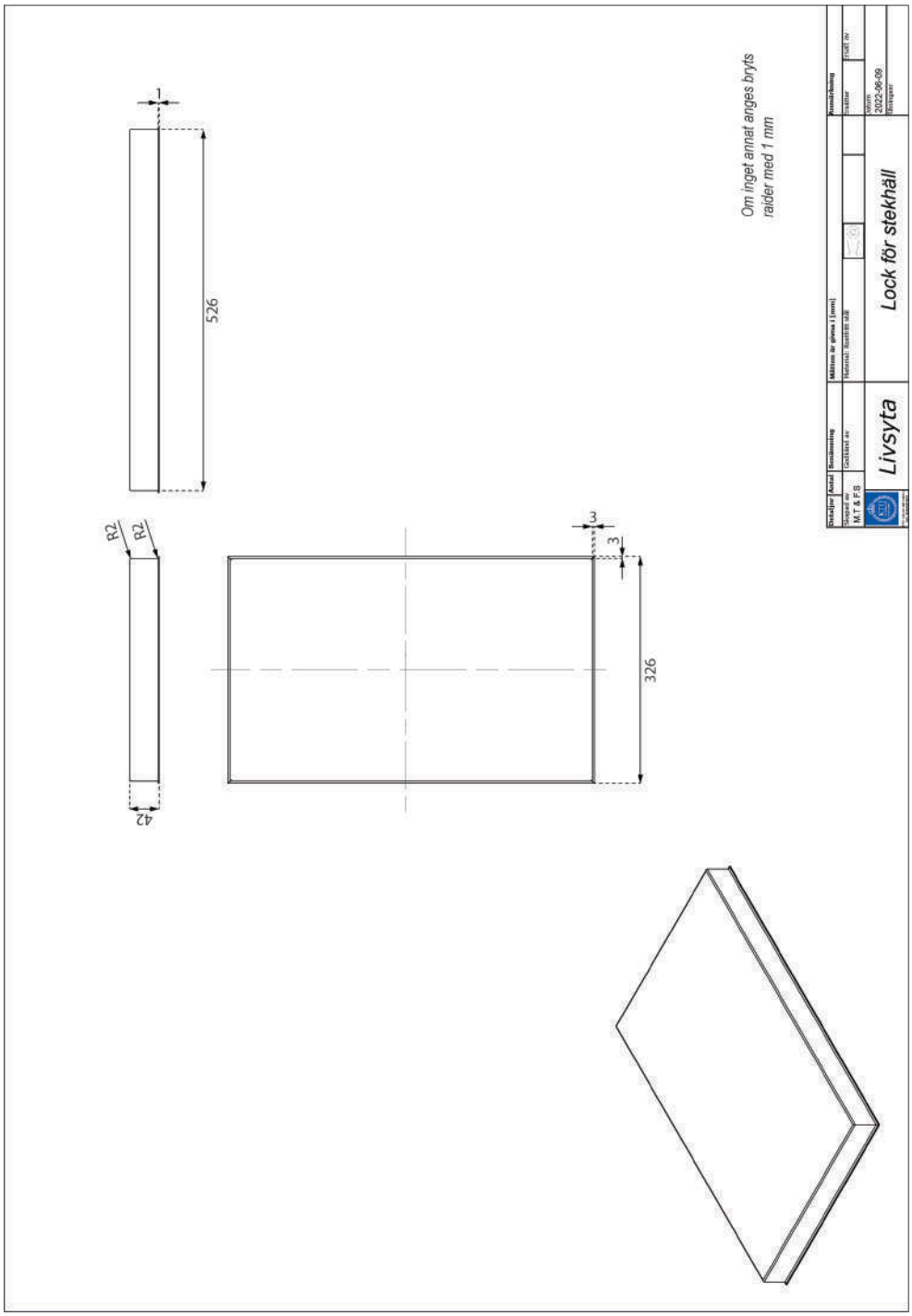

A-D har fjocklek 20 mm
E har fjocklek 8 mm
Om inget annat anges bryts
raider med 1 mm
A Yttre sidovägg
B Yttre sidovägg
C Framsida beklädnad
D Ryggbeklädnad
E Bänkskiva
Notering: B = A
Livsyta
Beklädnad arbetsbänkmodul
2022-06-09
855
700
682
118
490
640
720
650
R5
20
55
40
25


A-D har fjocklek 10 mm
E har fjocklek 8 mm
Om inget annat anges bryts
raider med 1 mm
A Yttre sidovägg
B Yttre sidovägg
C Framsida beklädnad
D Ryggbeklädnad
E Bänkskiva
Notering: B = A
Livsyta
Beklädnad diskhomodul
855
700
40
682
118
640
490
55
20
720
650
340,5
400,5
Ø35
75
R5
59
25
25
40


A-D har fjocklek 10 mm
E har fjocklek 8 mm
Om inget annat anges bryts raider med 1 mm
A Yttre sidovägg
B Yttre sidovägg
C Framsida beklädnad
D Ryggbeklädnad
E Bänkskiva
Notering: B = A
Livsyta
Beklädnad stekhällmodul
2022-06-09
855
700
40
494
75
R5
650
292
720
855
20
55
490
640
682
118
640
20
25
25
40

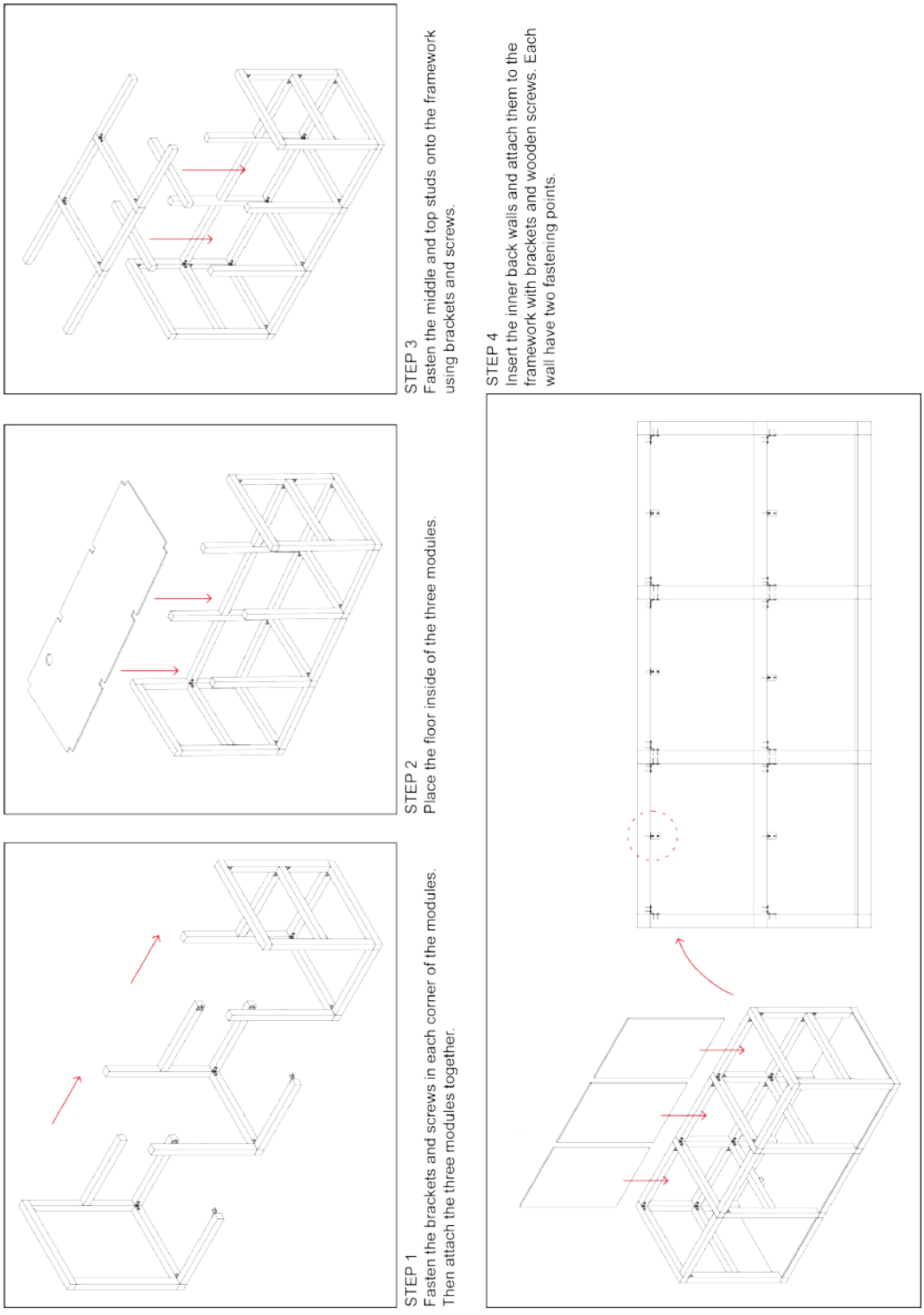

A Yttre sidovägg
B Yttre sidovägg
C Beklädnad framsida
D Ryggbeklädnad
E Ryggbeklädnad
F Bänkskiva
Notering: B = A och E = D spegelvänt
A-E har tjocklek 20 mm
F har tjocklek 8 mm
Om inget annat anges bryts
raider med 1 mm
Livsyta
Beklädnad grillmodul
855
700
682
118
490
65
20
75
935
435
720
R5
597,5
40
25
25
20

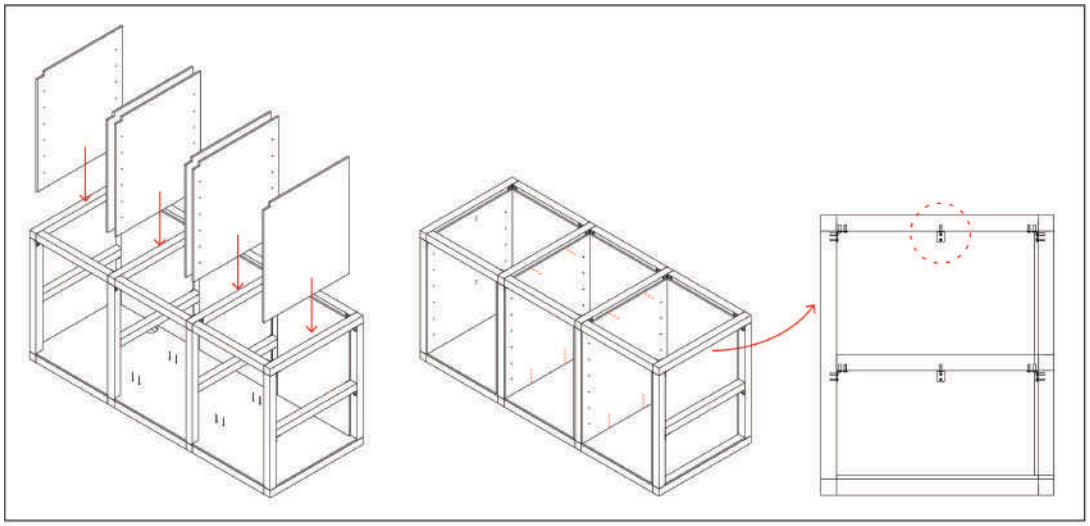

Om inget annat anges bryts raider med 1 mm
680
18
55
15
28
490
20
20
15
Livsyta
Beklädnad dörr


Om inget annat anges bryts
raider med 1 mm
526
R2
R2
42
3
326
Livsyta
Lock för stekhäll

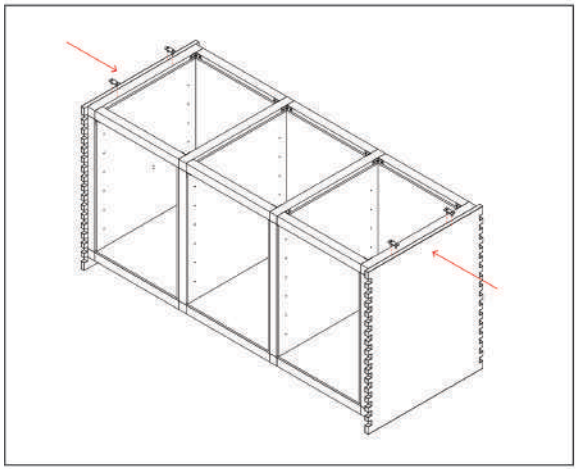
Appendix G

The step by step guide for building the outdoor kitchen.

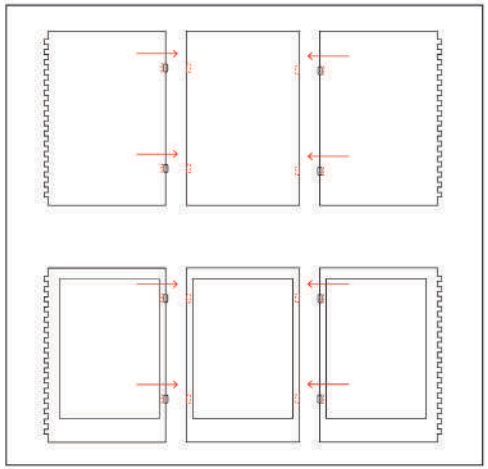




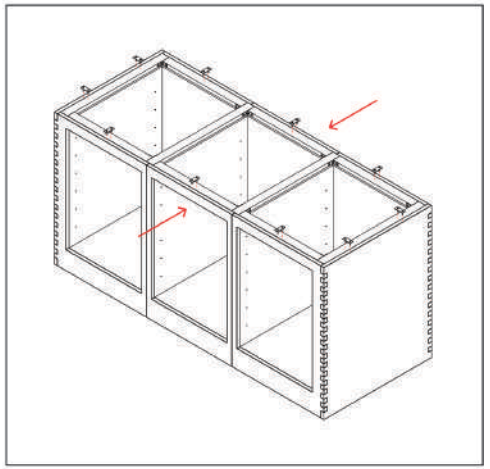
STEP 5
Insert the inner side walls, starting from the middle towards the ends of the outdoor kitchen. The walls in the middle are fastened with screws from the floor and screws at the top. The walls on the side of the outdoor kitchen are attacthed in the framework with brackets and screws. The walls in the middle have four fastening points each meanwhile the two walls on the side have two fastening points.



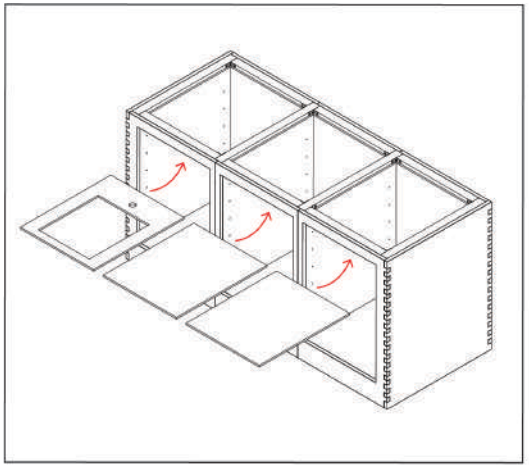
STEP 6
Fasten the side walls of the outer cladding onto the framework using flat brackets and screws. The walls have two fastening points.



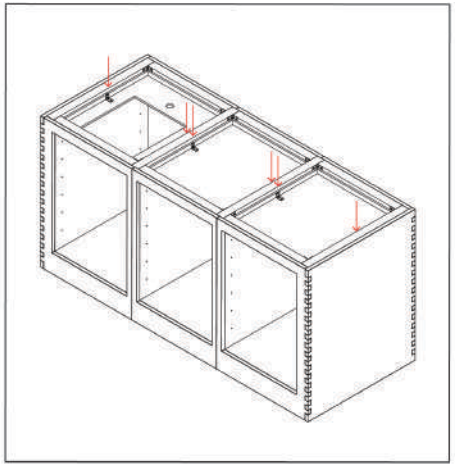
STEP 7
Fasten the back and front outer cladding with wooden chips and screws.



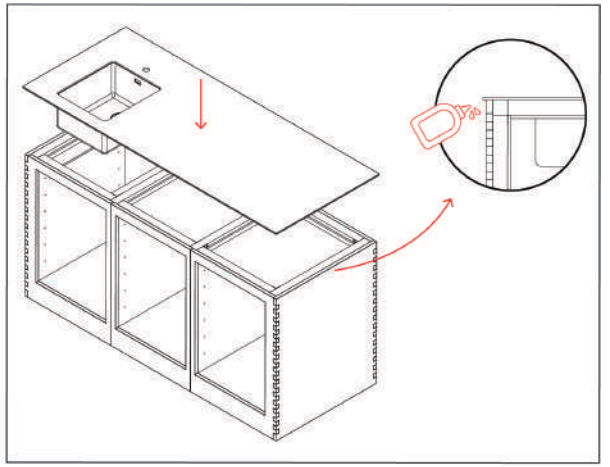
STEP 8
Mount the back and front outer cladding onto the framework using flat brackets and screws.



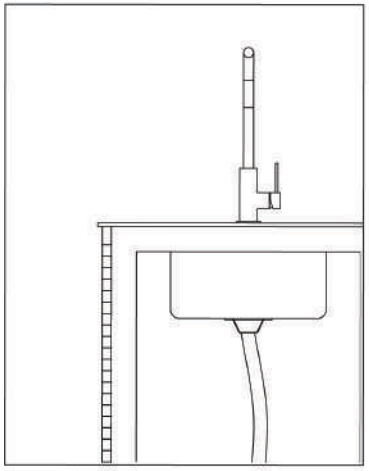
STEP 9
Insert the inner roofs.



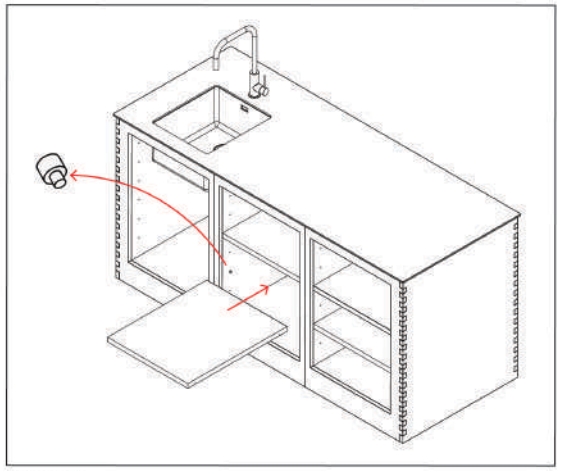
STEP 10
Fasten the inner roofs with brackets and screws.
Each inner roofs have two fastening points.



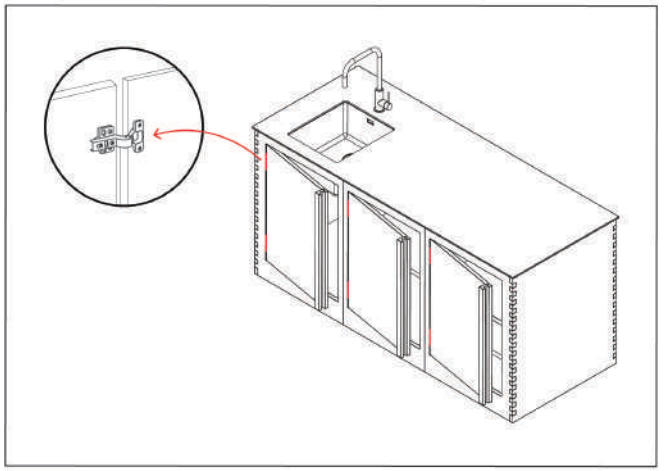
STEP 11
Fasten the counter top with glue around the whole framework top.
When the glued have hardened, use a seleant around the countertop.



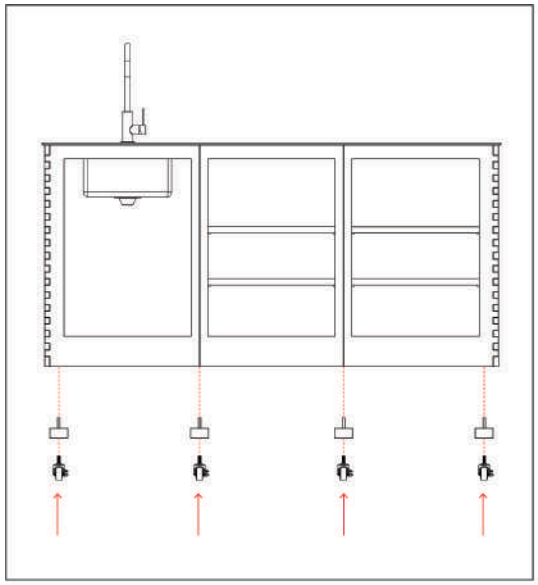
STEP 12
Attach the brackets and wood screws in each corner of the modules. Then attach the three modules together.



STEP 13
Insert the shelves using the shelf-pins.

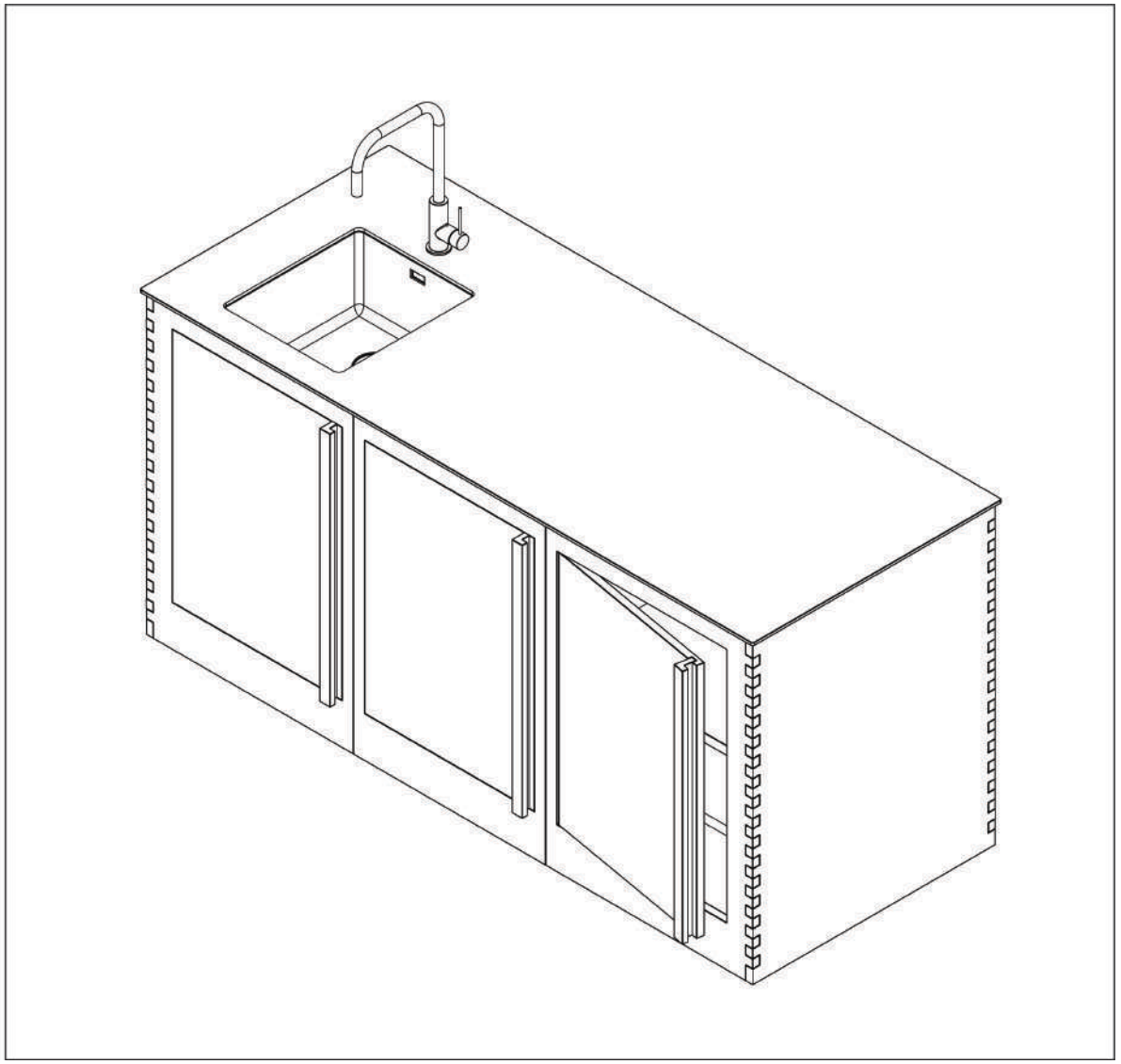


STEP 14
Install the hinges for the doors. The doors have two hinges each.



STEP 15
Install the wheels on the bottom of the framework. Using the small feet first that are screwed into the framework. Then attach the wheels to the small feet.

FINISHED
The outdoor kitchen is finalized.



Appendix H

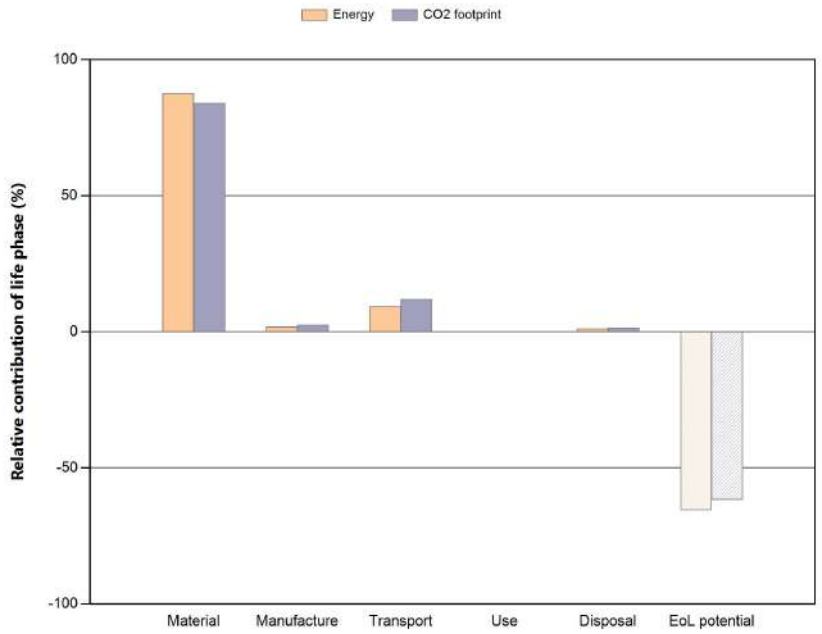
The sustainability report created in *GRANTA EduPack* for one module of the final concept design.

Ansys
GRANTA EDUPACK

Eco Audit Report

Product name	Utekök
Country of use	Europe
Product life (years)	10

Summary:

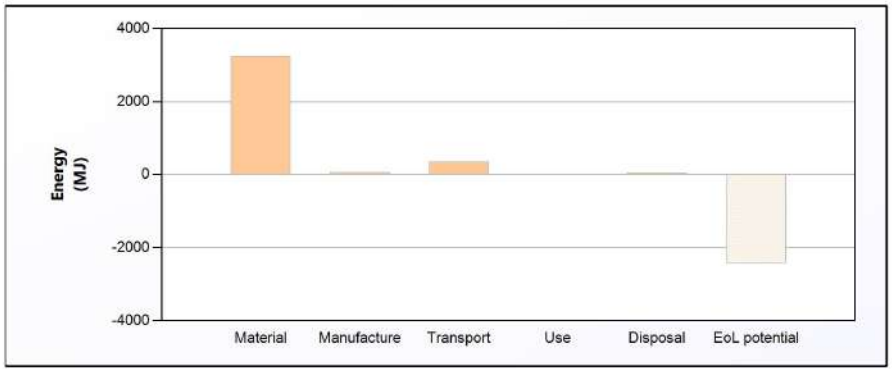


Energy details CO2 footprint details

Phase	Energy (MJ)	Energy (%)	CO2 footprint (kg)	CO2 footprint (%)
Material	3,24e+03	87,7	172	84,0
Manufacture	67,5	1,8	5,06	2,5
Transport	342	9,2	24,6	12,0
Use	0	0,0	0	0,0
Disposal	44,6	1,2	3,12	1,5
Total (for first life)	**3,7e+03**	**100**	**205**	**100**
End of life potential	-2,42e+03		-127	

Energy Analysis

Summary



	Energy (MJ/year)
Equivalent annual environmental burden (averaged over 10 year product life):	370

Detailed breakdown of individual life phases

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Ramverk	Pine (pinus sylvestris) (l)	Virgin (0%)	11	1	11	1,2e+02	3,7
Tak och golv	Pine (pinus sylvestris) (l)	Virgin (0%)	2,8	2	5,6	62	1,9
Innervägg	Pine (pinus sylvestris) (l)	Virgin (0%)	5,4	3	16	1,8e+02	5,5
Yttervägg	Cedar (chamaecyparis lawsoniana) (l)	Virgin (0%)	22	4	88	9,7e+02	29,9
Bänkskiva	Stainless steel, austenitic, AISI 304, annealed	Virgin (0%)	28	1	28	1,9e+03	57,5
Ben	Pine (pinus sylvestris) (l)	Virgin (0%)	1	4	4	44	1,4
Total				**15**	**1,5e+02**	**3,2e+03**	**100**

*Typical: Includes 'recycle fraction in current supply'

Manufacture:

Summary

Component	Process	Amount processed	Energy (MJ)	%
Bänkskiva	Forging	28 kg	67	100,0
Total			**67**	**100**

Transport:

Summary

Breakdown by transport stage

Stage name	Transport type	Distance (km)	Energy (MJ)	%
Cedar 1	Ocean freight	6,2e+03	1,7e+02	50,0
Cedar 2	40 tonne (6 axle) truck	5e+02	63	18,3
Fura 1	55 tonne (8 axle) truck	5e+02	54	15,9
Stål 1	55 tonne (8 axle) truck	5e+02	54	15,9
Total		**7,7e+03**	**3,4e+02**	**100**

Breakdown by components

Component	Mass (kg)	Energy (MJ)	%
Ramverk	11	25	7,2
Tak och golv	5,6	13	3,7
Innervägg	16	36	10,6
Ytterväggg	88	2e+02	57,6
Bänkskiva	28	63	18,3
Ben	4	9	2,6
Total	**1,5e+02**	**3,4e+02**	**100**

Use:

Summary

Relative contribution of static and mobile modes

Mode	Energy (MJ)	%
Static	0	
Mobile	0	
Total	**0**	**100**

Disposal:

Summary

Component	End of life option	Energy (MJ)	%
Ramverk	Landfill	2,2	4,9
Tak och golv	Landfill	1,1	2,5
Innervägg	Landfill	3,2	7,3
Yttervägg	Reuse	18	39,5
Bänkskiva	Recycle	20	44,0
Ben	Landfill	0,8	1,8
Total		**45**	**100**

EoL potential:

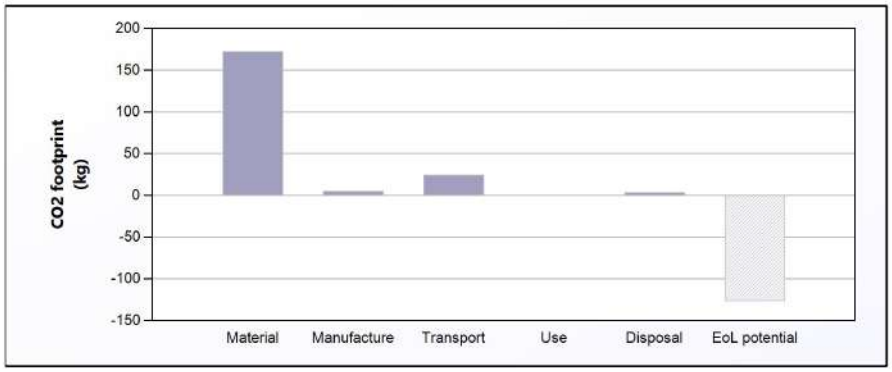
Component	End of life option	Energy (MJ)	%
Ramverk	Landfill	0	0,0
Tak och golv	Landfill	0	0,0
Innervägg	Landfill	0	0,0
Yttervägg	Reuse	-9,7e+02	40,1
Bänkskiva	Recycle	-1,5e+03	59,9
Ben	Landfill	0	0,0
Total		**-2,4e+03**	**100**

Notes:

Summary

CO2 Footprint Analysis

Summary



	CO2 (kg/year)
Equivalent annual environmental burden (averaged over 10 year product life):	20,5

Detailed breakdown of individual life phases

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	CO2 footprint (kg)	%
Ramverk	Pine (pinus sylvestris) (l)	Virgin (0%)	11	1	11	4	2,3
Tak och golv	Pine (pinus sylvestris) (l)	Virgin (0%)	2,8	2	5,6	2	1,2
Innervägg	Pine (pinus sylvestris) (l)	Virgin (0%)	5,4	3	16	5,9	3,4
Yttervägg	Cedar (chamaecyparis lawsoniana) (l)	Virgin (0%)	22	4	88	32	18,7
Bänkskiva	Stainless steel, austenitic, AISI 304, annealed	Virgin (0%)	28	1	28	1,3e+02	73,5
Ben	Pine (pinus sylvestris) (l)	Virgin (0%)	1	4	4	1,5	0,8
Total				**15**	**1,5e+02**	**1,7e+02**	**100**

*Typical: Includes 'recycle fraction in current supply'

Manufacture:

Summary

Component	Process	Amount processed	CO2 footprint (kg)	%
Bänkskiva	Forging	28 kg	5,1	100,0
Total			**5,1**	**100**

Transport:

Summary

Breakdown by transport stage

Stage name	Transport type	Distance (km)	CO2 footprint (kg)	%
Cedar 1	Ocean freight	6,2e+03	12	50,0
Cedar 2	40 tonne (6 axle) truck	5e+02	4,5	18,3
Fura 1	55 tonne (8 axle) truck	5e+02	3,9	15,9
Stål 1	55 tonne (8 axle) truck	5e+02	3,9	15,9
Total		**7,7e+03**	**25**	**100**

Breakdown by components

Component	Mass (kg)	CO2 footprint (kg)	%
Ramverk	11	1,8	7,2
Tak och golv	5,6	0,9	3,7
Innervägg	16	2,6	10,6
Ytterväg	88	14	57,6
Bänkskiva	28	4,5	18,3
Ben	4	0,64	2,6
Total	**1,5e+02**	**25**	**100**

Use:

Summary

Relative contribution of static and mobile modes

Mode	CO2 footprint (kg)	%
Static	0	
Mobile	0	
Total	**0**	**100**

Disposal:

Summary

Component	End of life option	CO2 footprint (kg)	%
Ramverk	Landfill	0,15	4,9
Tak och golv	Landfill	0,078	2,5
Innervägg	Landfill	0,23	7,3
Yttervägg	Reuse	1,2	39,5
Bänkskiva	Recycle	1,4	44,0
Ben	Landfill	0,056	1,8
Total		**3,1**	**100**

EoL potential:

Component	End of life option	CO2 footprint (kg)	%
Ramverk	Landfill	0	0,0
Tak och golv	Landfill	0	0,0
Innervägg	Landfill	0	0,0
Yttervägg	Reuse	-32	25,4
Bänkskiva	Recycle	-94	74,6
Ben	Landfill	0	0,0
Total		**-1,3e+02**	**100**

Notes:

Summary

TRITA – ITM-EX 2022:377